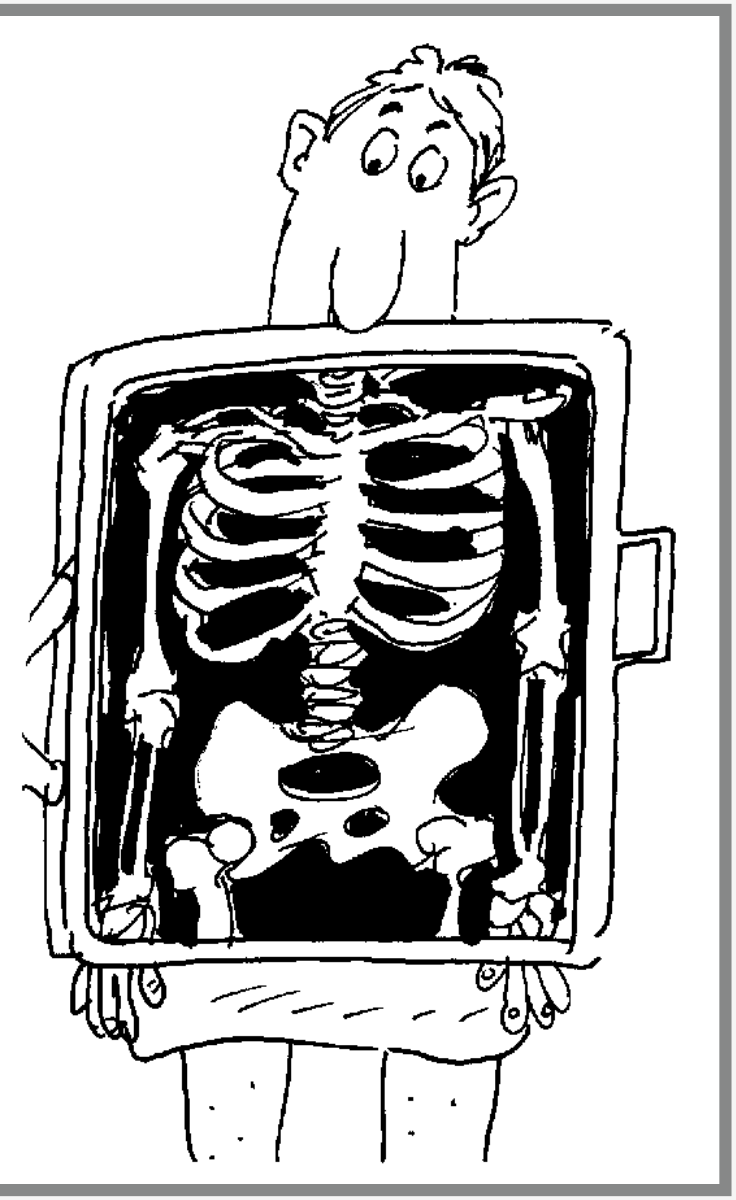

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Chest Radiography: The Basics

Joseph Mazzie D.O.
Department of Clinical Specialties, Radiology

- The Key: A systematic approach

  - My approach:

    - Assess study technique and quality
    - Tubes and lines/Hardware/devices**
    - Look at the partially imaged abdomen
    - Bones and soft tissues
    - Heart/Mediastinum
    - Lungs and pleura- each one individually and then together for symmetry

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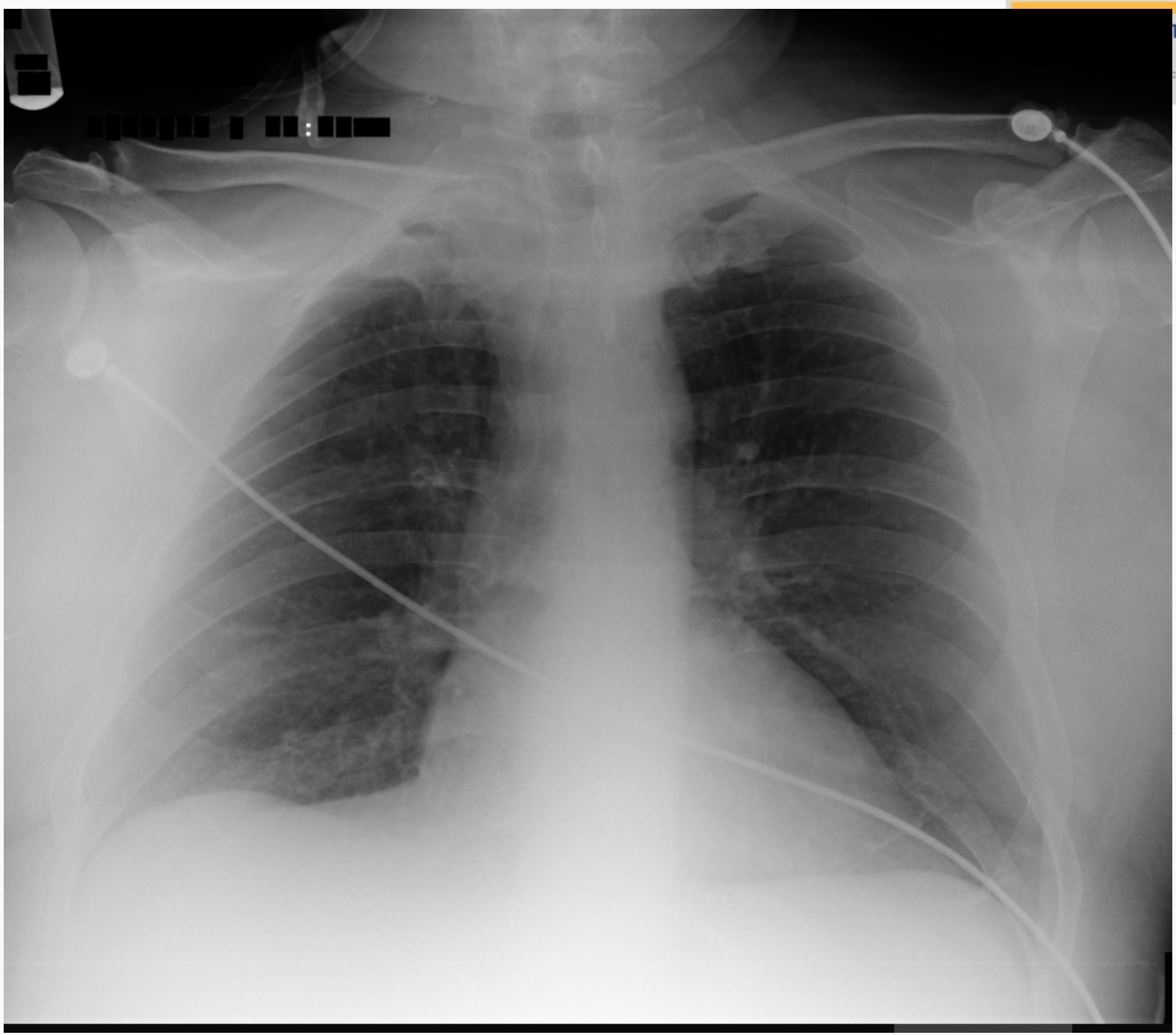
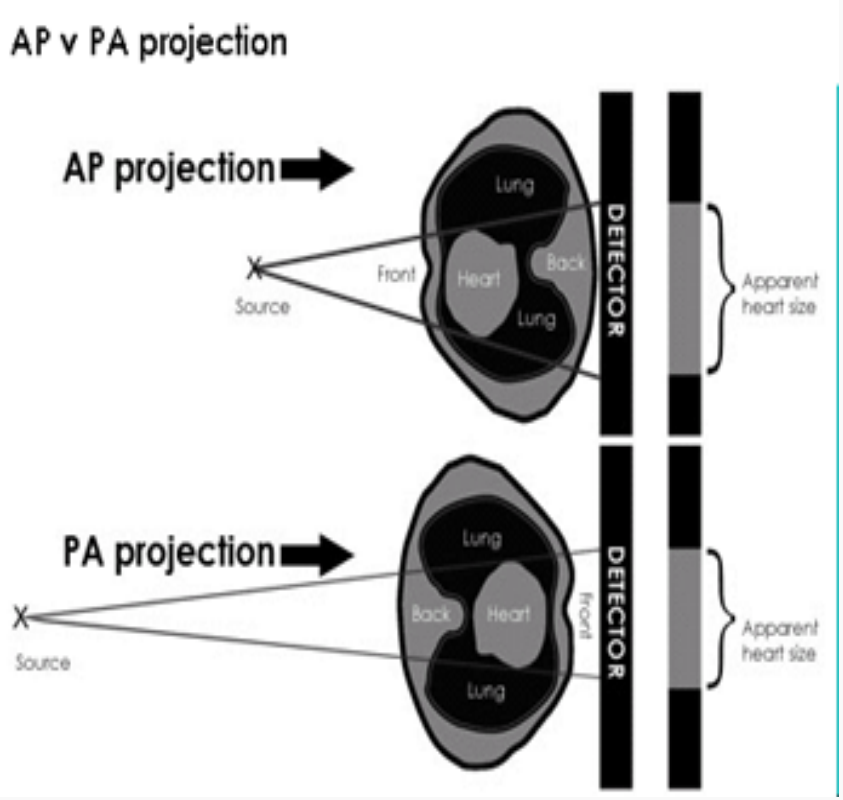


AP vs. PA Radiograph



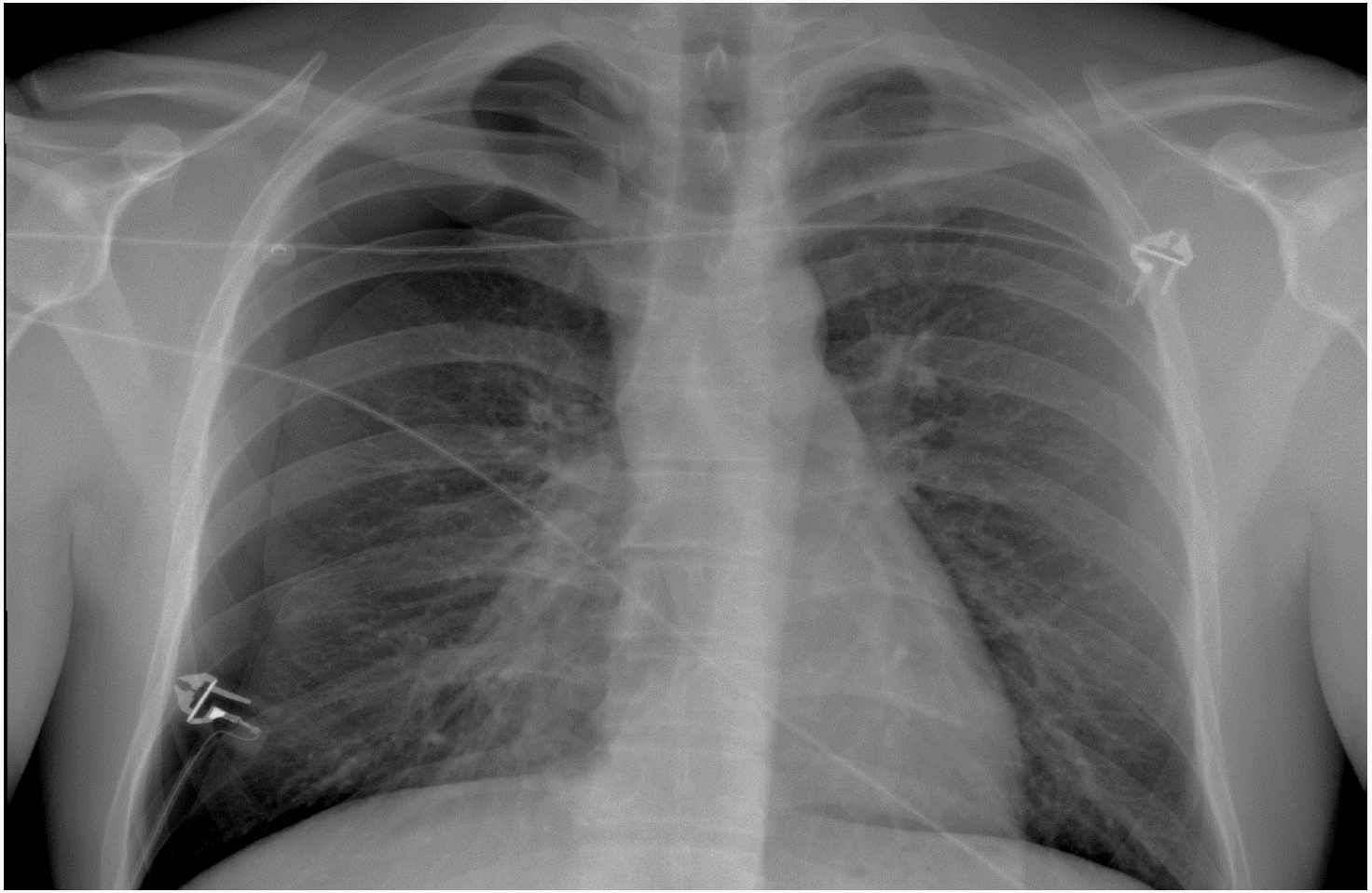
In an AP radiograph, the heart will therefore appear larger, and the mediastinum will be magnified

- Pneumothorax
- Pneumomediastinum
- Misplaced ET tube
- Misplaced NG tube
- Other misplaced tubes and devices
- Findings suggestive of active TB
- Pericardial effusion
- Rib fractures
- Pneumonia
- Free Air under the diaphragm
- Complete lung collapse

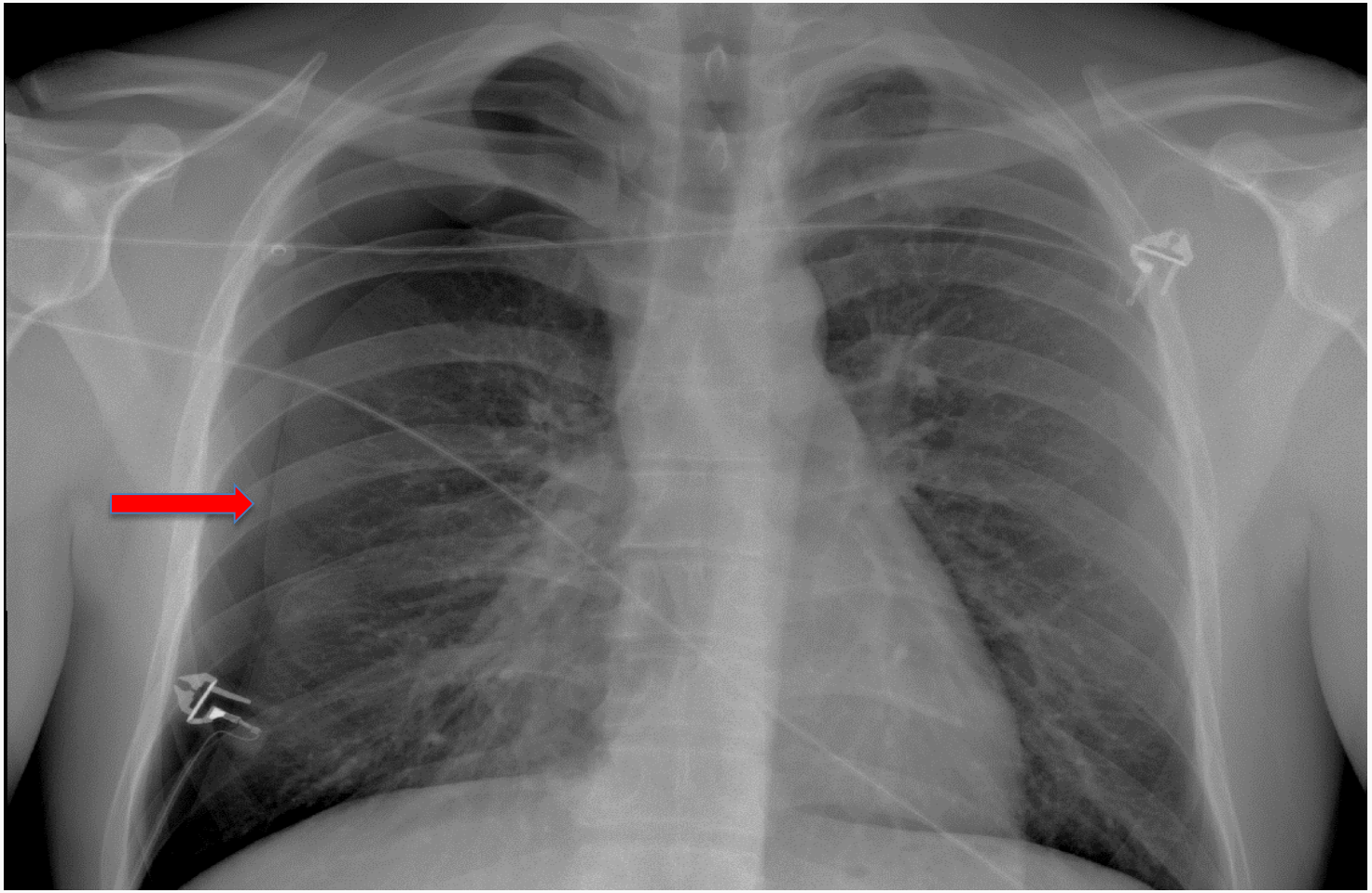
*Pneumothorax

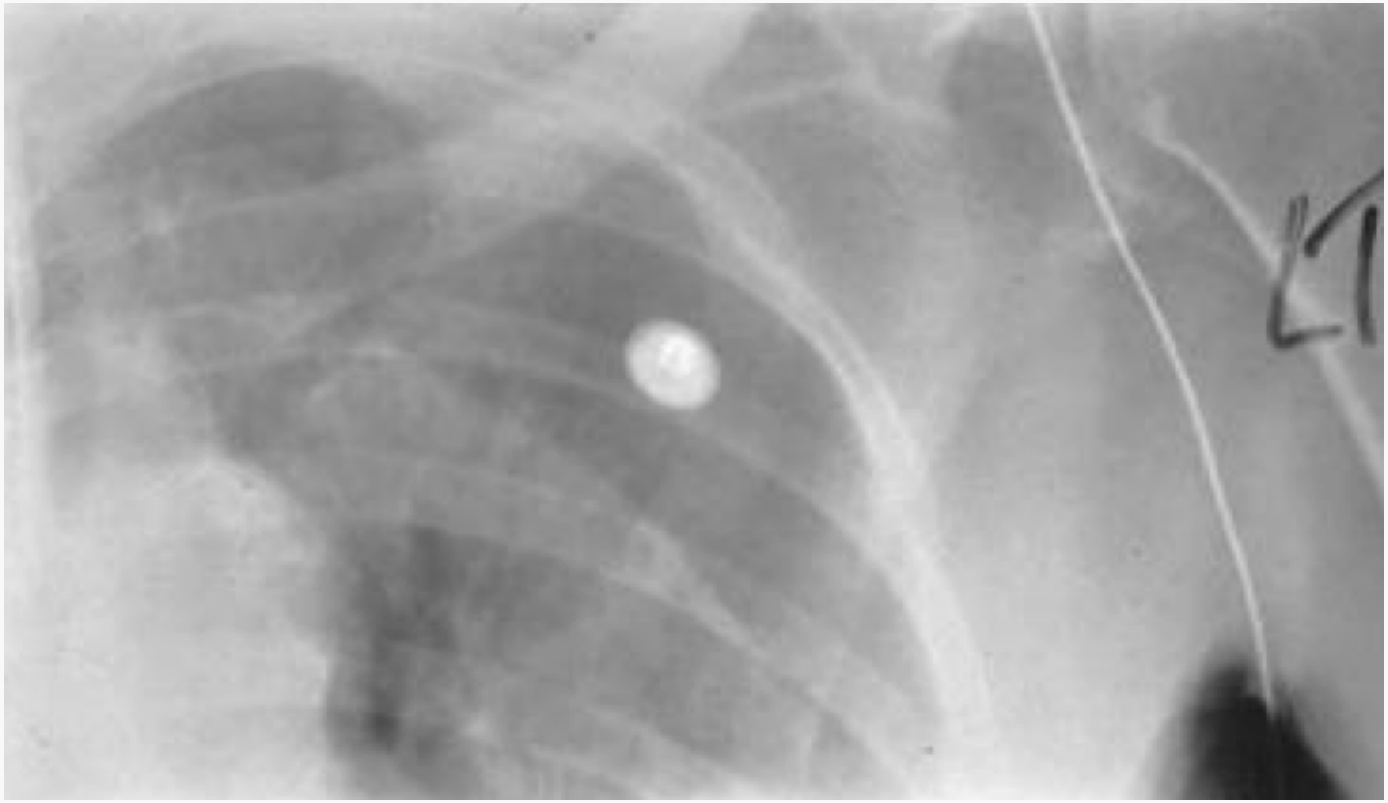
- Air in the pleural space-may be spontaneous (rupture of a bleb), traumatic (esp. secondary to rib fractures) or iatrogenic (status post tube placement/removal or biopsy)
- Key findings: a THIN pleural line which can not extend beyond the confines of the thorax with absence of lung markings peripheral to the visceral (inner) pleura
- Pitfalls: Skin folds, scapula/ribs, overlying sheets, supine positioning
- Look out for mediastinal shift suggestive of tension

*Pneumothorax

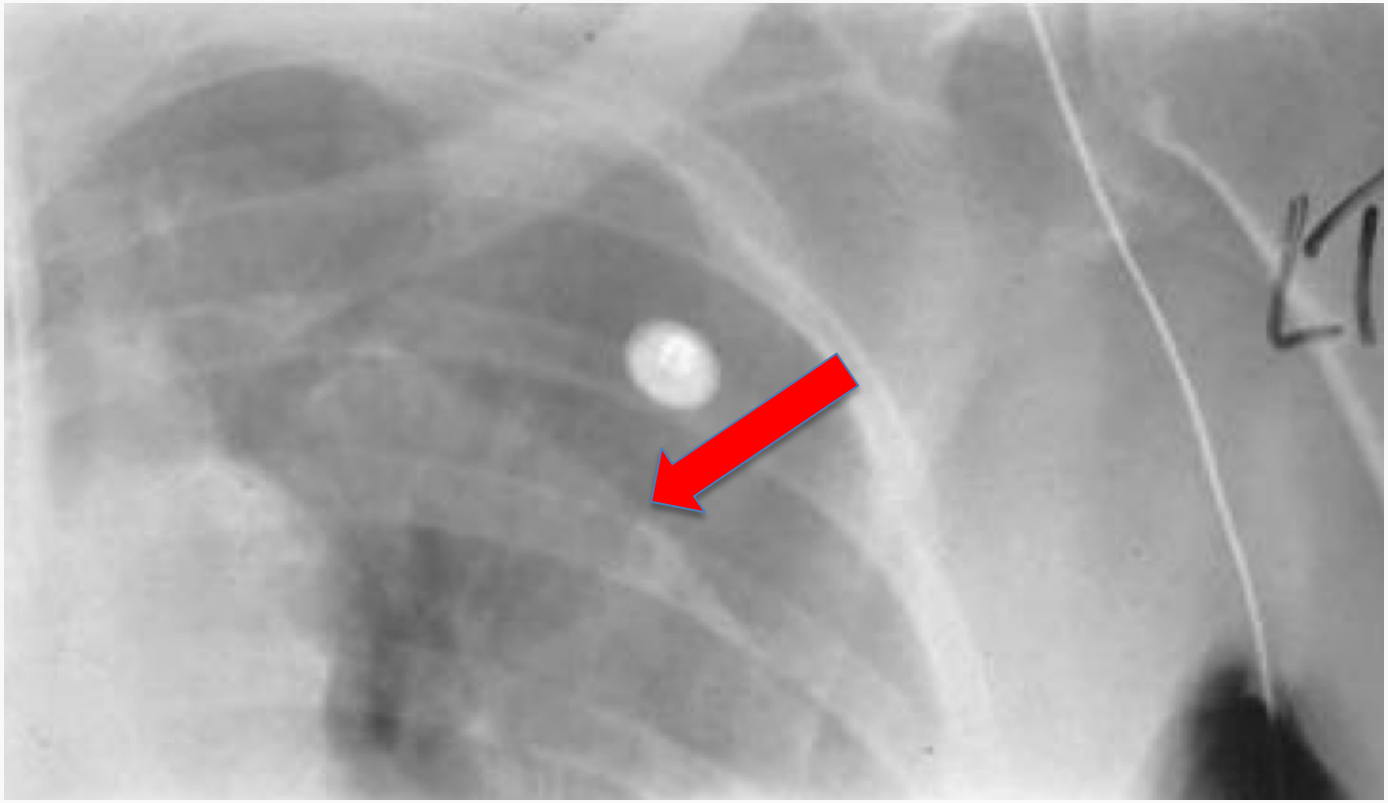


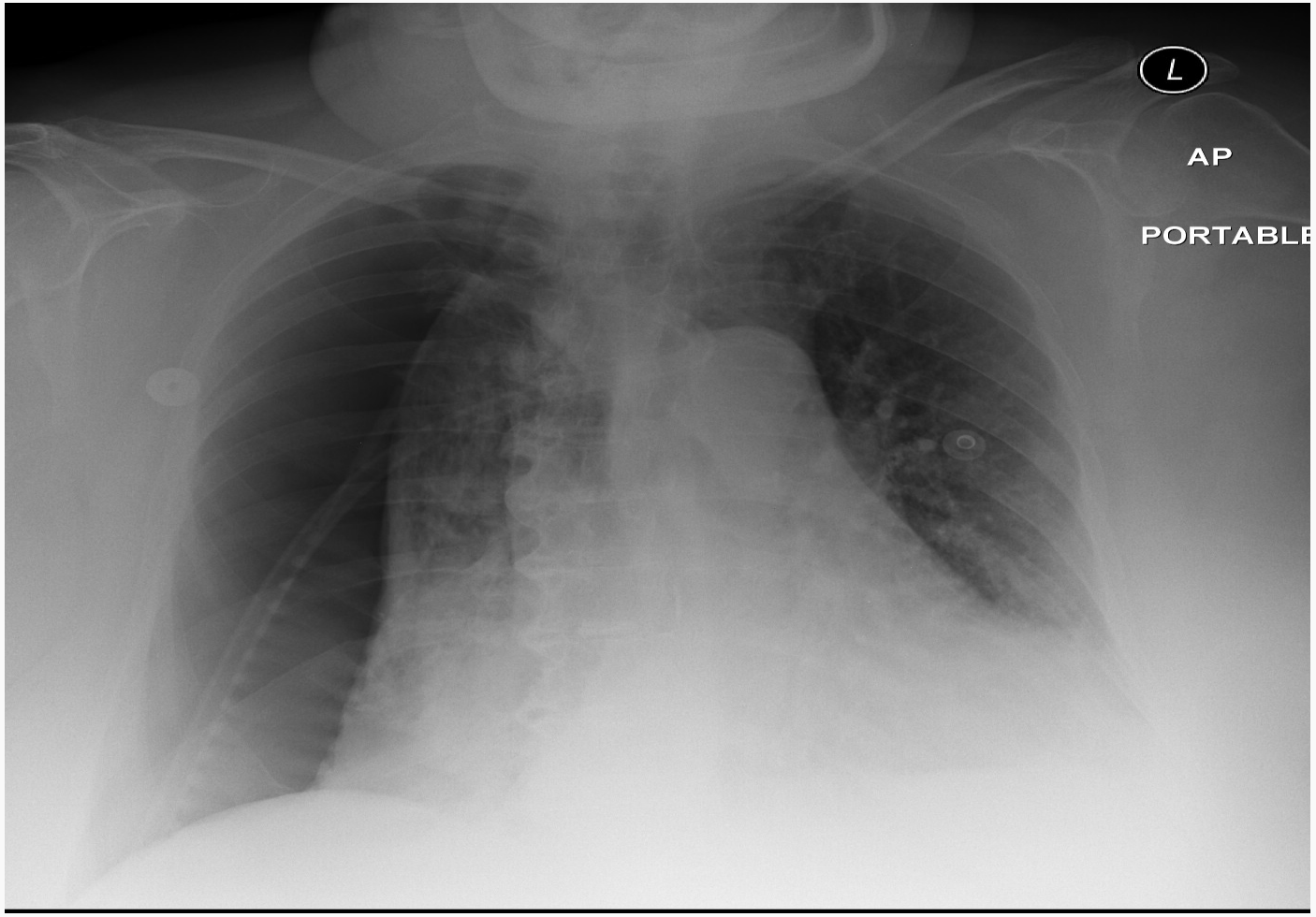
Pneumothorax



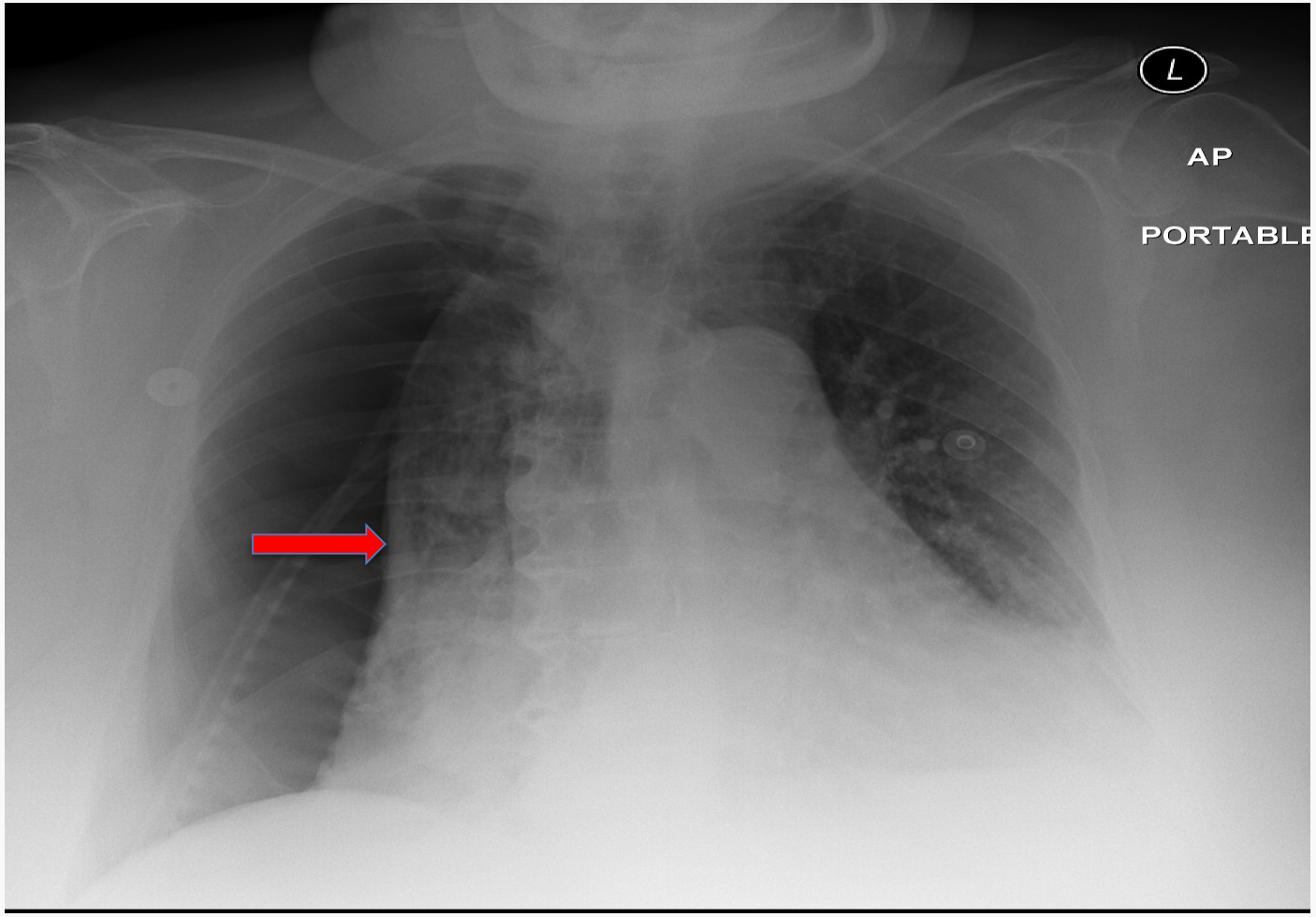


Skin fold




L
AP
PORTABLE

Tension Pneumothorax



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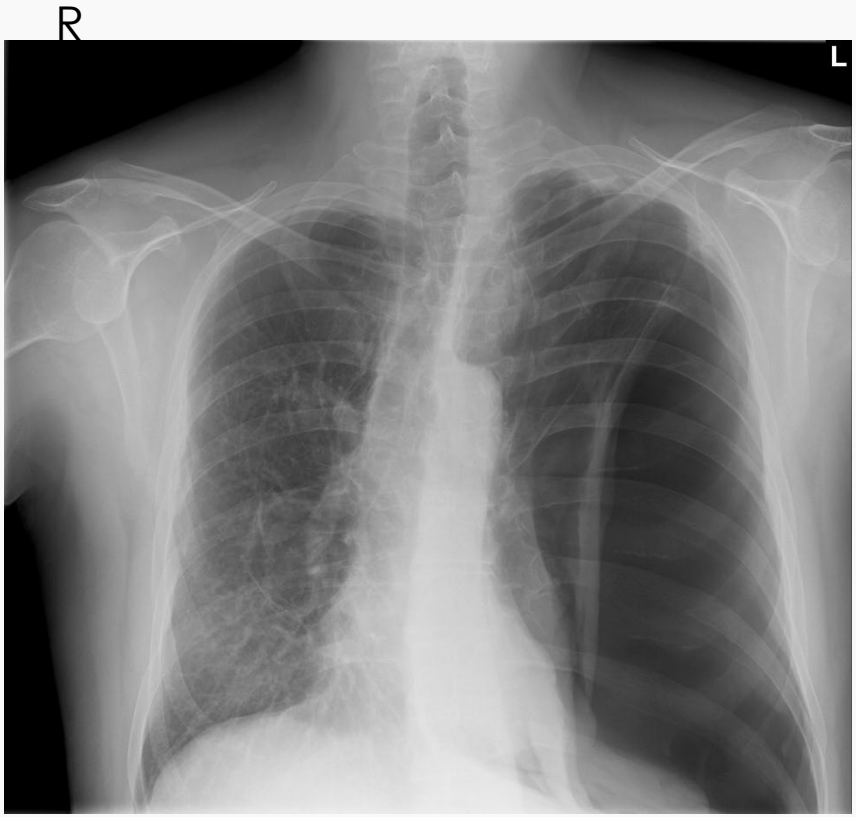

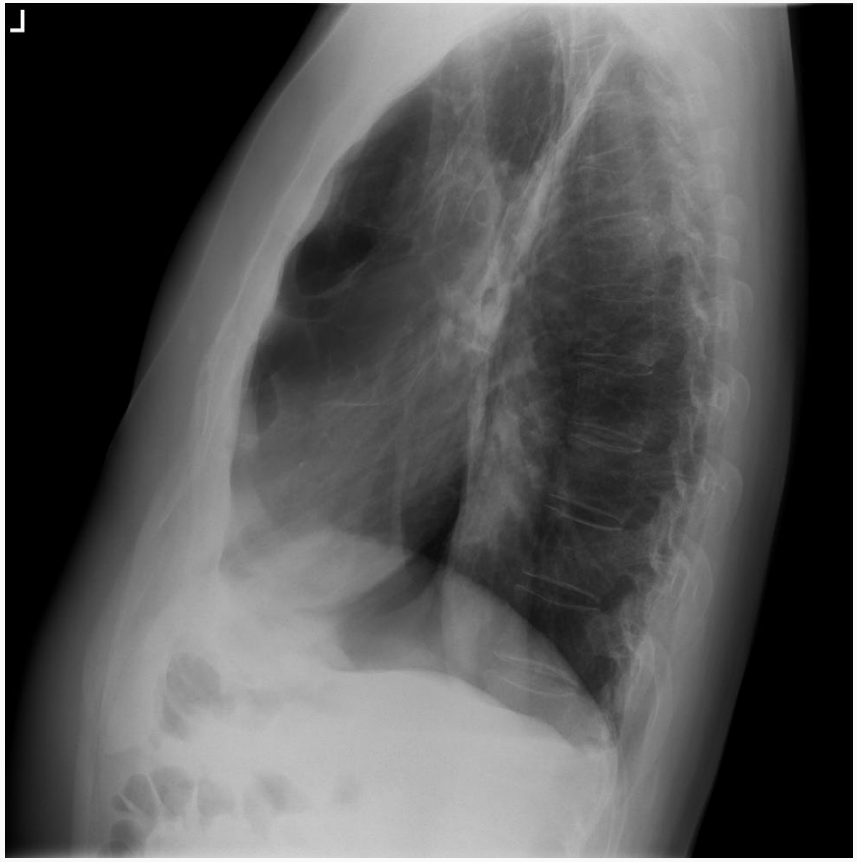

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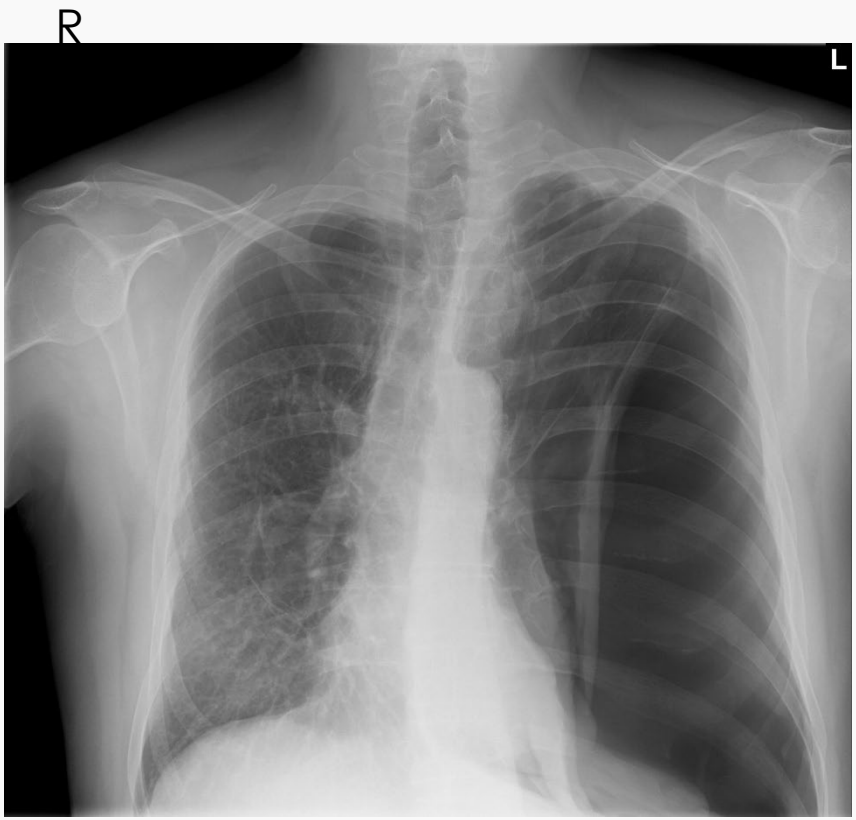
Left tension pneumothorax



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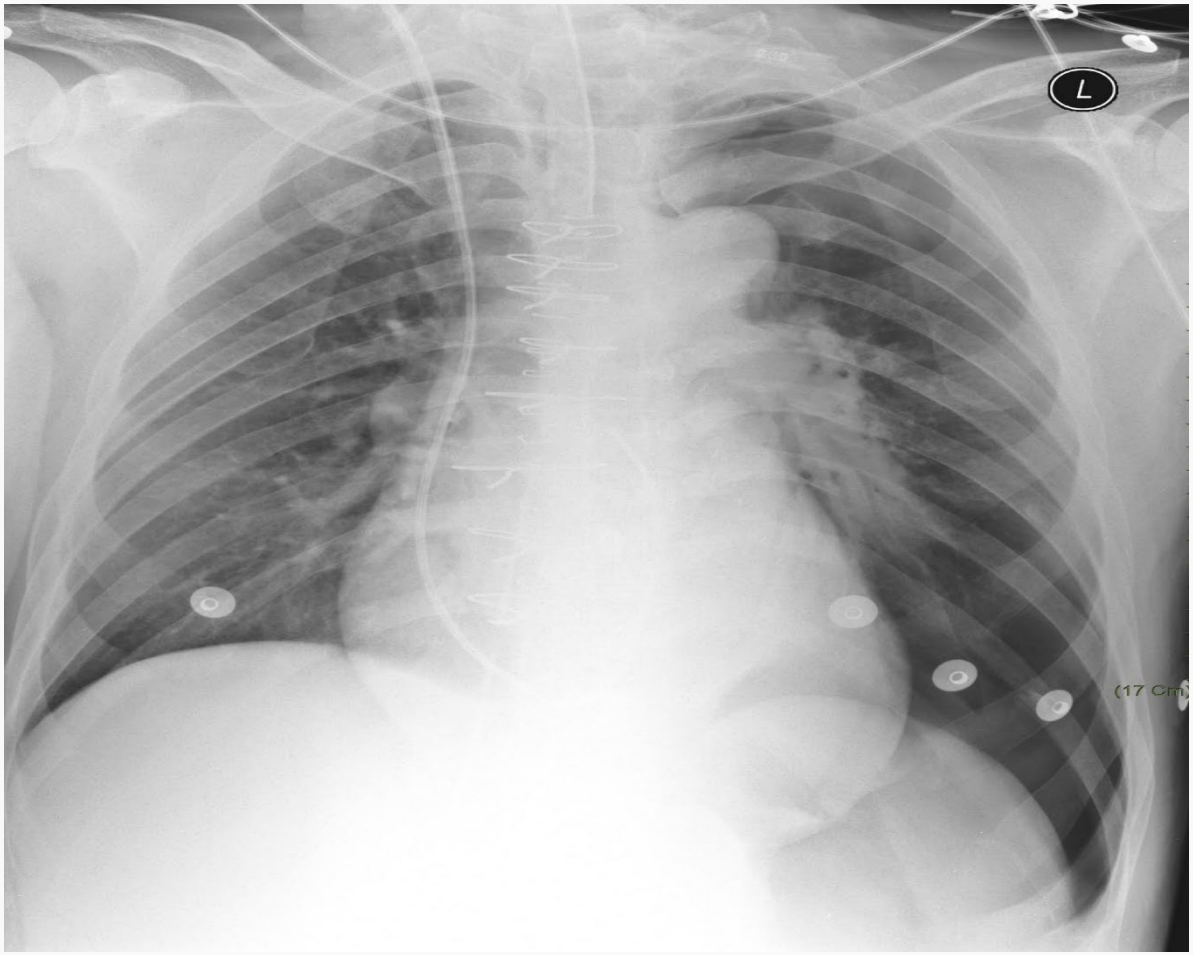
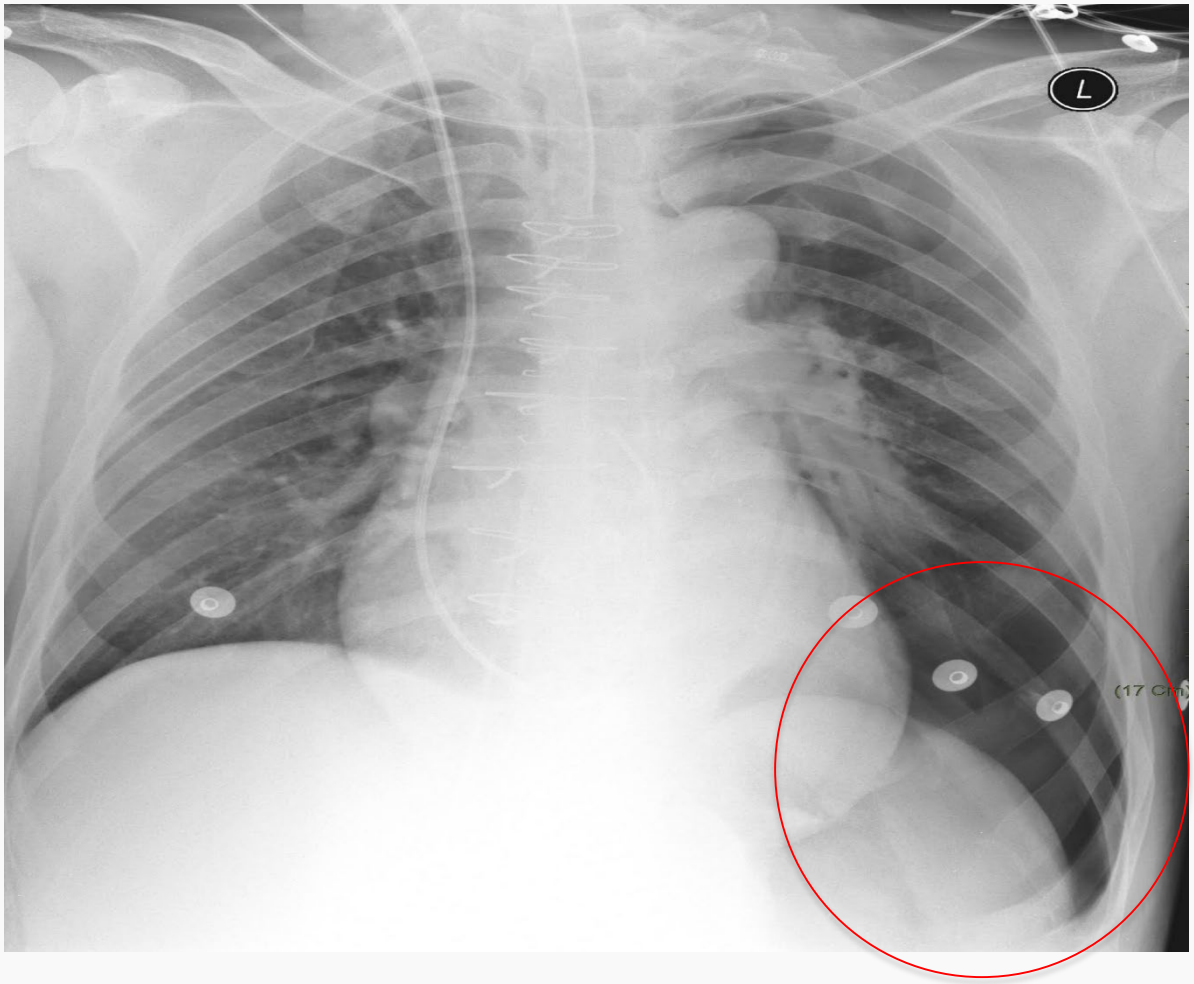

L
(17 Cm)

*Deep sulcus sign-on supine x-ray



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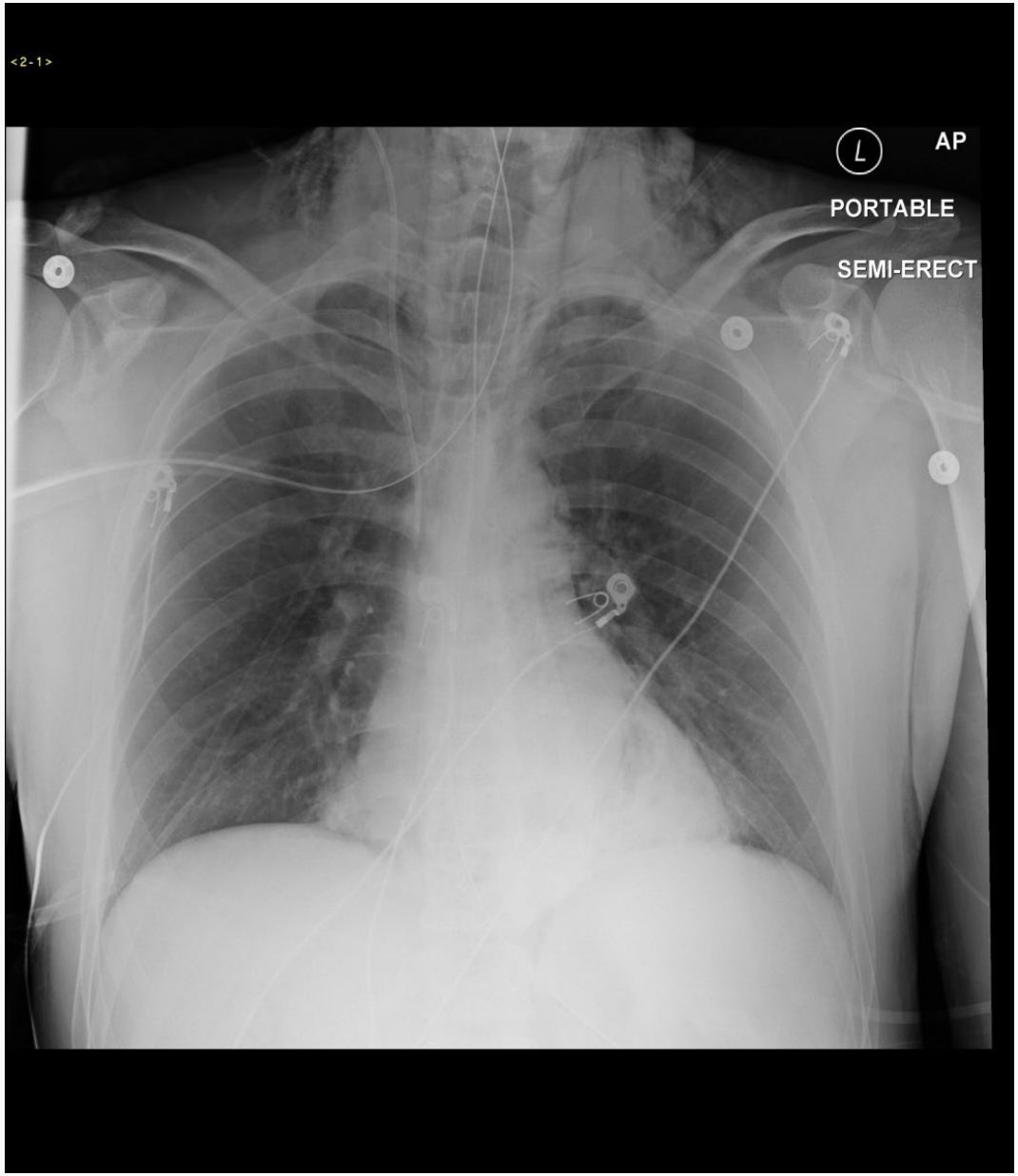
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L
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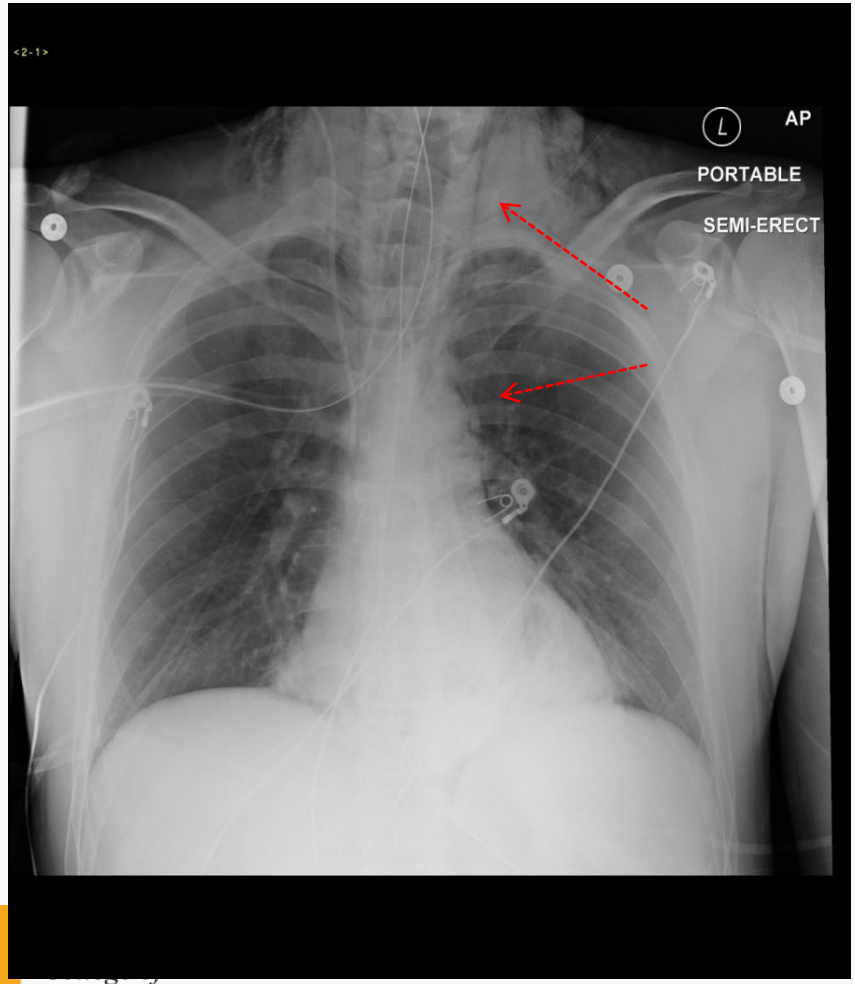
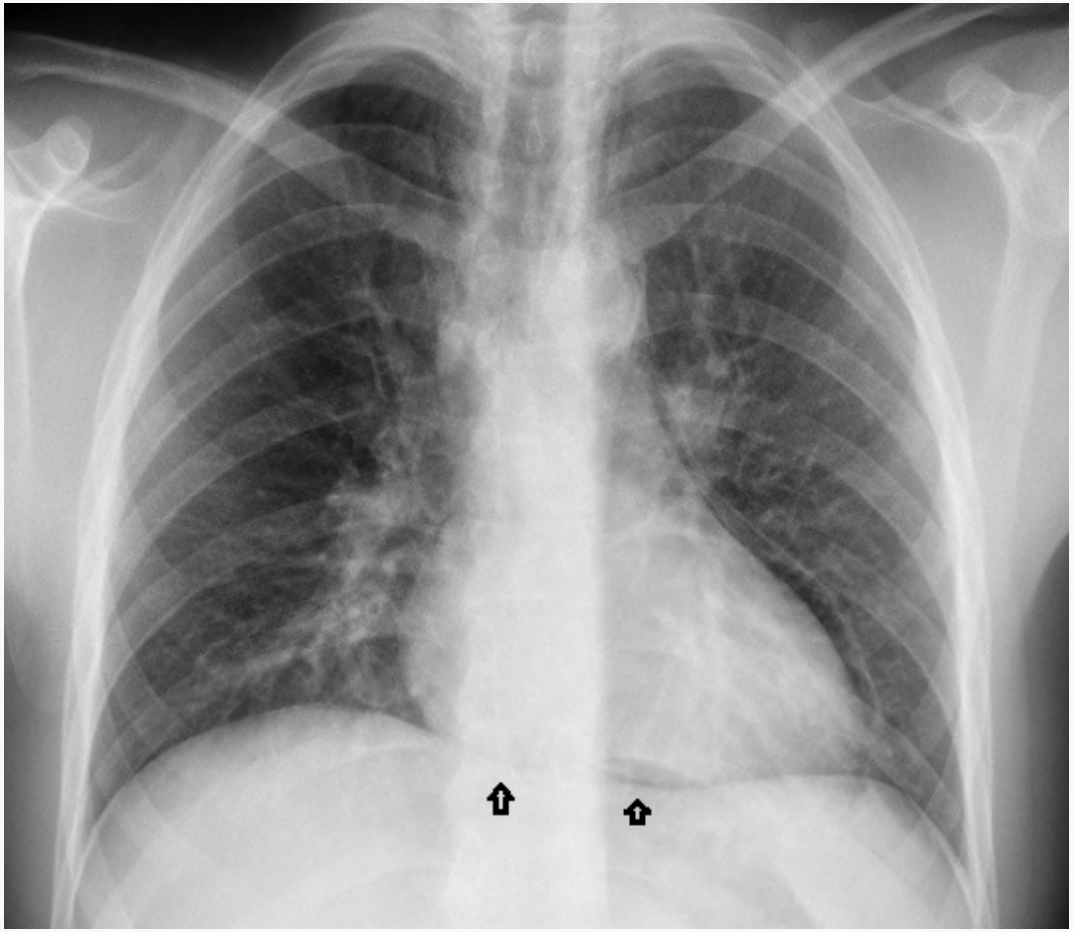
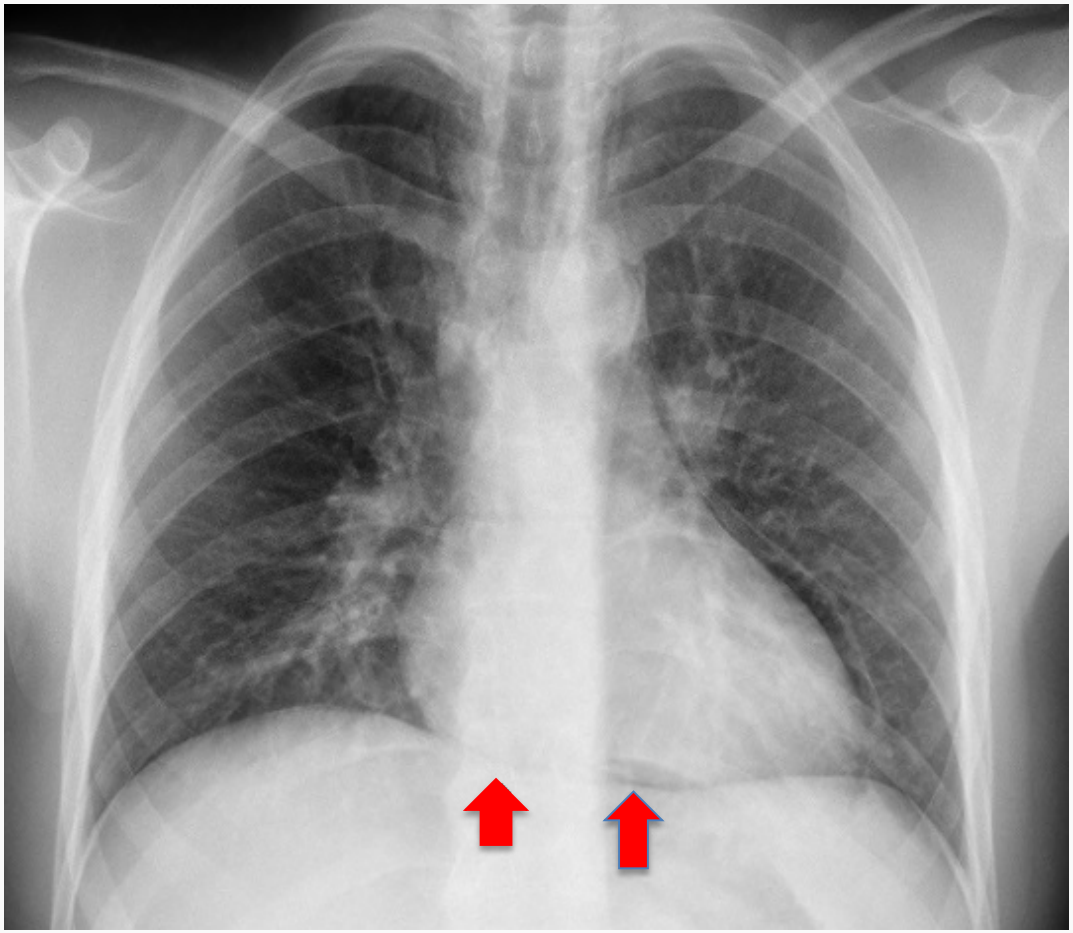
Pneumomediastinum





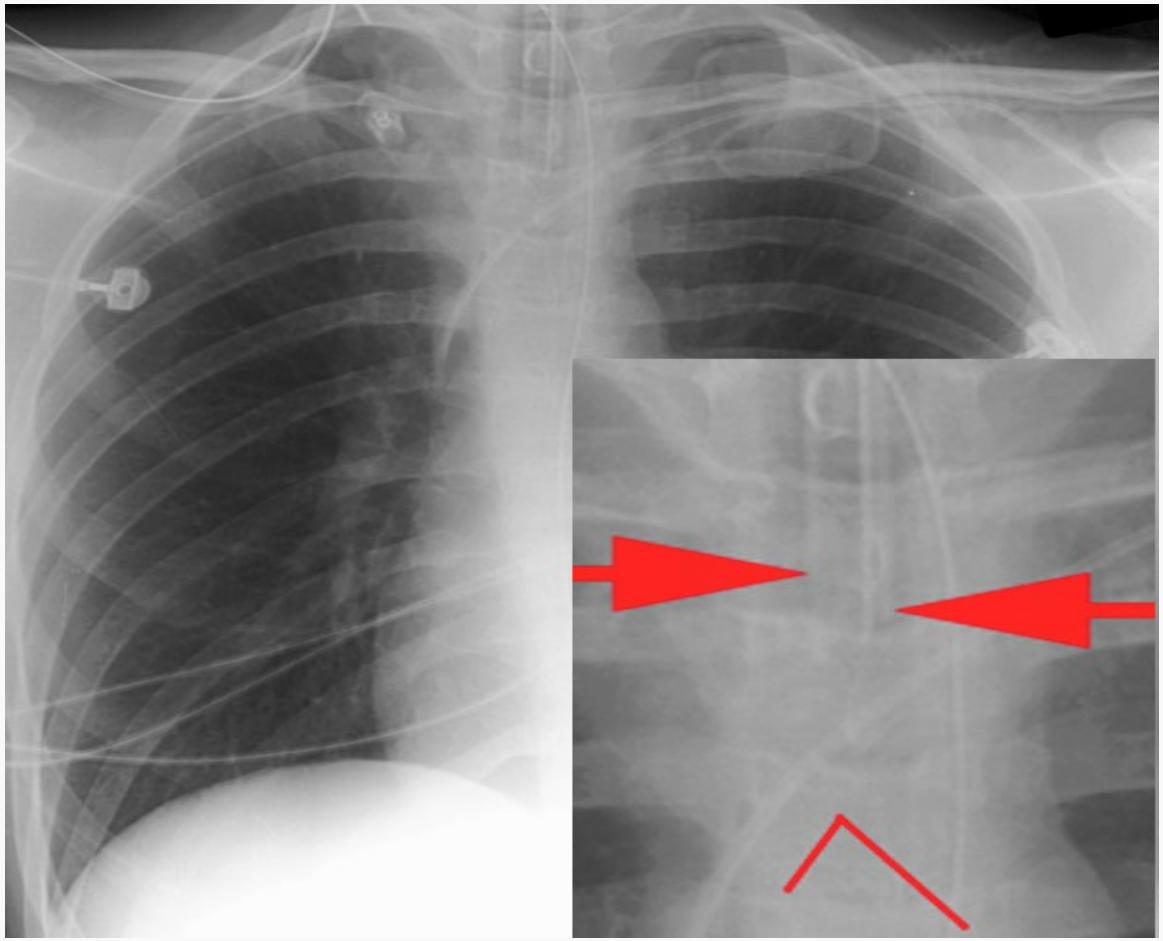
Pneumomediastinum



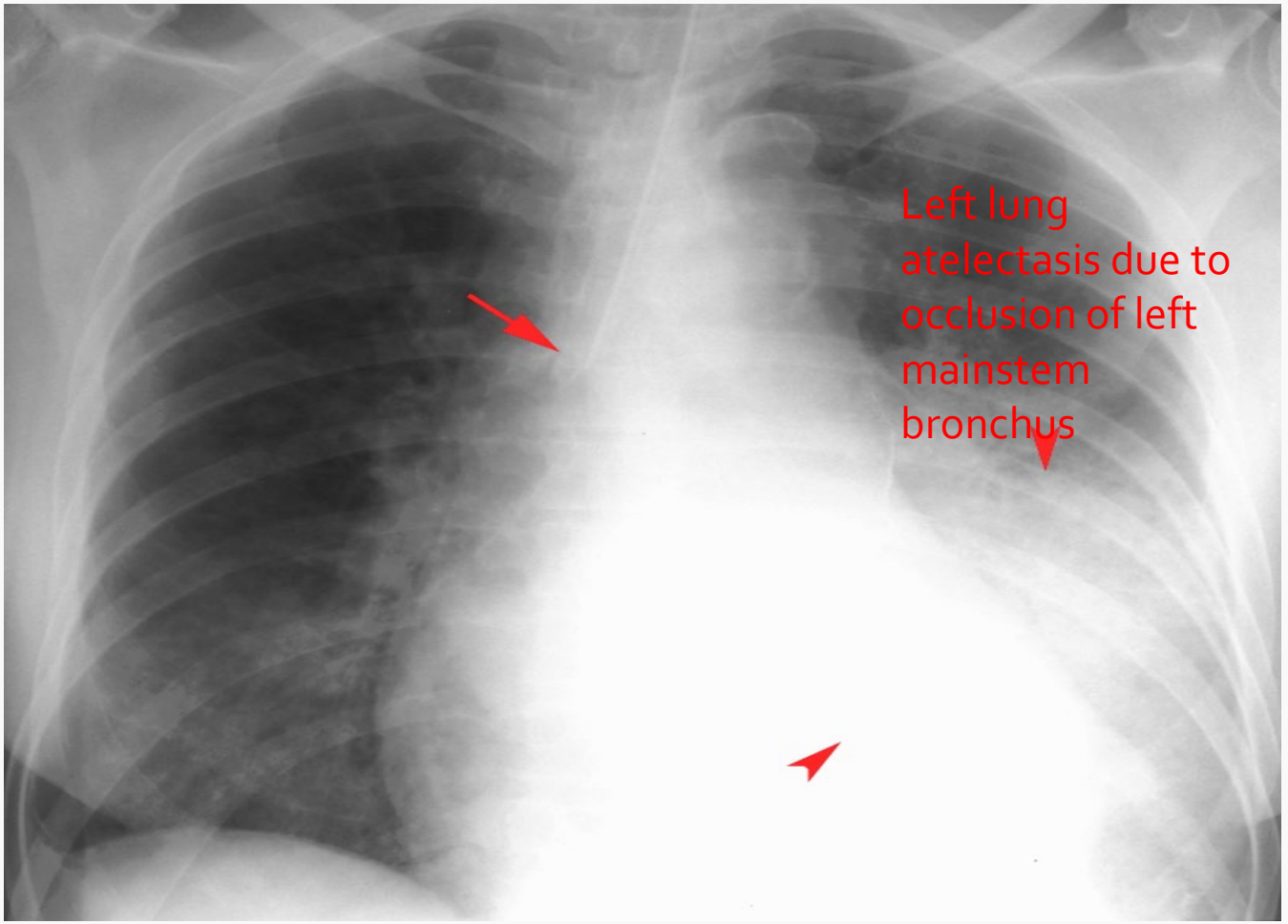
Lines and Tubes: What is normal?

- Endotracheal Tube
  - With normal neck movement, an ET tube can move down 2 cm
  - ET tubes therefore must be at least 2 cm above carina, ideally 4-5 cm above
- Central lines
  - Tip ideally in the superior vena cava.
- Swan-Ganz catheters
  - Tip ideally in right or left main pulmonary artery; further than this can cause pulmonary infarction
- Feeding Tube
  - Tip ideally in stomach/proximal duodenum; however any NG/OG tube clearly terminating below the diaphragm is acceptable

Satisfactory ET tube placement



ET tube in Right mainstem bronchus



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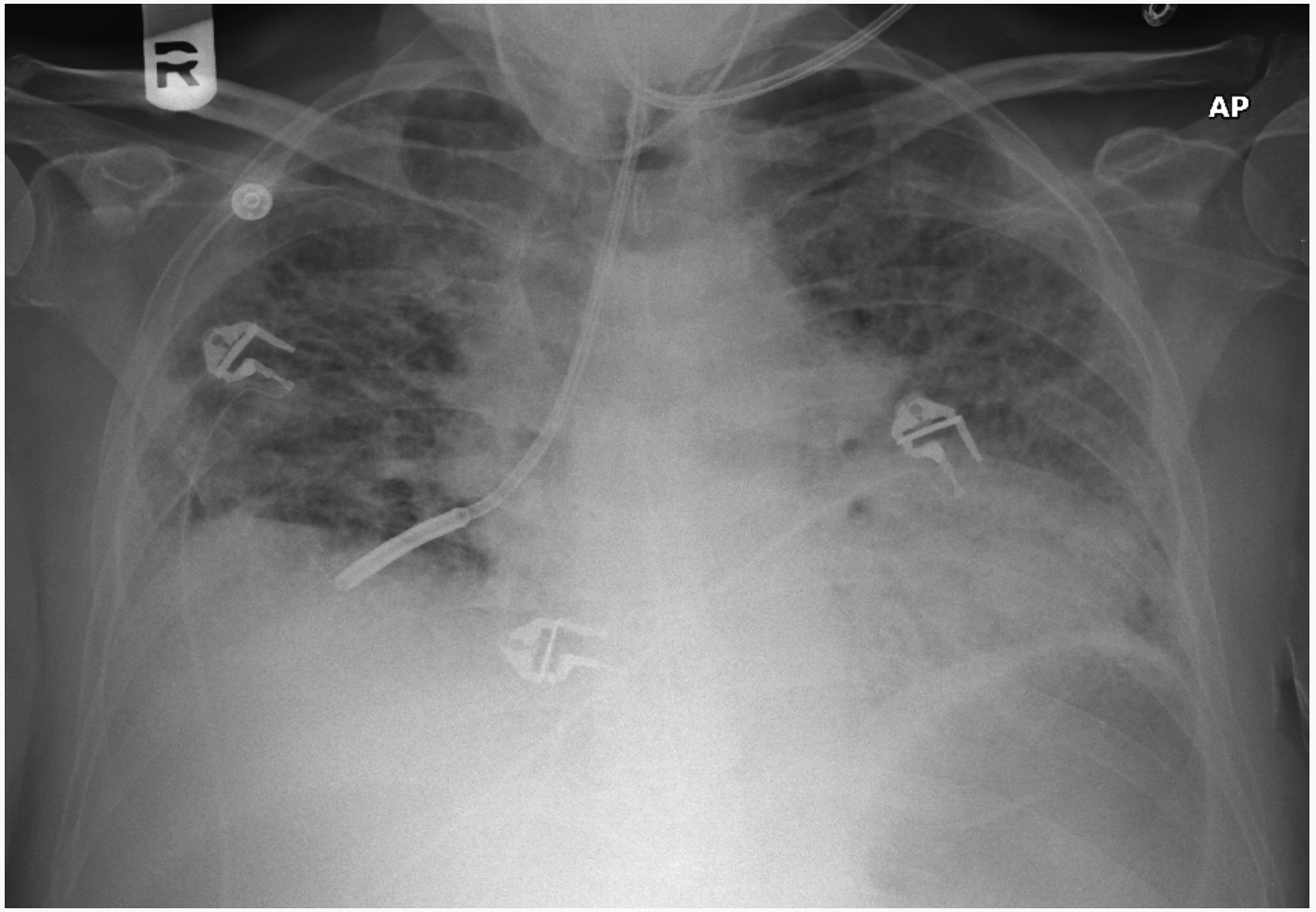
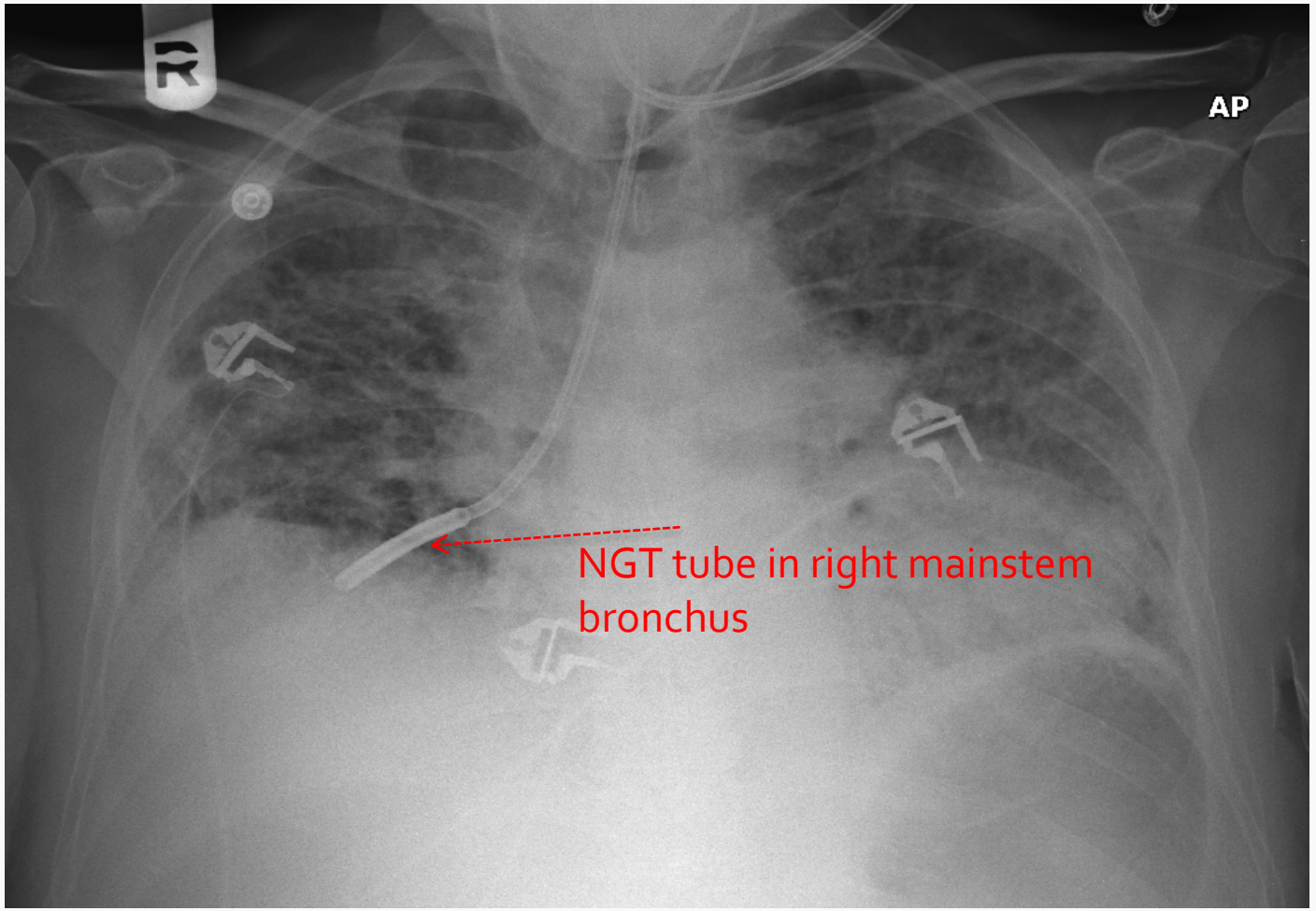
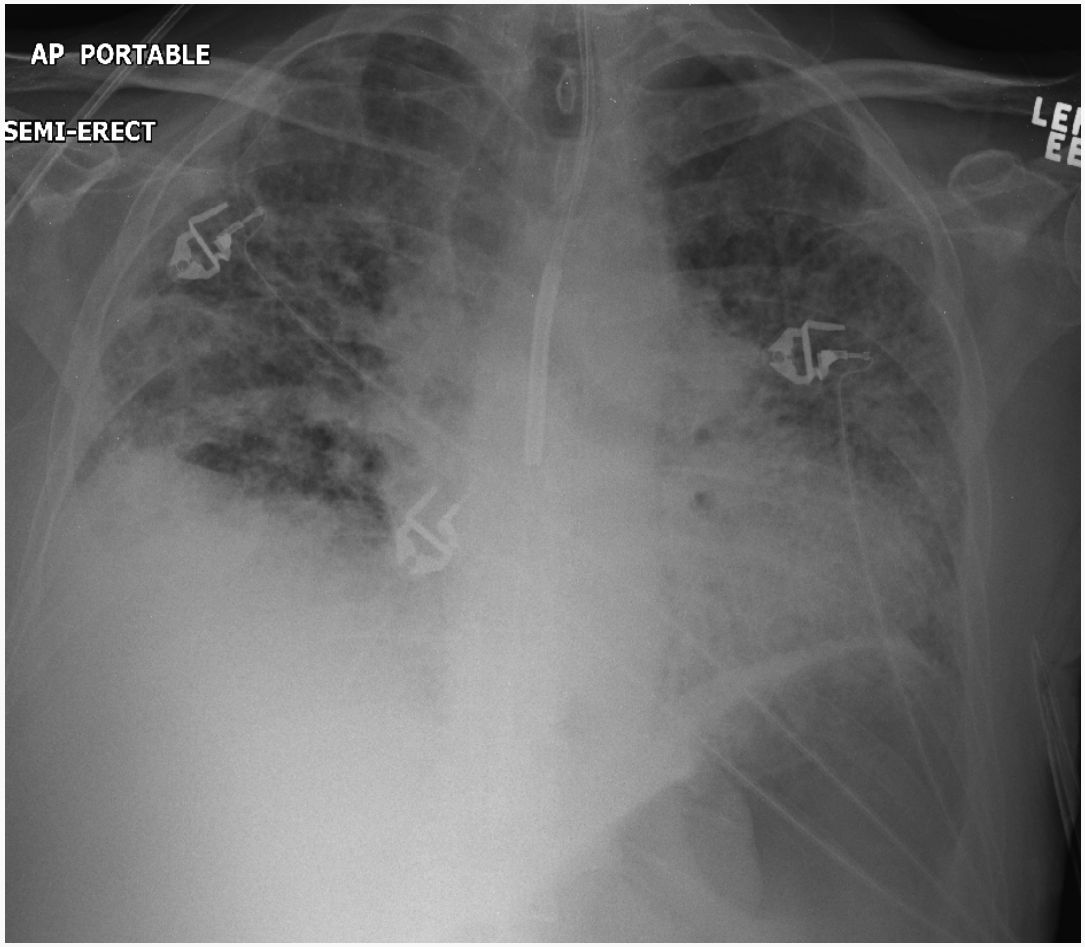
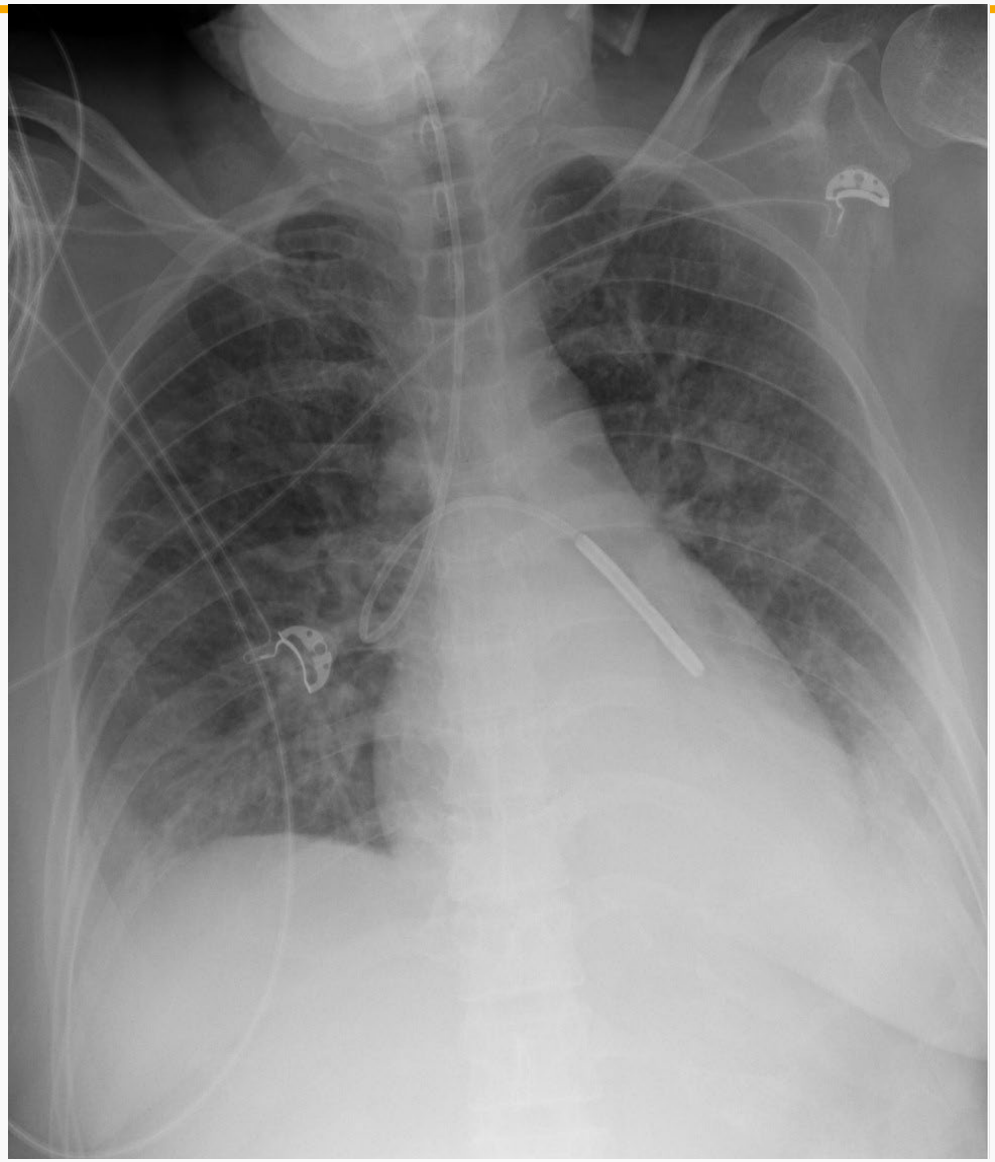
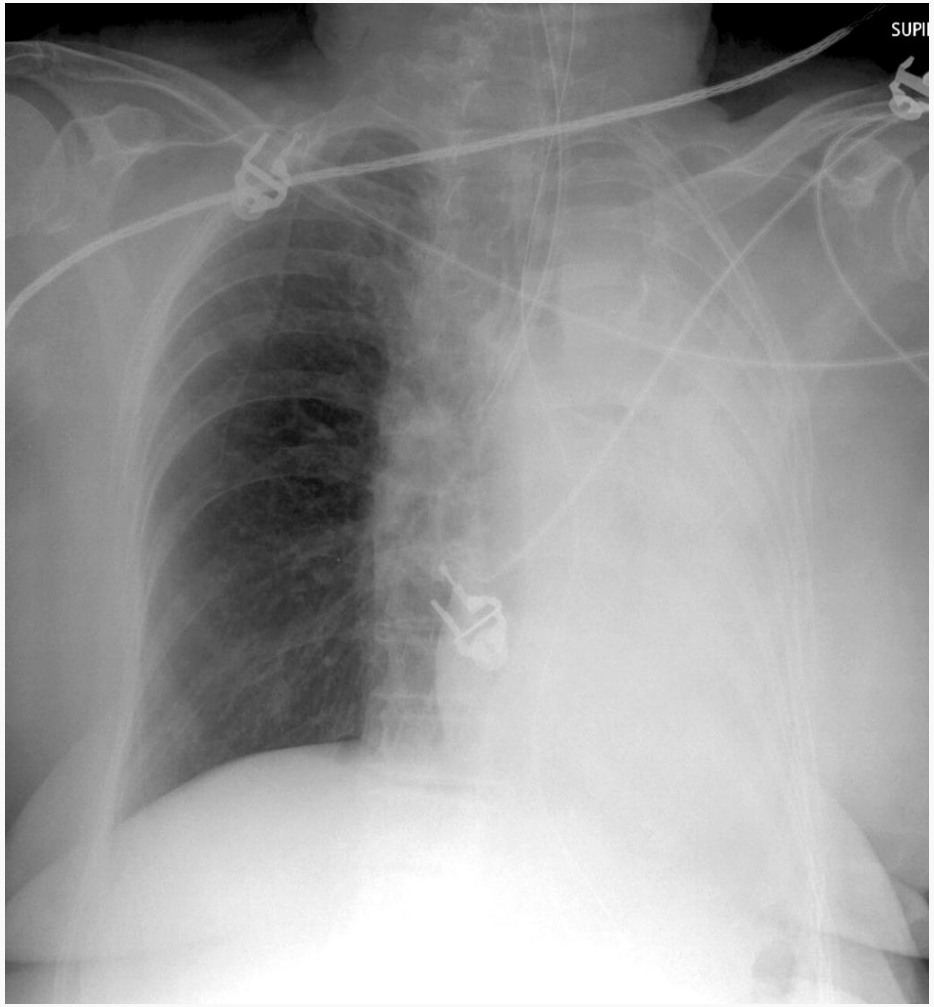



R
AP
NGT tube in right mainstem bronchus

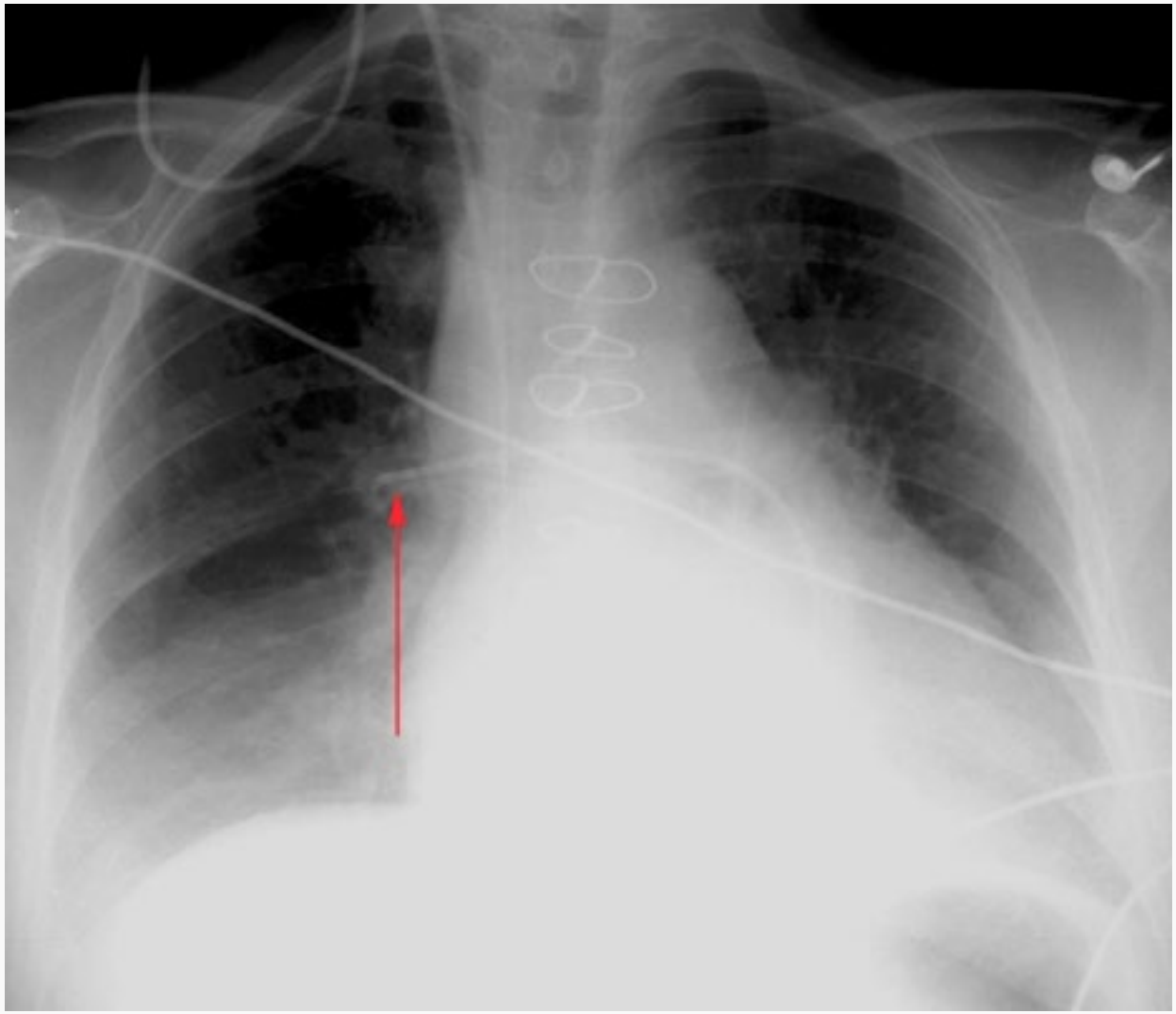
*S/P repositioning-is this better?



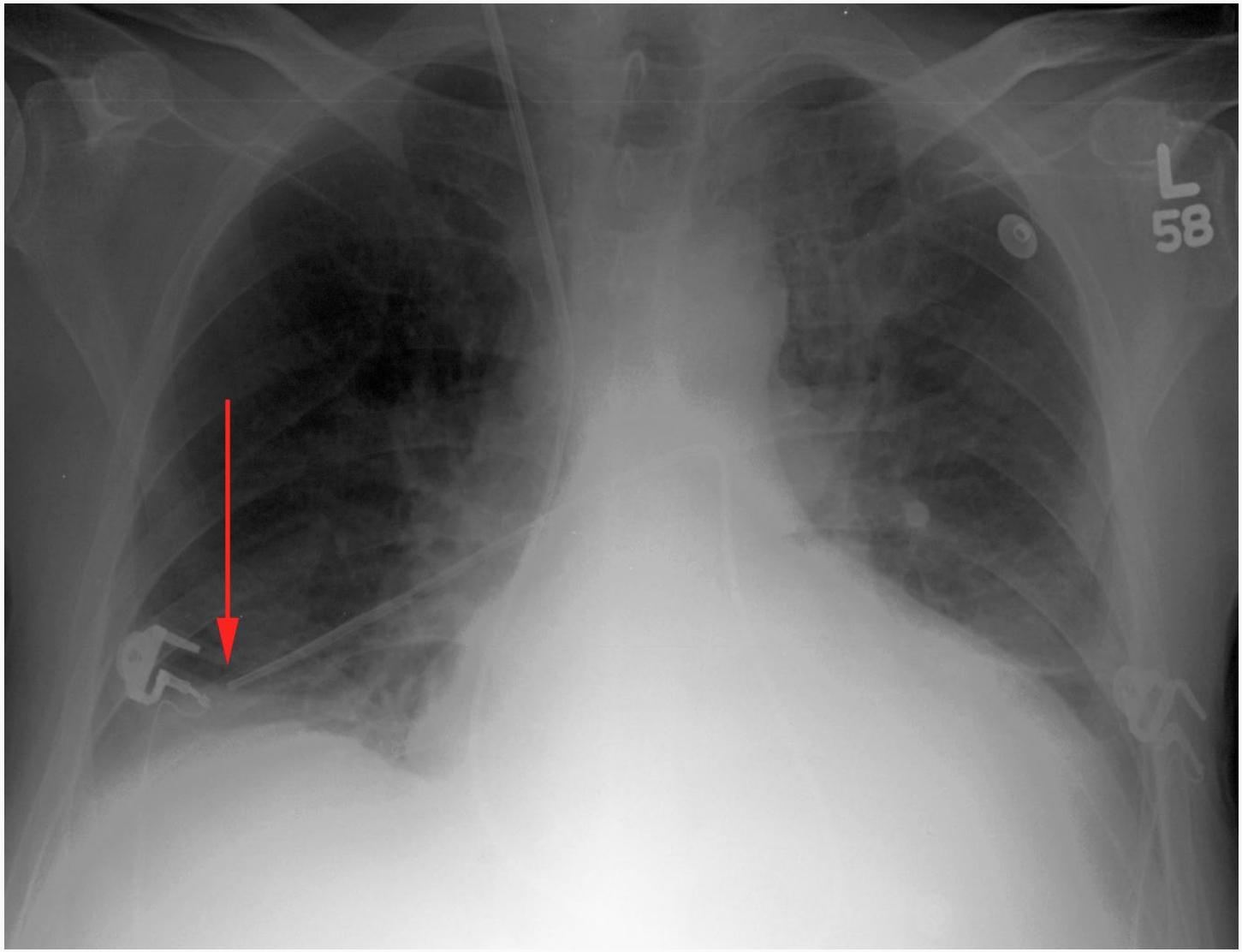




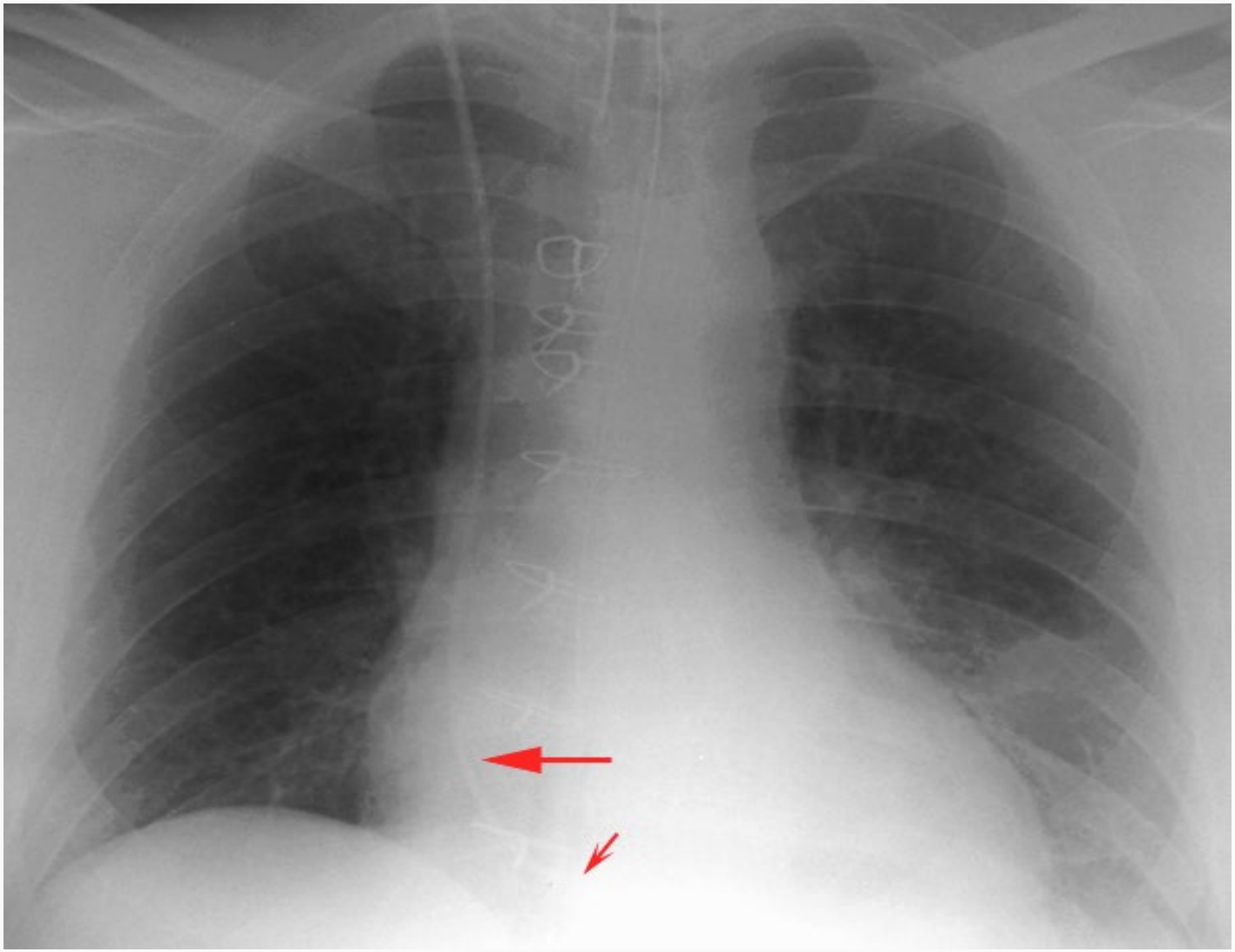
Satisfactory Swan Ganz placement



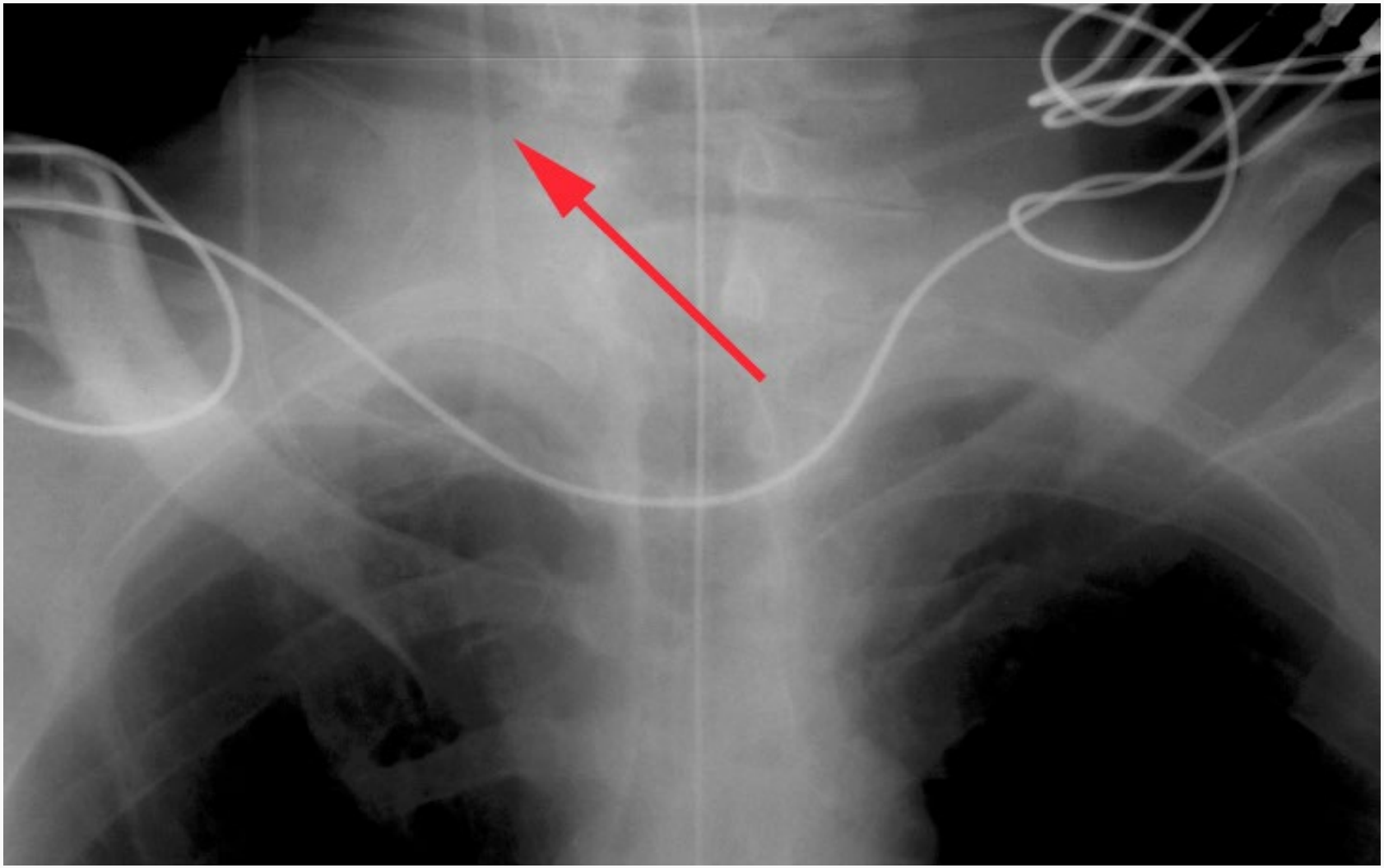
Swan Ganz in too far, distal to main pulmonary artery



This Swan Ganz catheter does not go in far enough



Right subclavian line going into Internal jugular instead of SVC

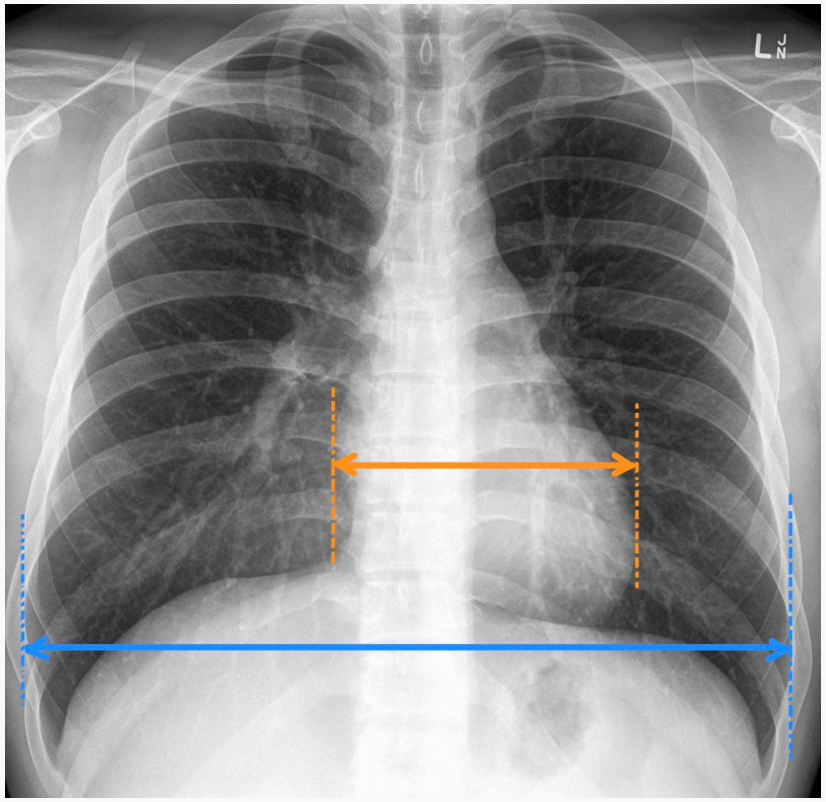


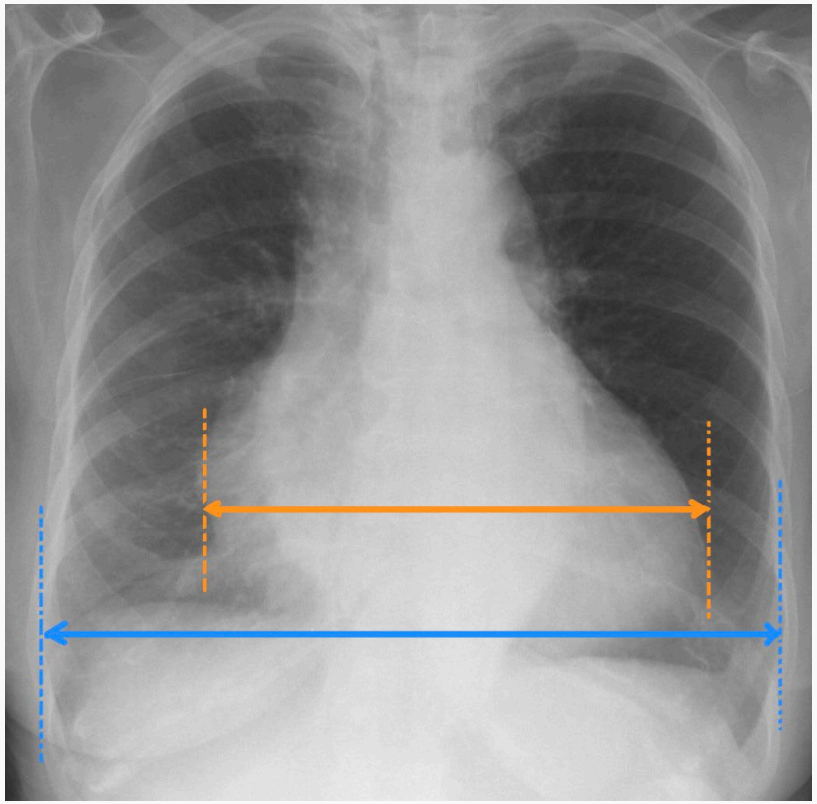
L SUPINE MOBILE

Severe subcutaneous emphysema

*Cardiomegaly

- The **cardiothoracic ratio** (**CTR**) aids in the detection of enlargement of the cardiac silhouette
  - Most commonly from cardiomegaly **BUT** can be due to other processes such as a pericardial effusion!
- CTR is measured on a PA chest x-ray, and is the ratio of maximal horizontal cardiac diameter to maximal horizontal thoracic diameter (inner edge of ribs/edge of pleura).
  - A normal measurement should be less than 0.5.
- The CTR should approached with caution an AP chest x-ray!
  - Objects nearer the x-ray tube appear artificially enlarged due to divergence of the x-ray beam, resulting in the heart appearing artificially large on AP radiographs.



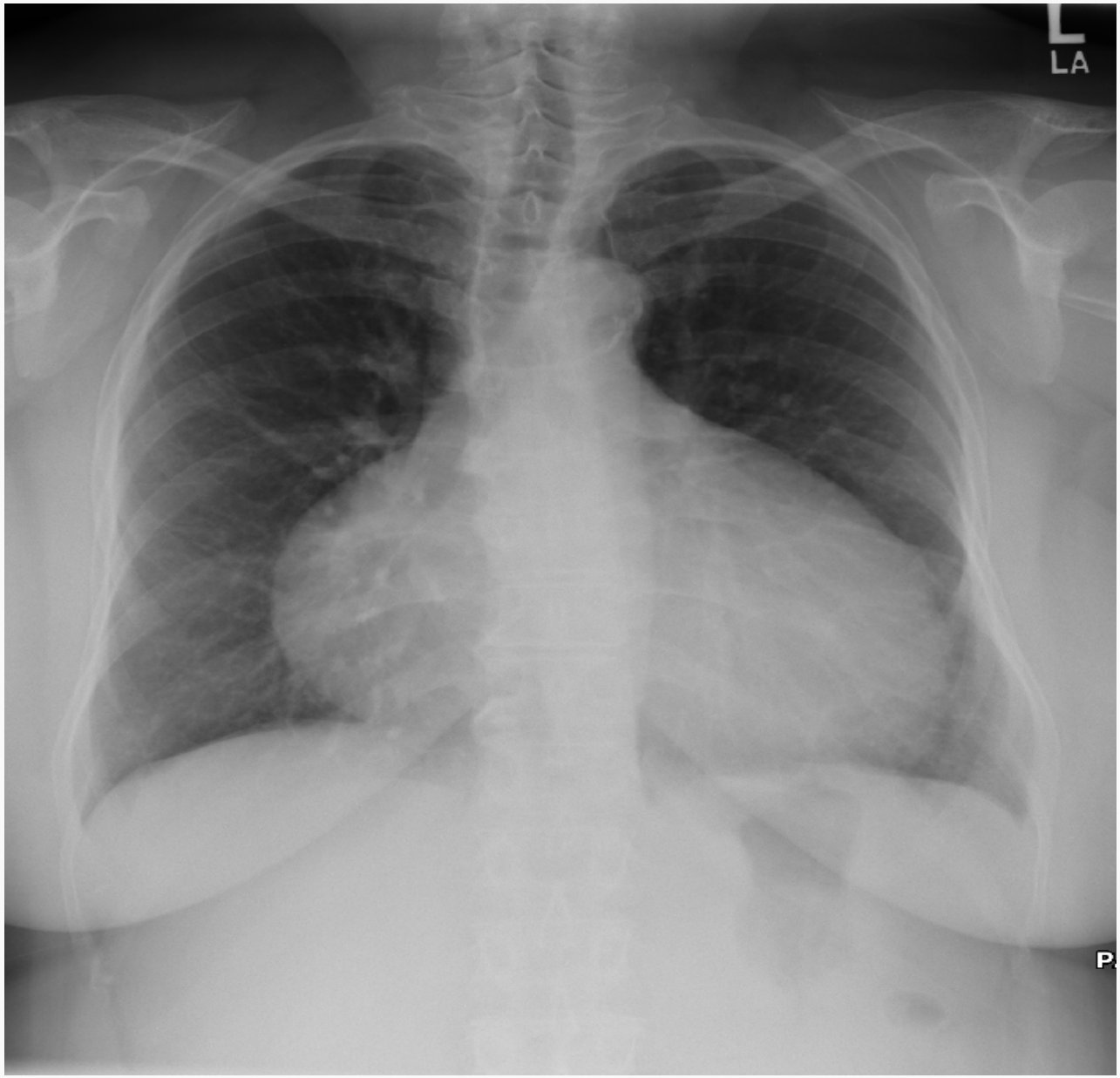


Normal

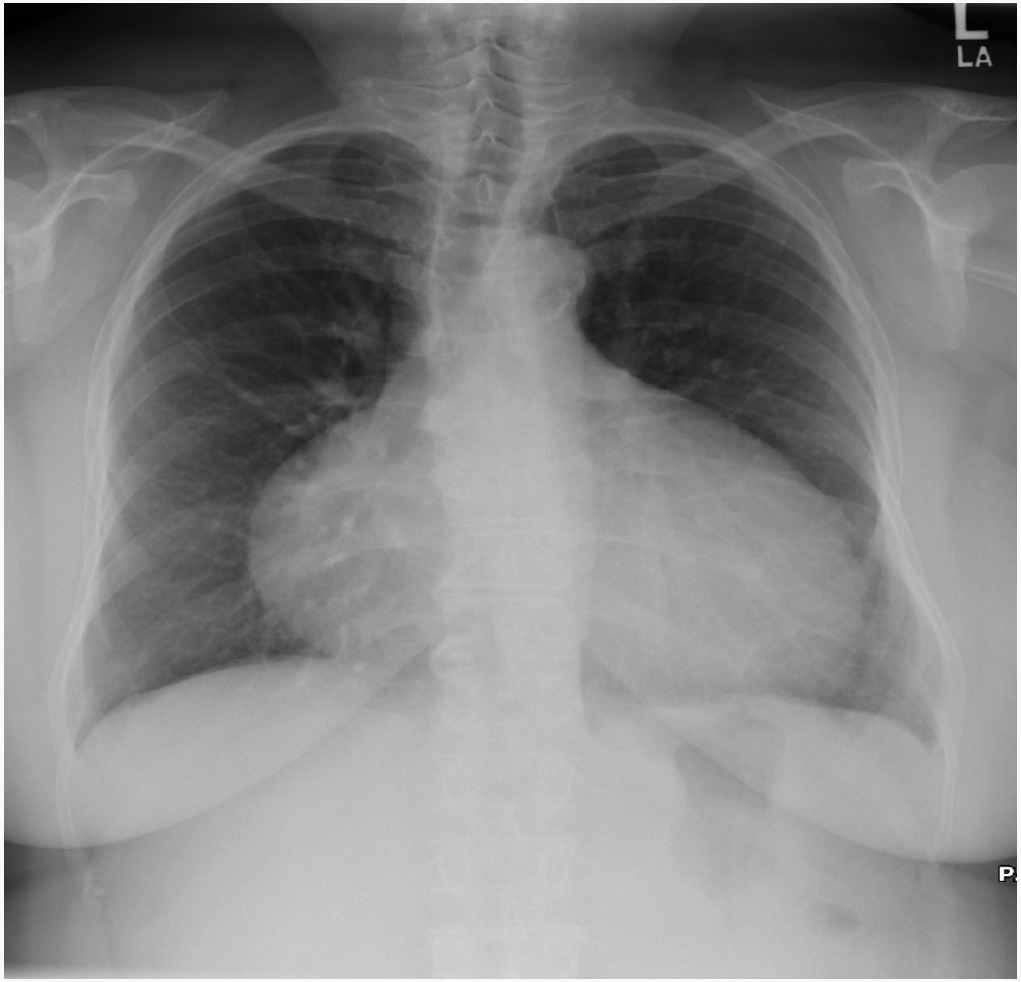
Cardiomegaly

Case courtesy of A.Prof Frank Gaillard, Radiopaedia.org. From the case rID: 12334

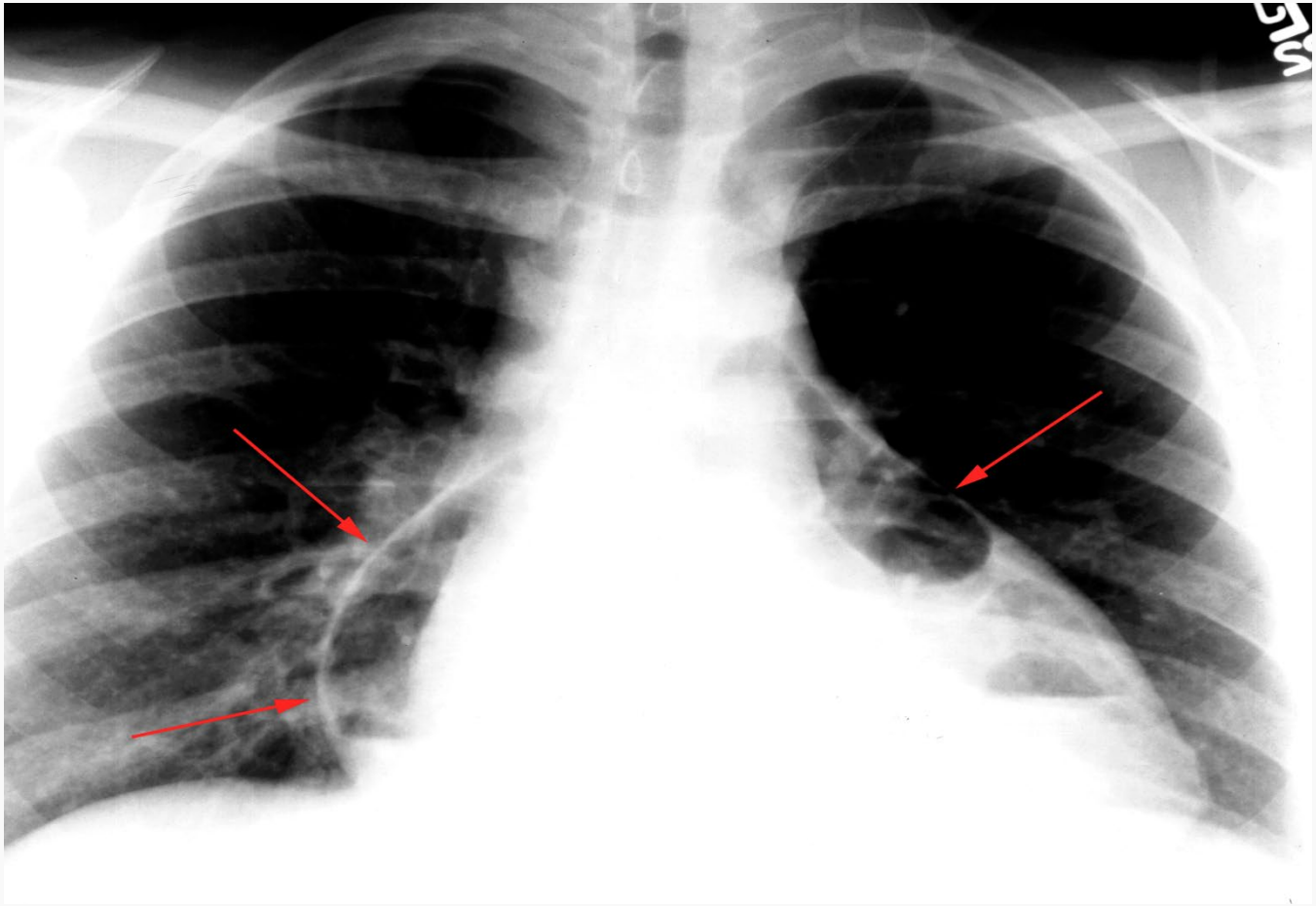
*Chest pain



Large pericardial effusion-”water bottle sign”

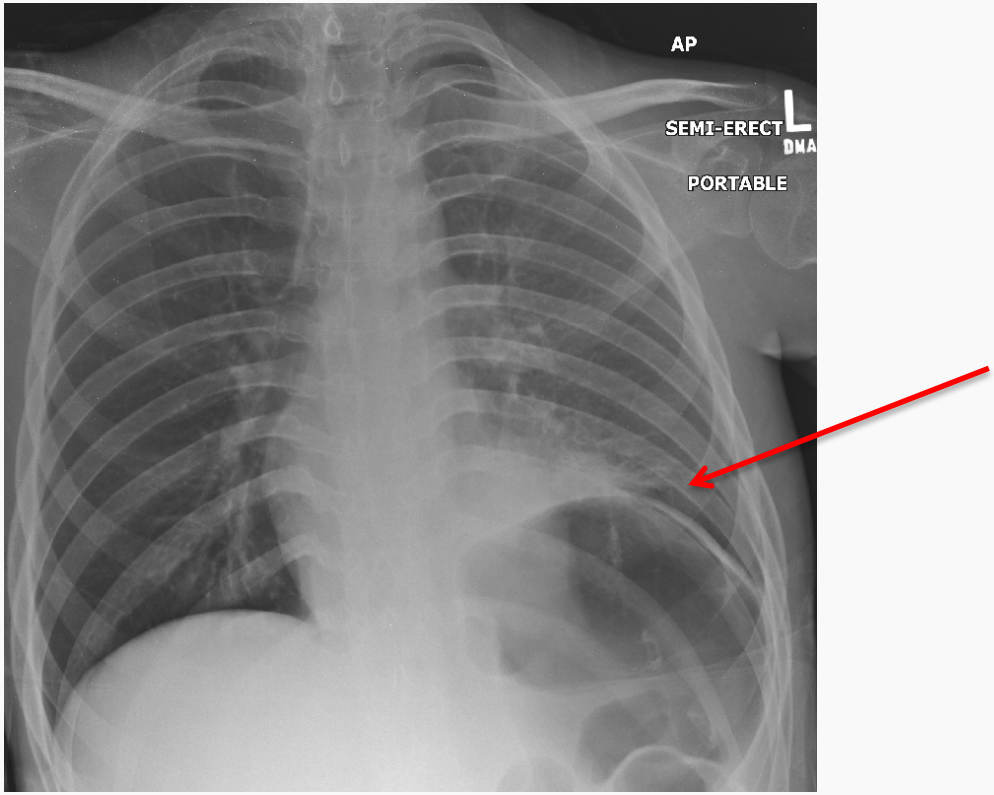


Pneumopericardium s/p pericardiocentesis



Left lower lobe atelectasis

- Note volume loss and diaphragmatic elevation



Lobar Patterns

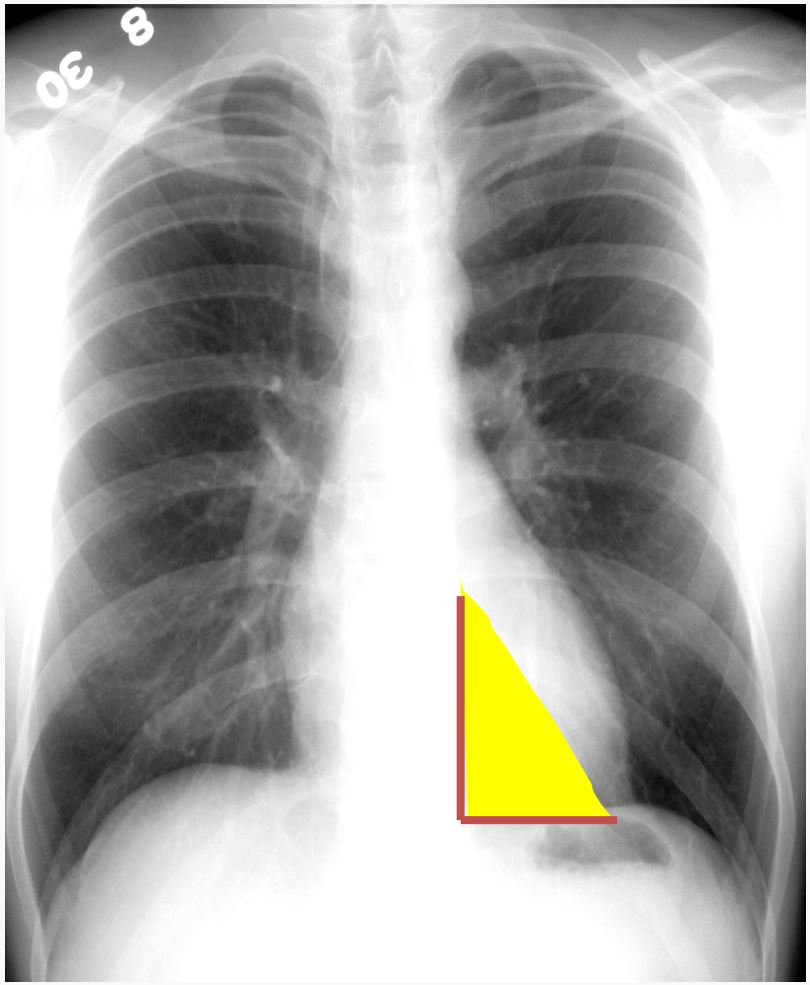
- Remember:
  - The whole lobe may not be involved
  - The patterns of atelectasis are very similar to consolidation, except:
  - Volume loss
  - Left upper lobe collapse
  - Check for your silhouette signs

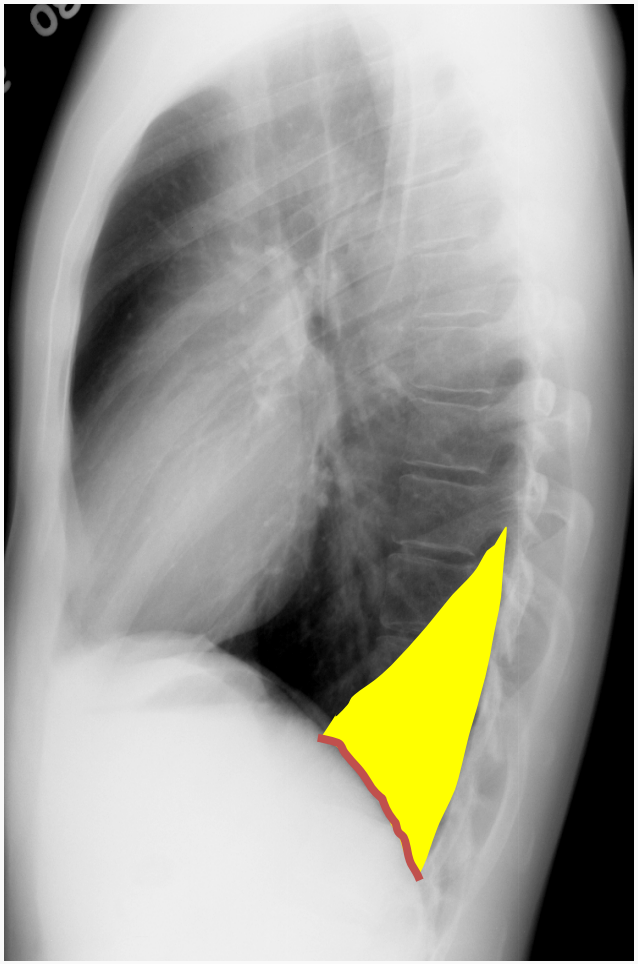
Signs of Consolidation

- Increased opacity
- Fluffy, “air space” opacity
- No/negligible volume loss signs
- Air bronchograms (pathognomonic)
- Not all consolidation is due to pus
  - Fluid, cells, blood, protein

Lobar patterns of consolidation

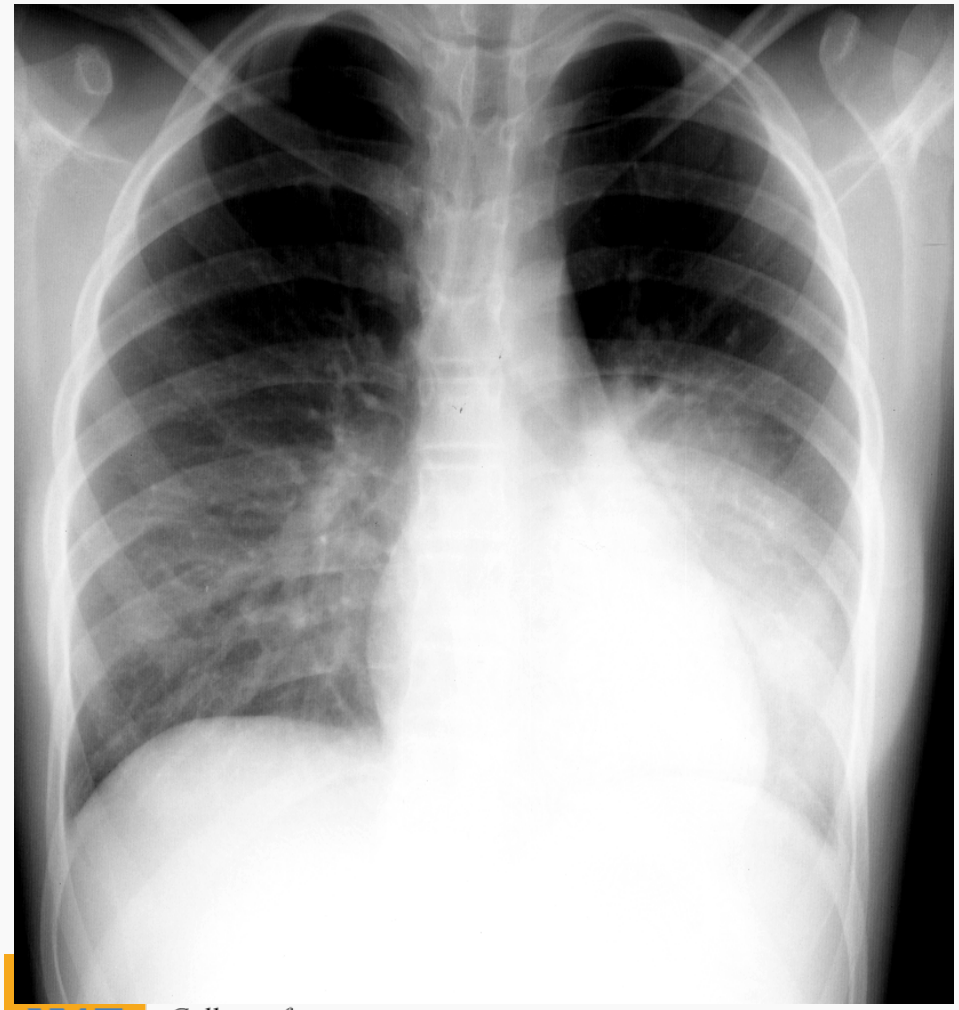
- On the following slides you will see:
  - A diagram of the lobar pattern of consolidatio
    - Yellow = opacity
    - Orange = site of silhouette sign
  - An example of lobar pneumonia

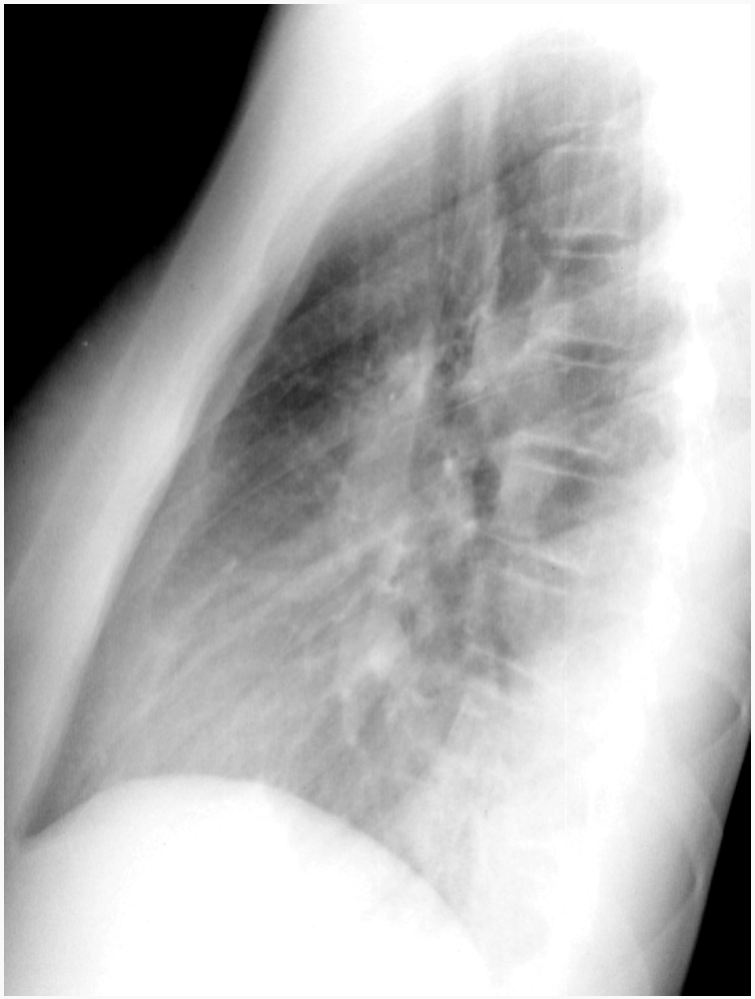




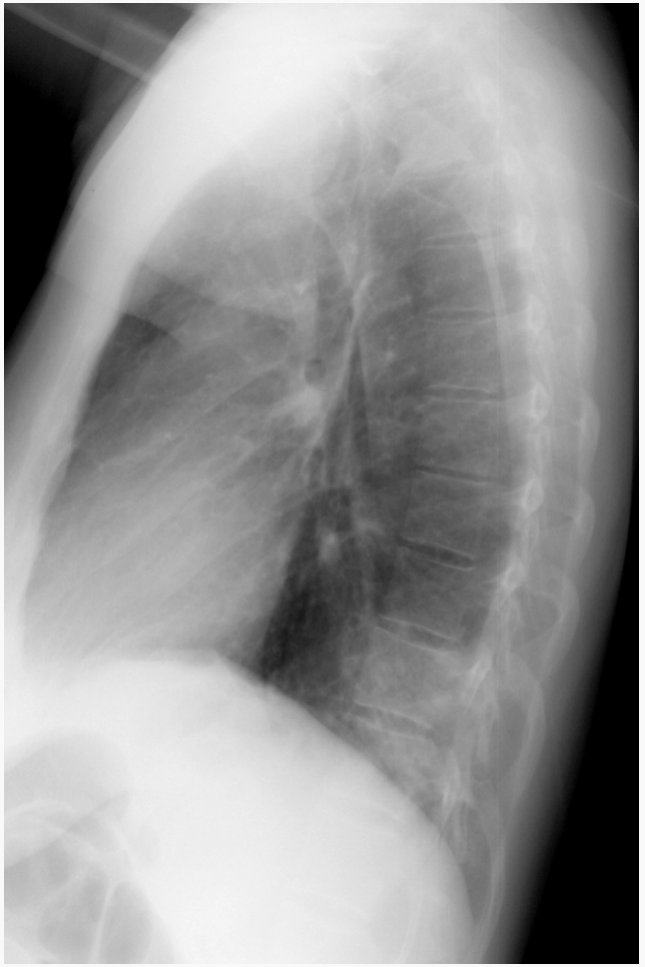
Left Lower Lobe Consolidation

Left Lower Lobe Pneumonia

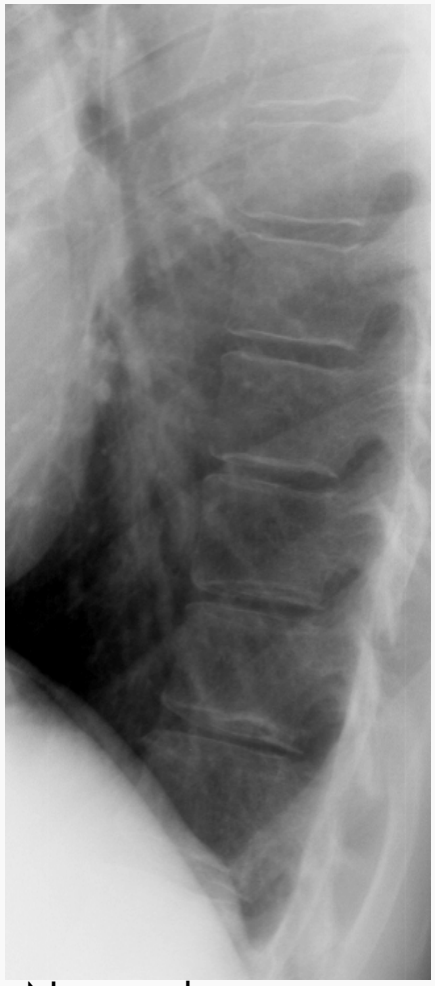




Left Lower Lobe Pneumonia

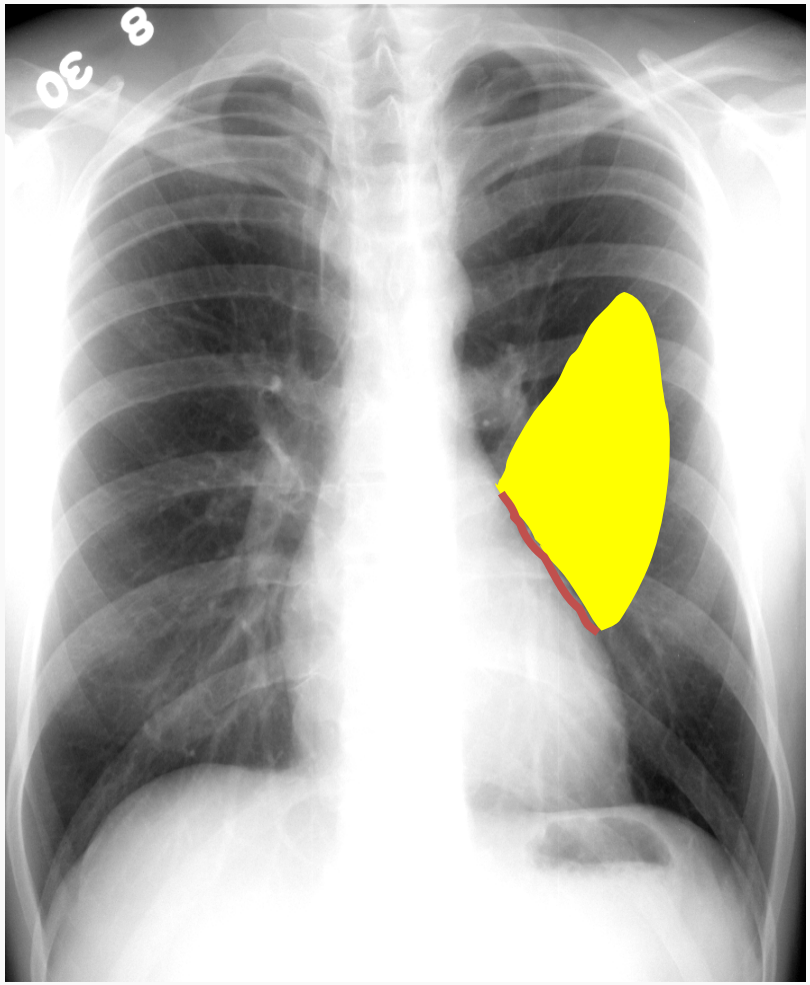


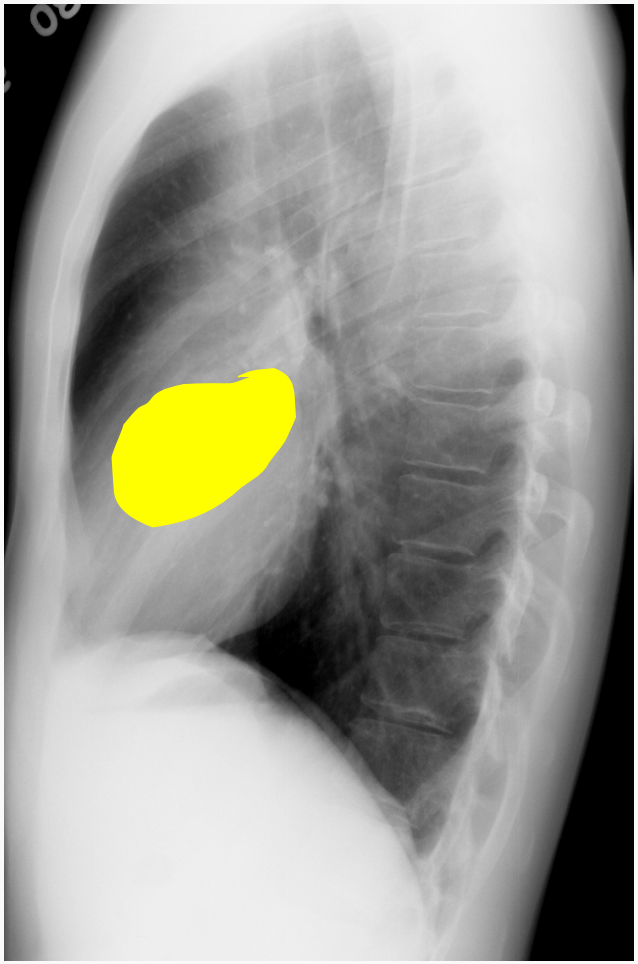
*The Spine Sign



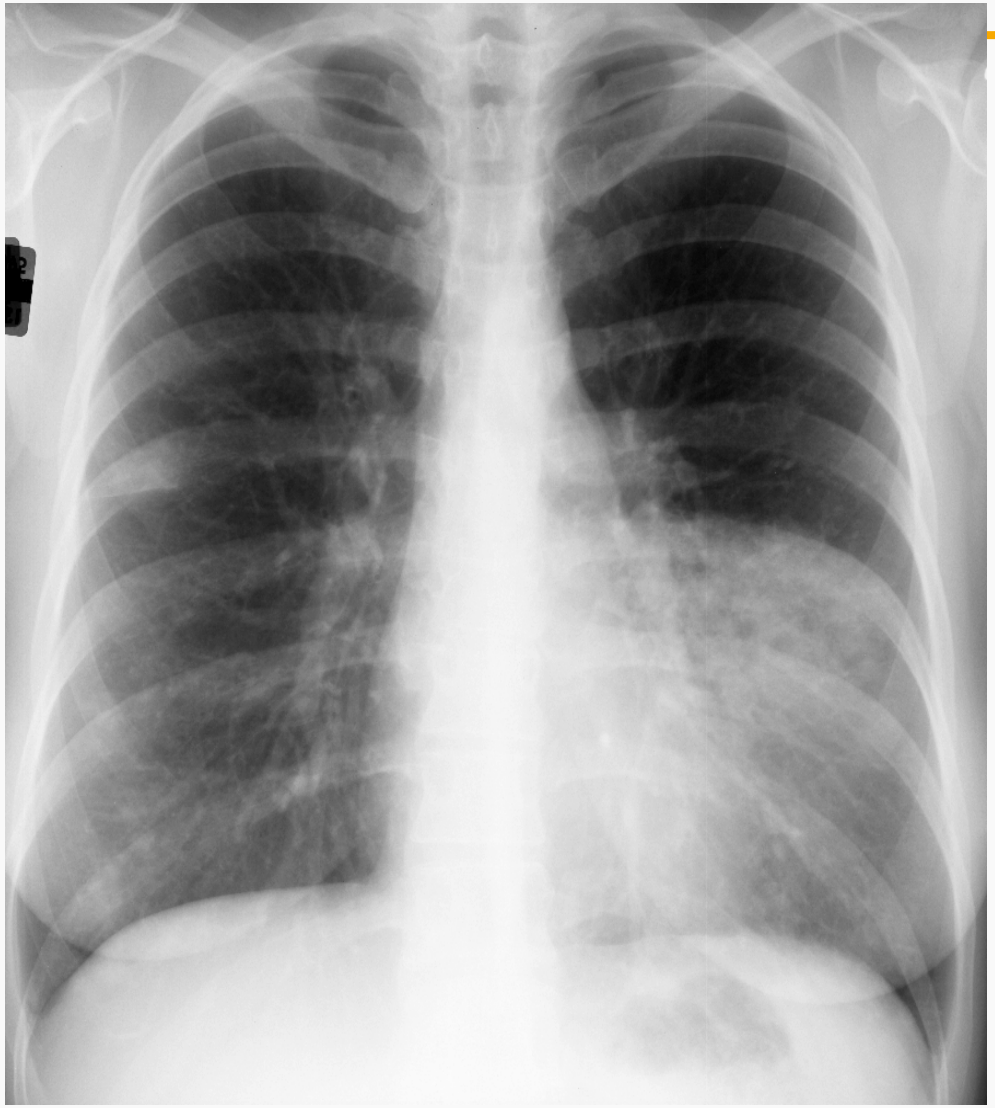
Normal

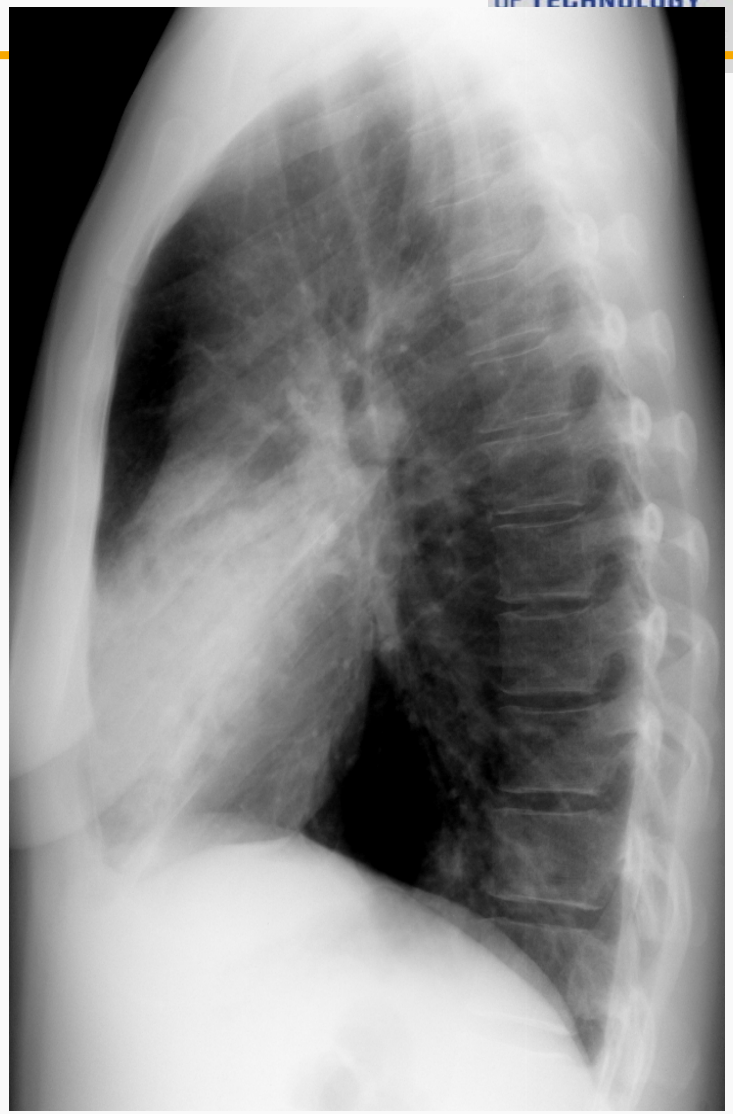
Spine Sign



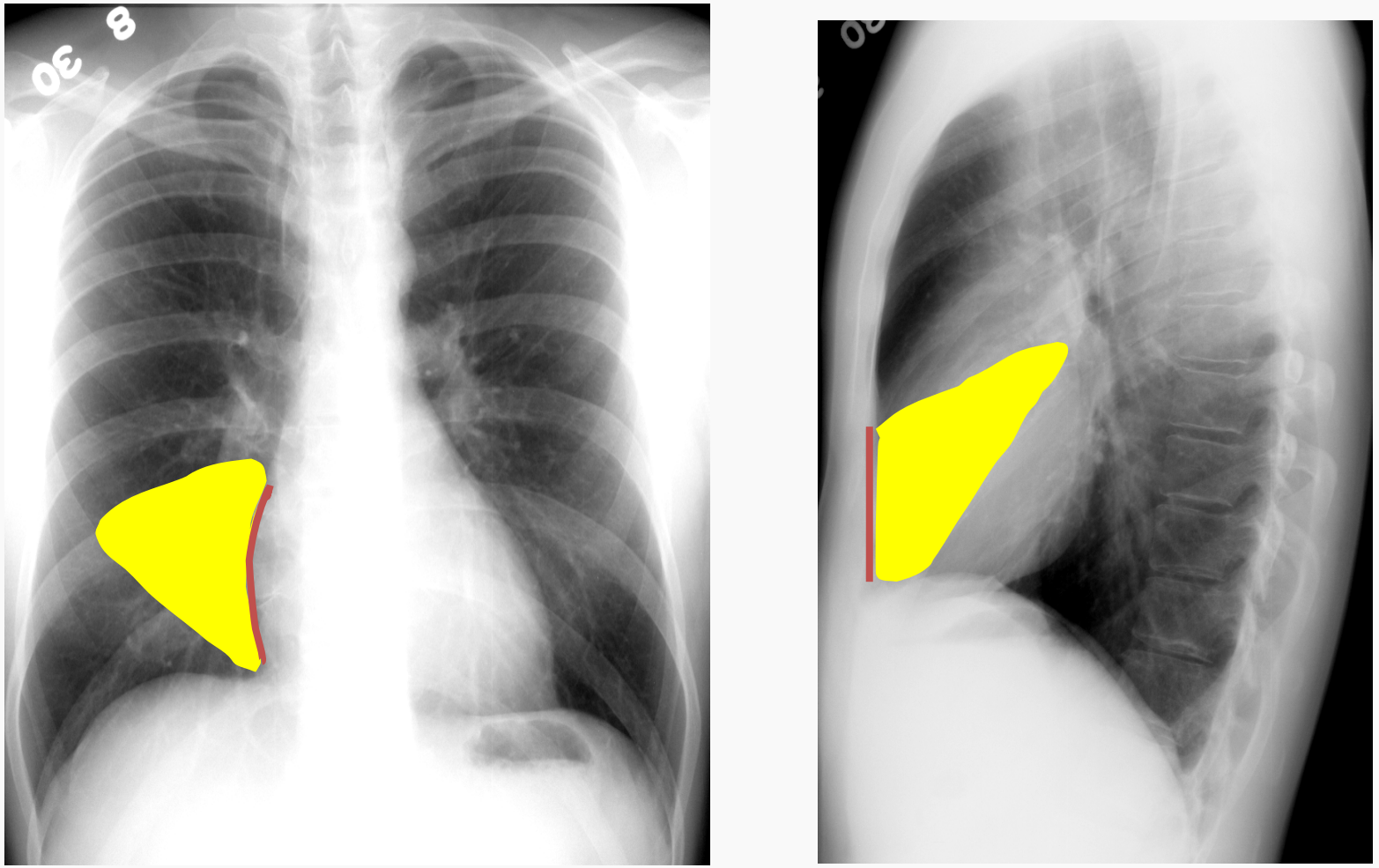


Left Upper Lobe/ Lingular Consolidation



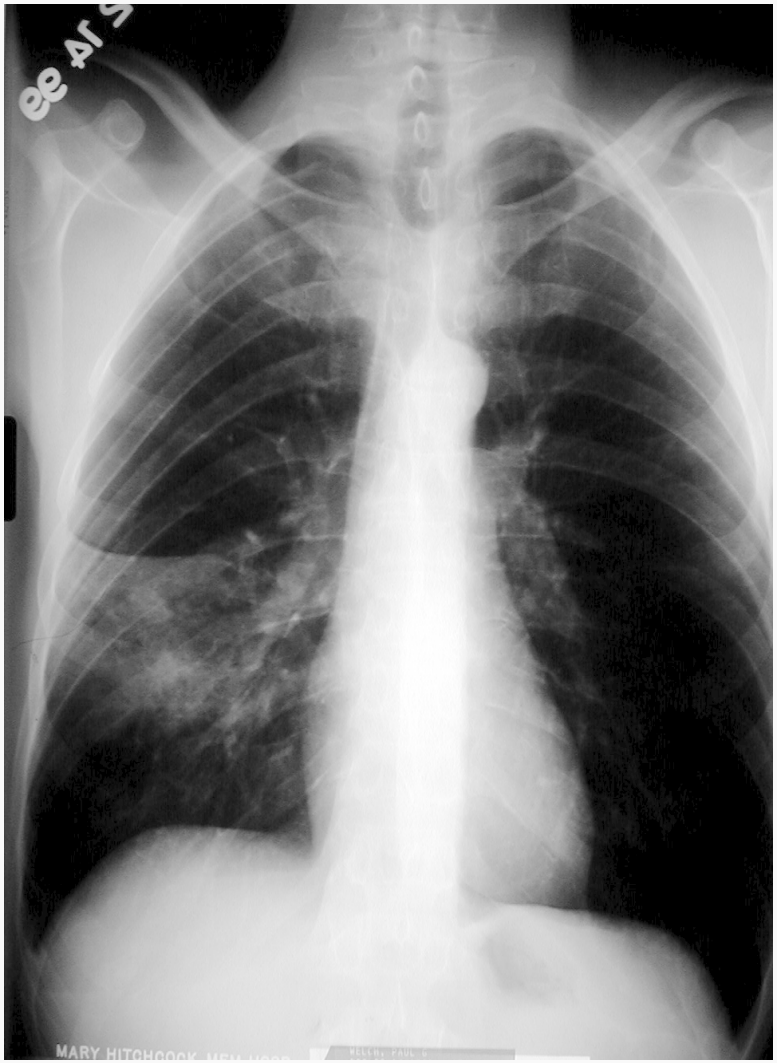


Left Upper Lobe/Lingular Consolidation



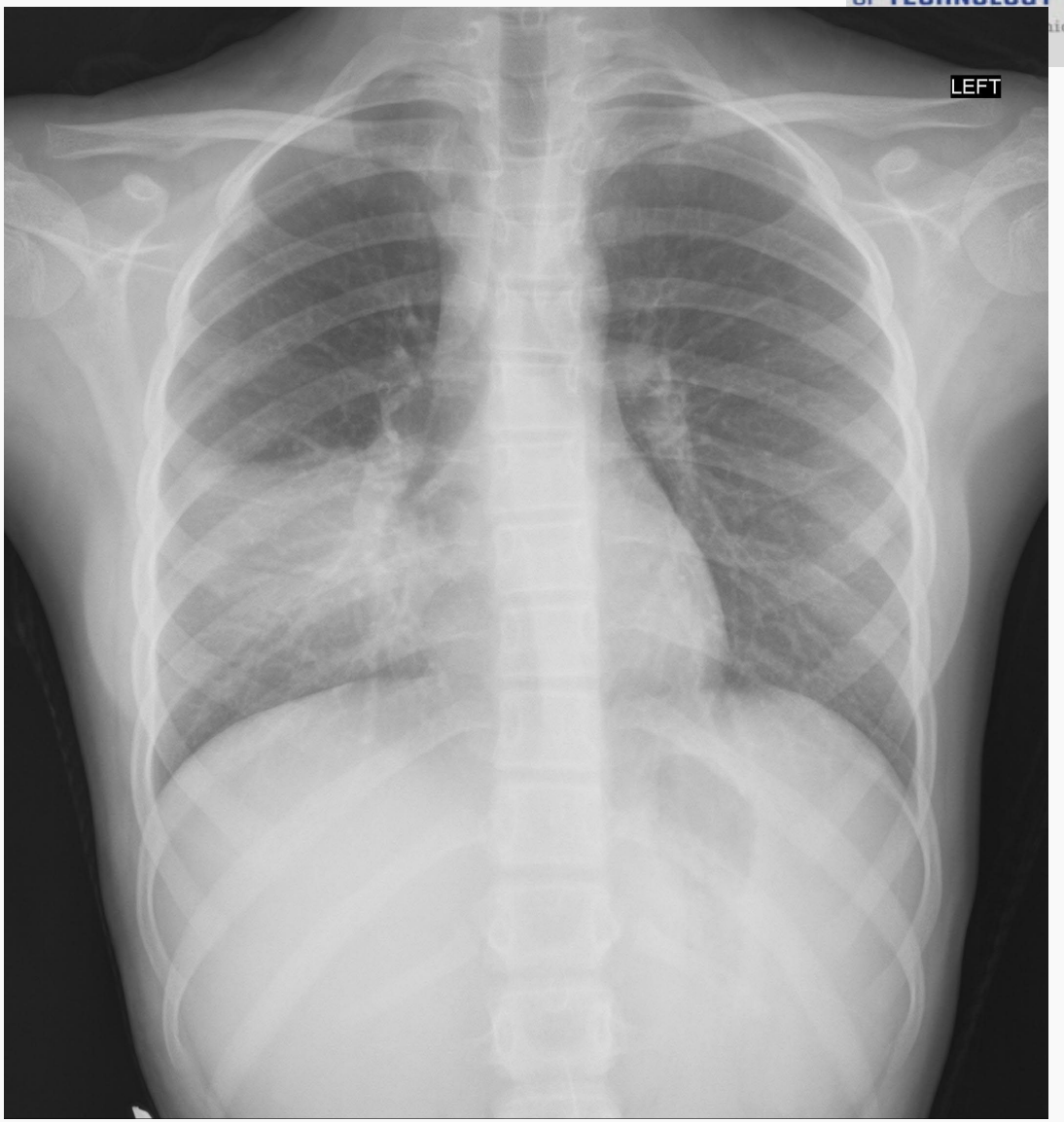
Right Middle Lobe Consolidation

Right Middle Lobe Consolidation



-Area of increased density on patient's right

-Obscures right heart boarder (silhouette sign)



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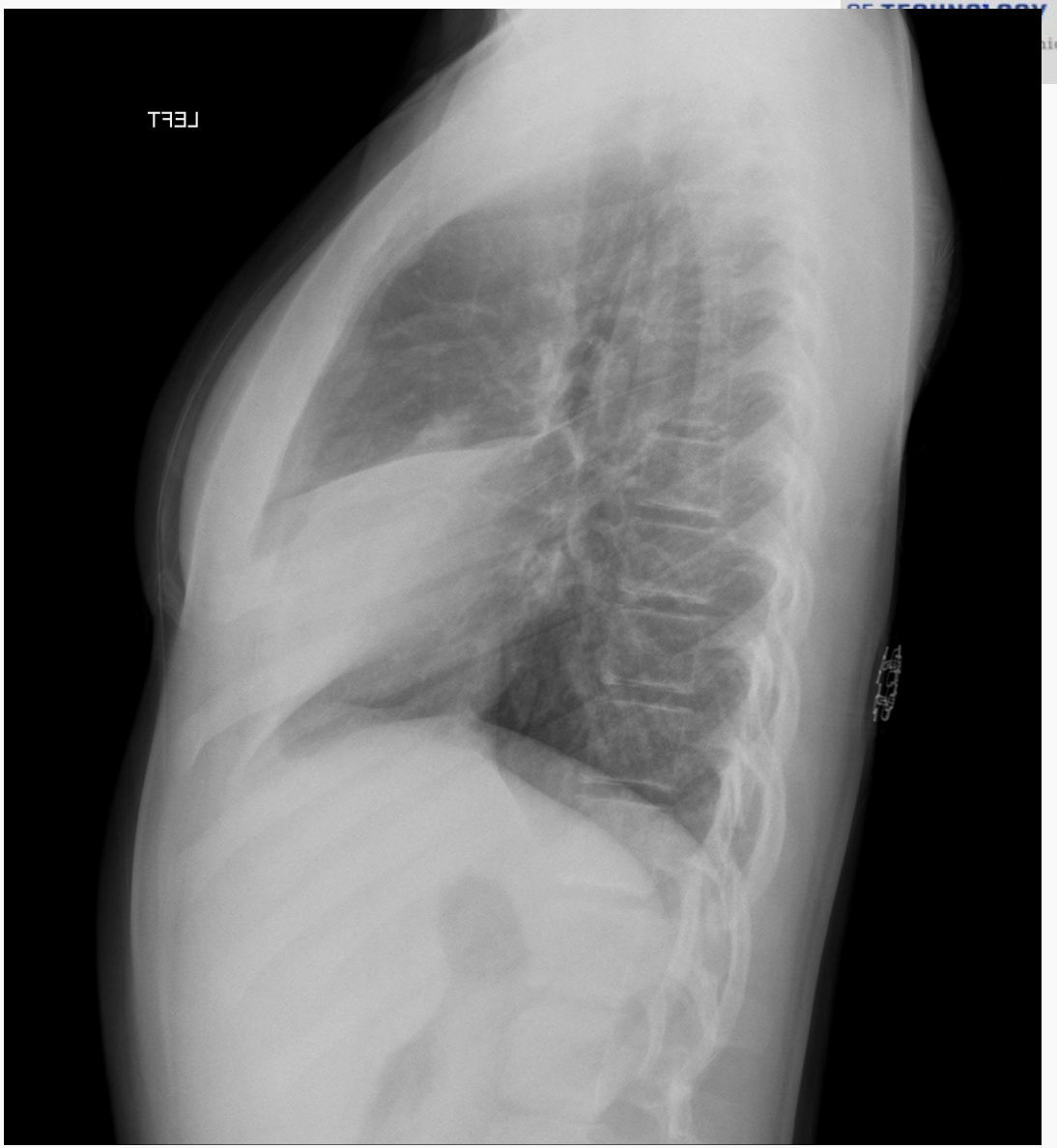
Area of increased density superimposed on heart

Visible minor fissure as superior boarder (silhouette sign)



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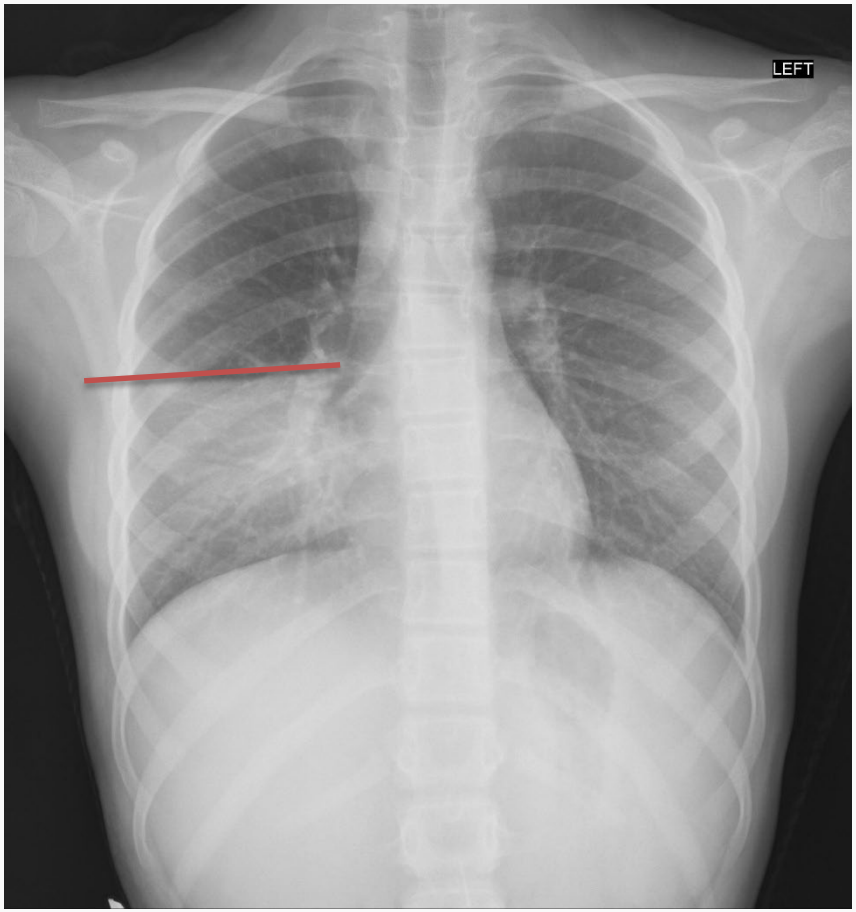
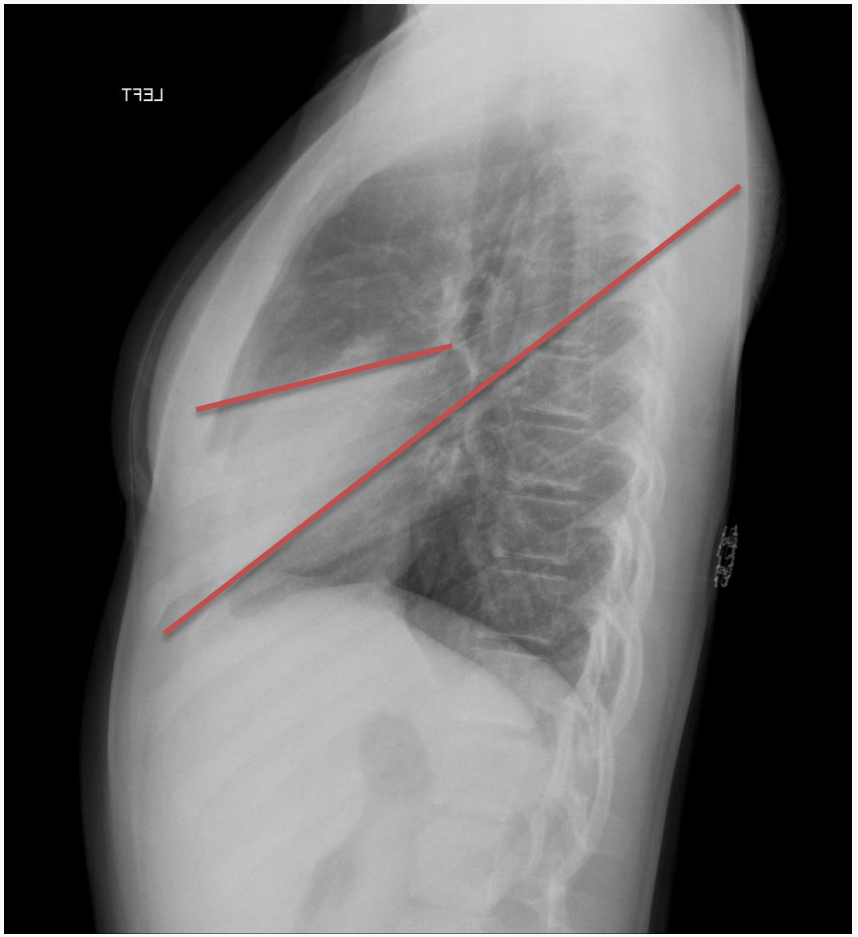
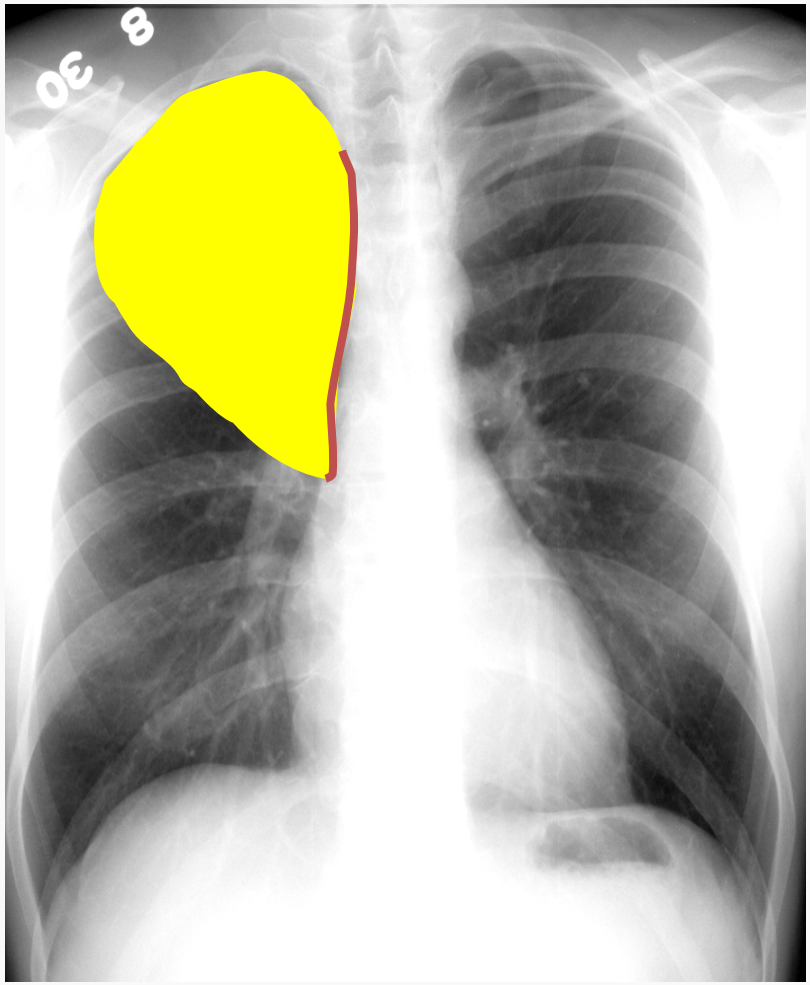
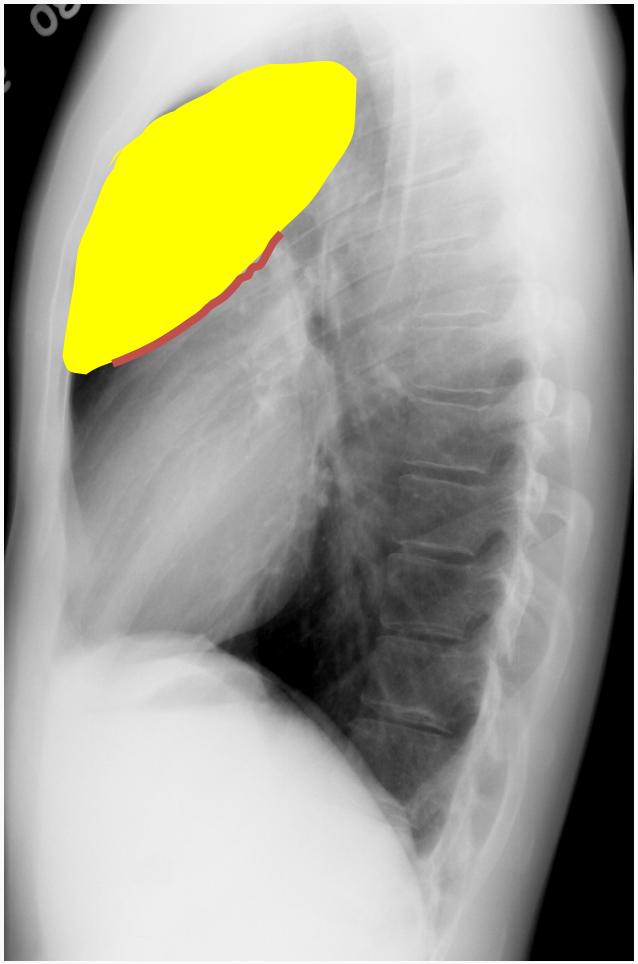
Medicine

RML pneumonia

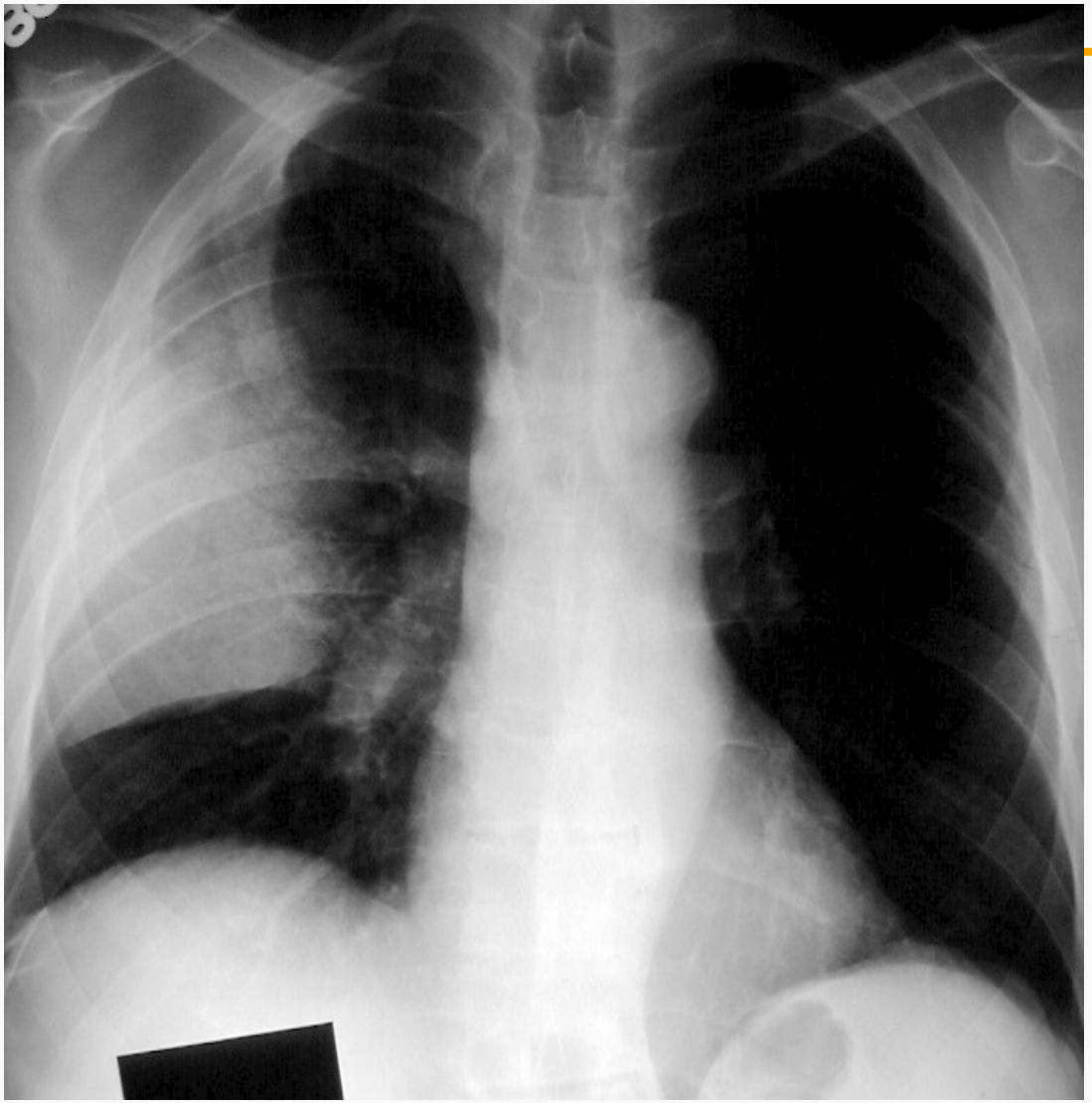


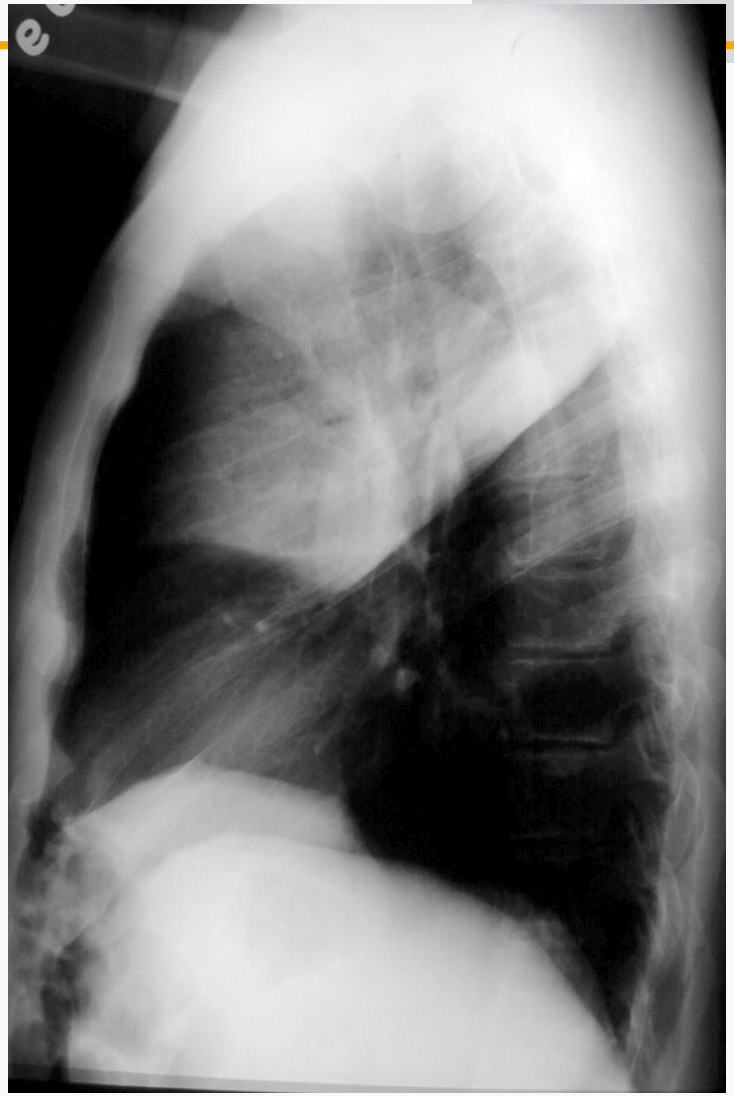




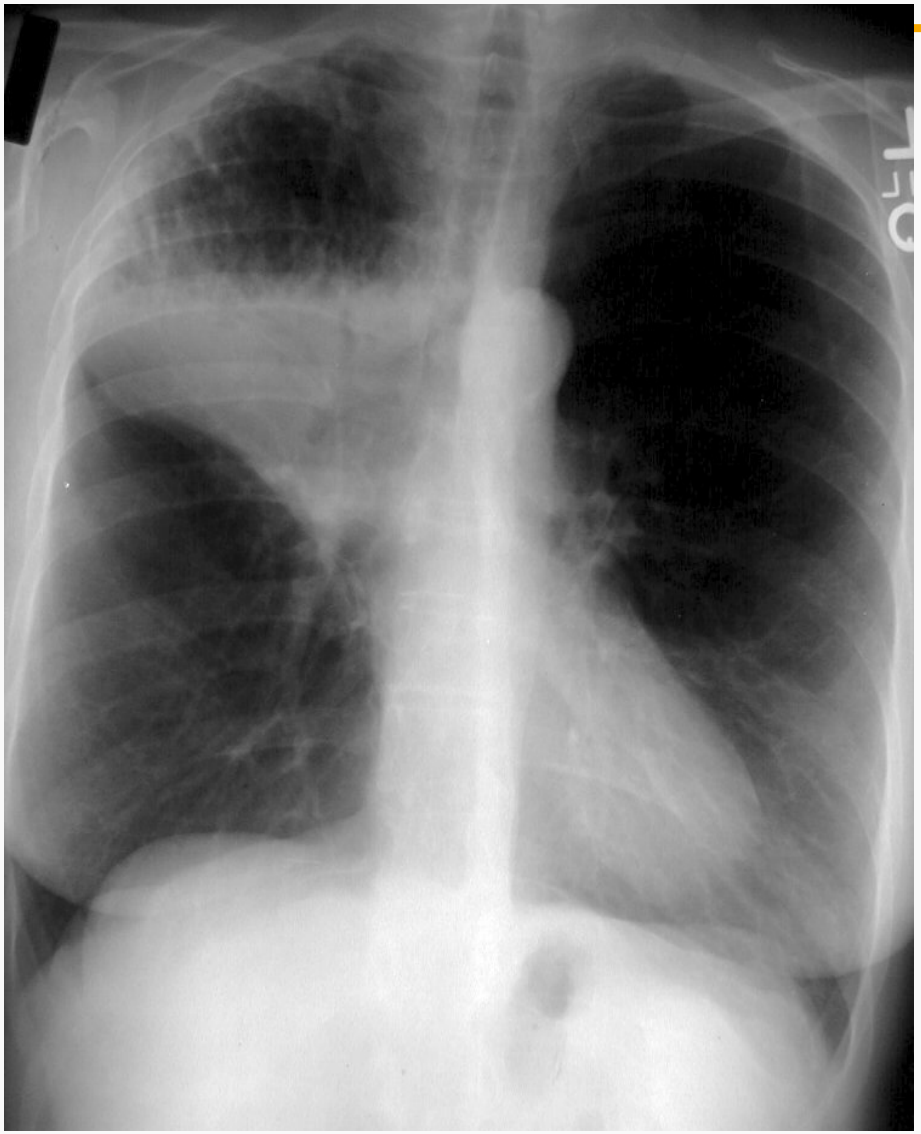


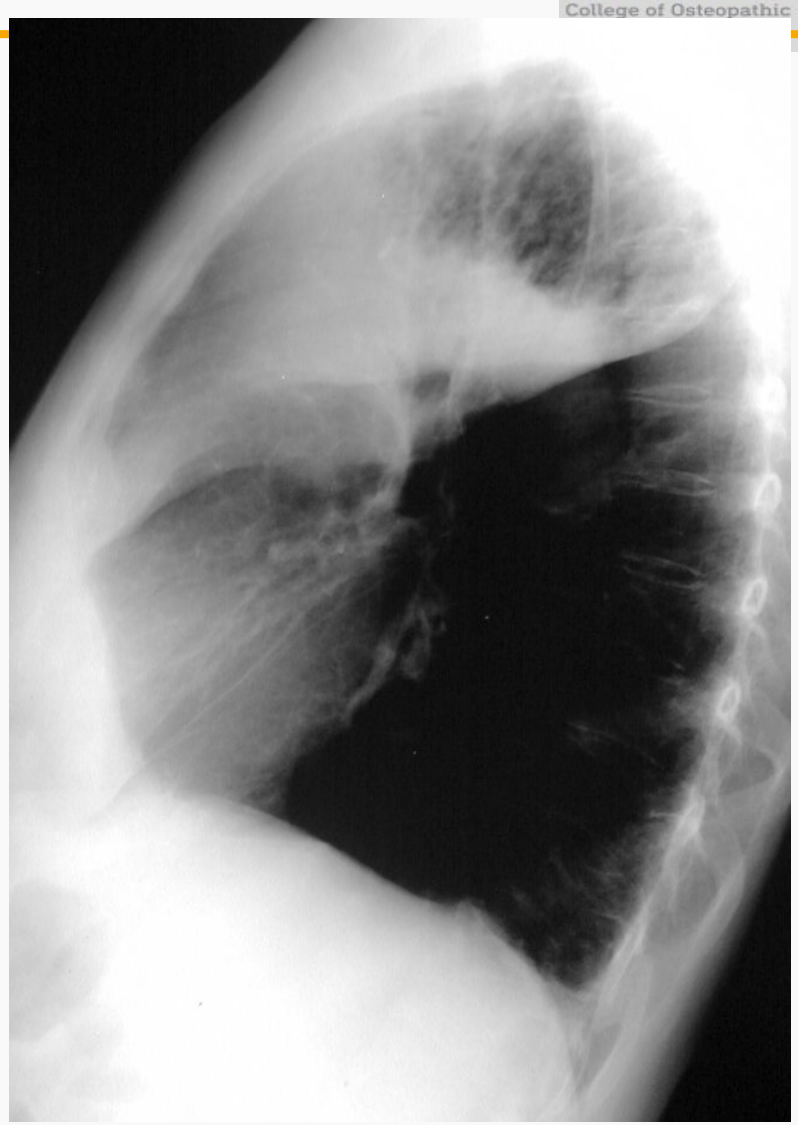
Right Upper Lobe Consolidation





Right Upper Lobe Consolidation





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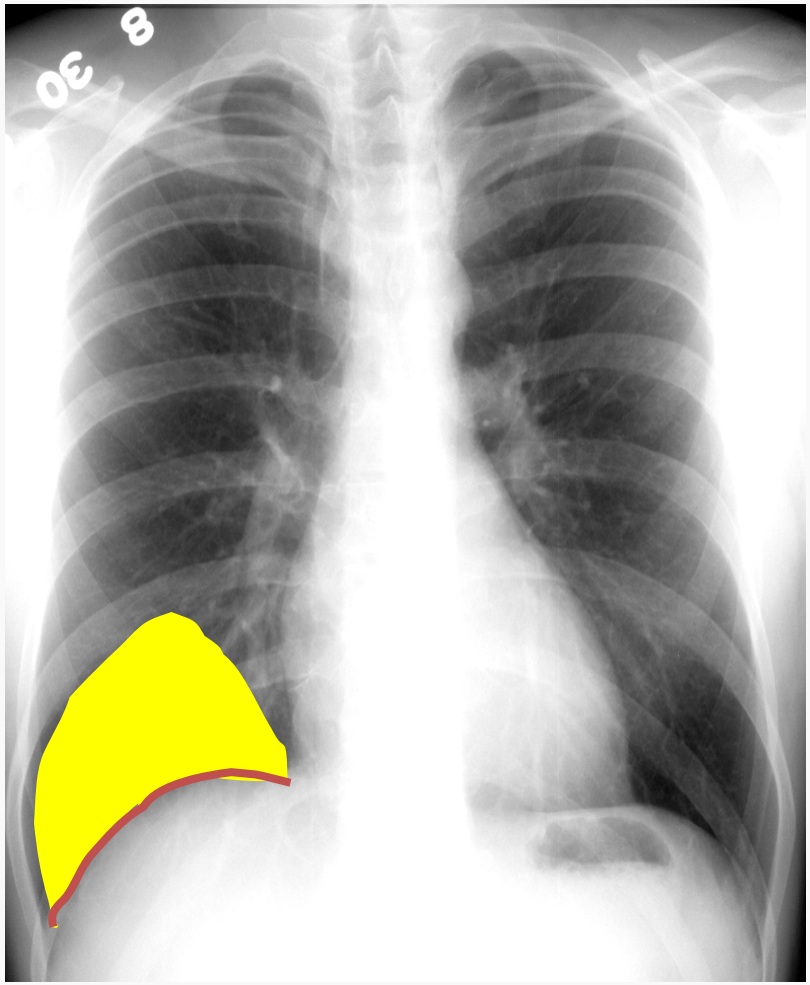
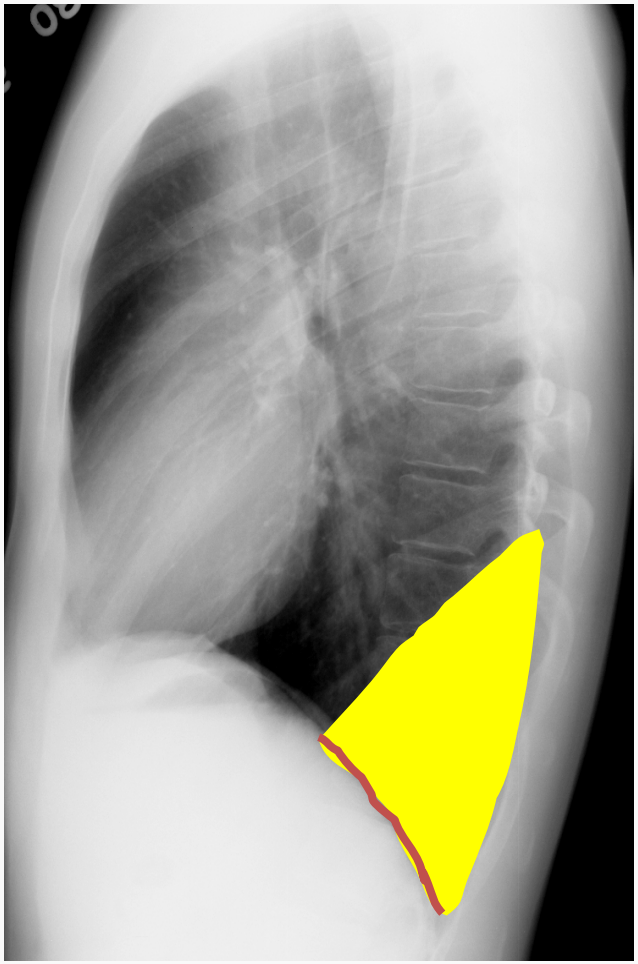
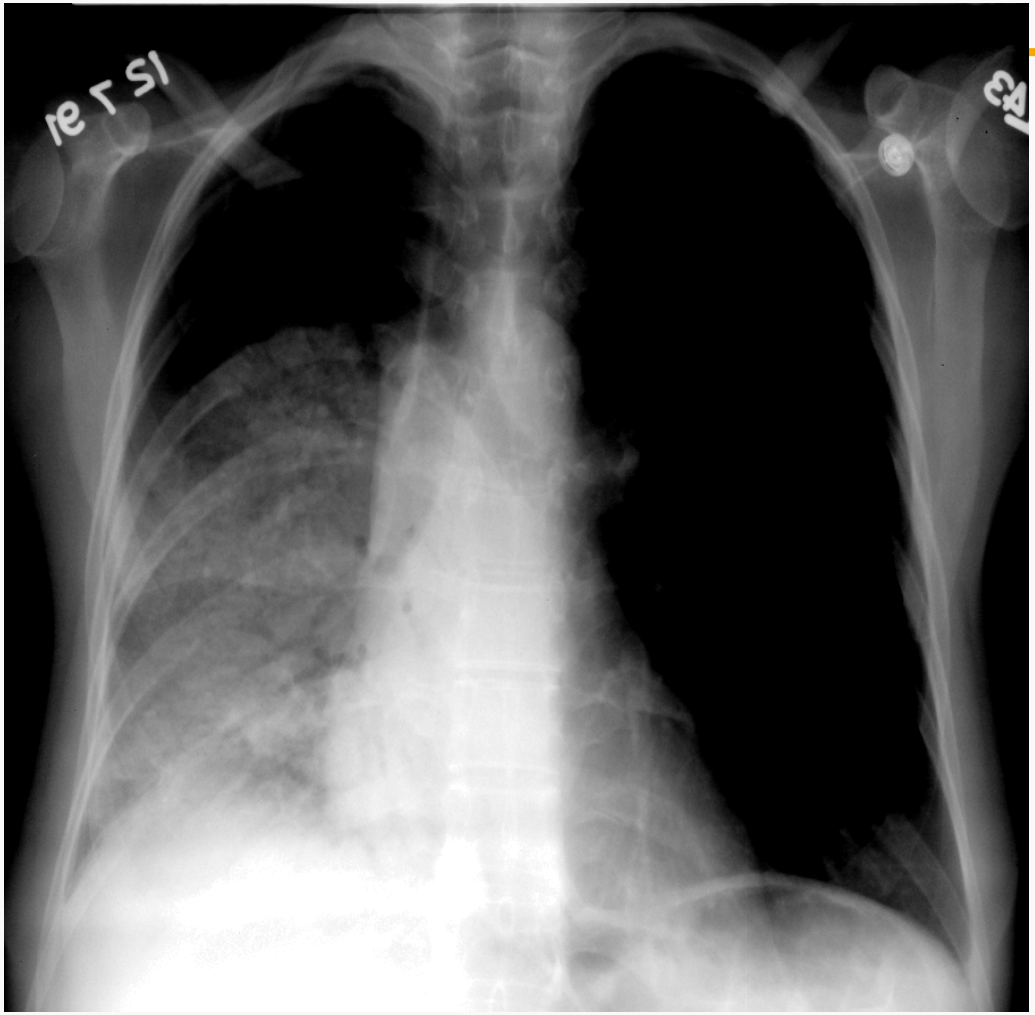
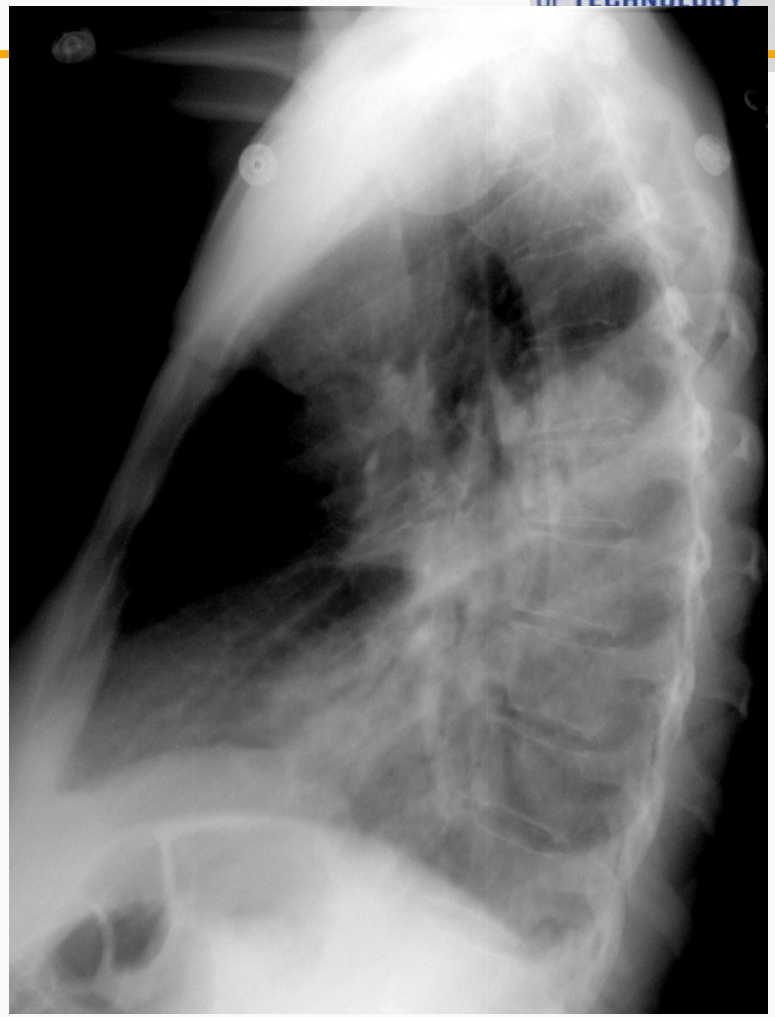




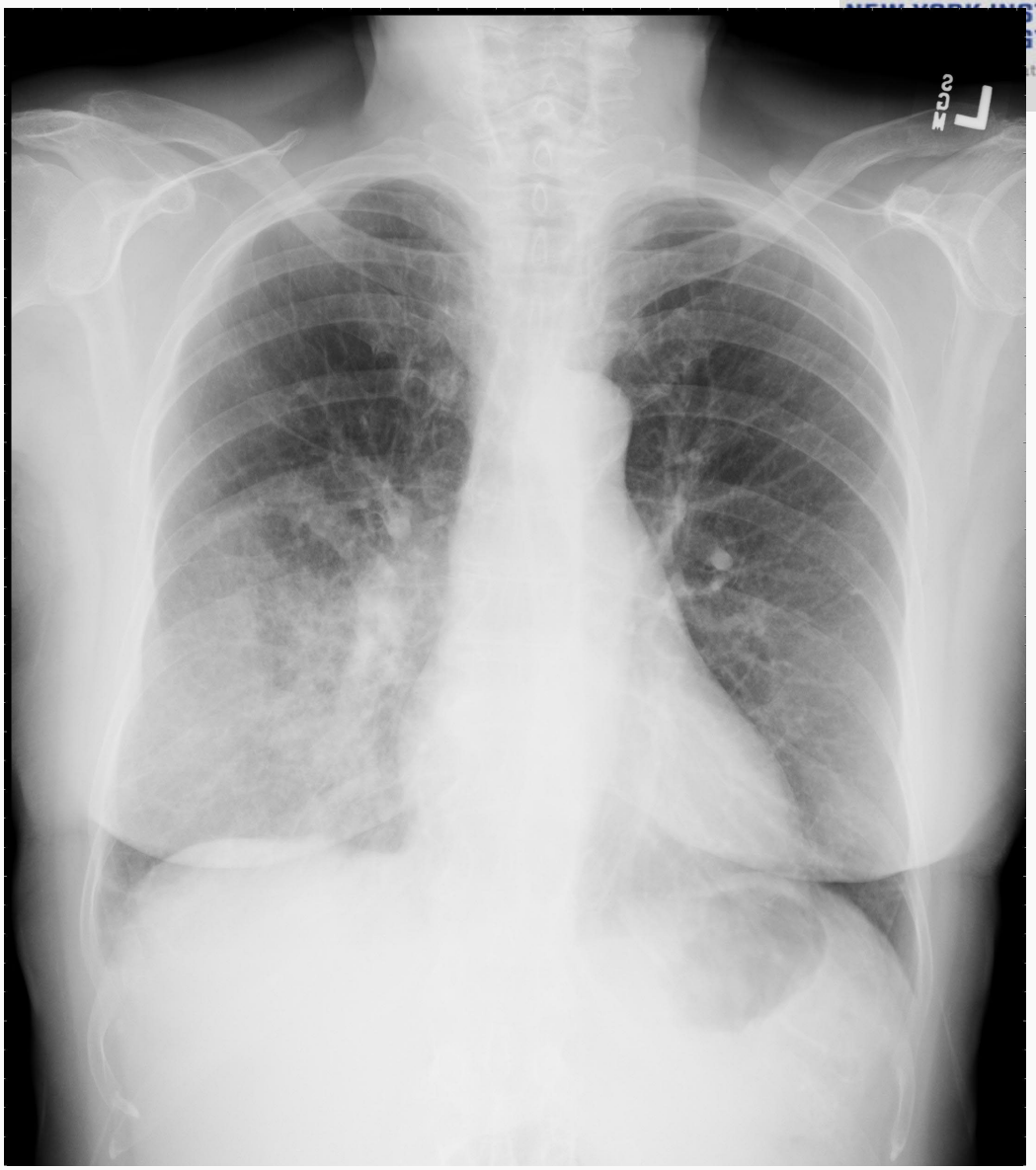
Right Lower Lobe Pneumonia





Right Lower Lobe Pneumonia

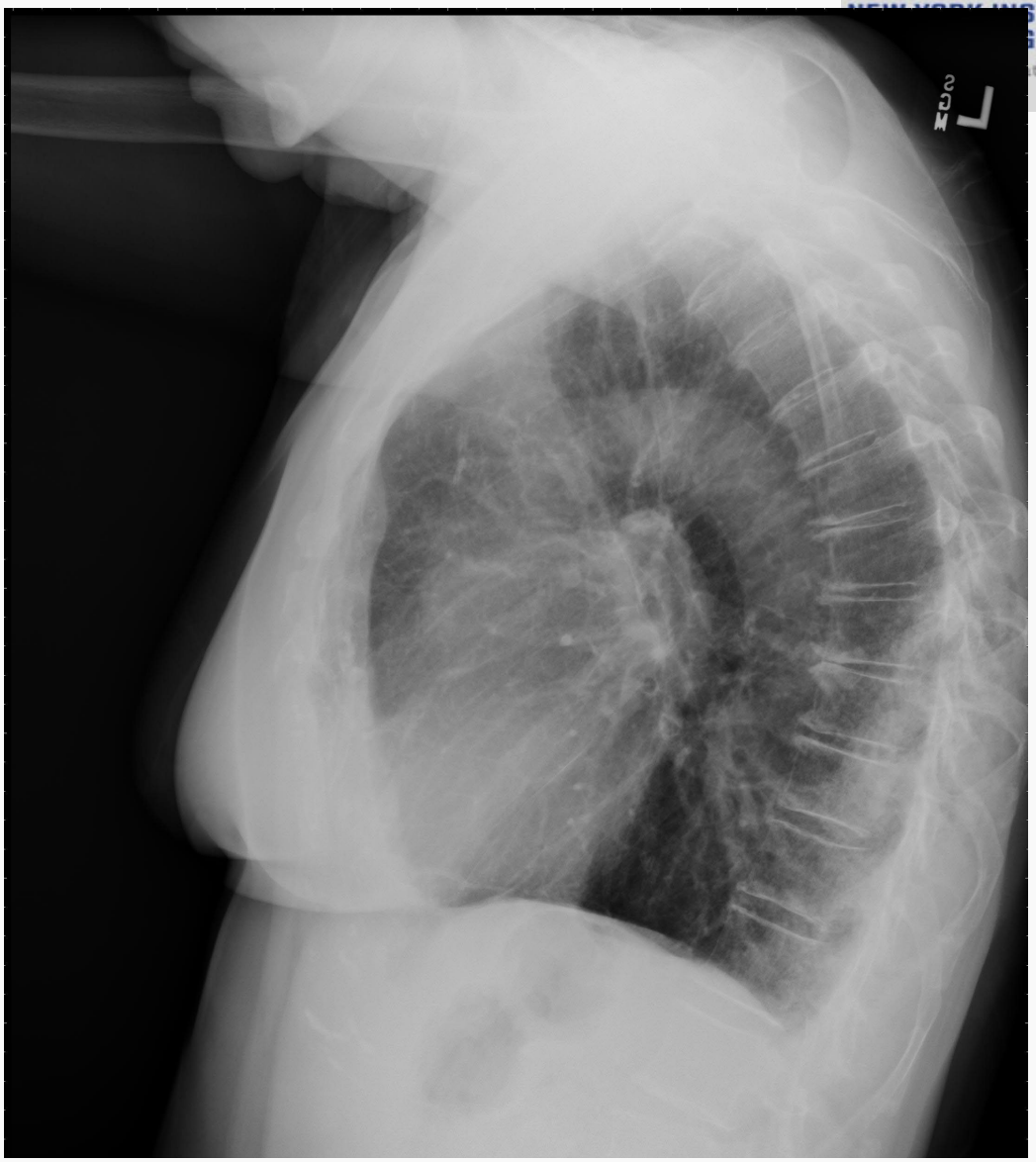
Area of increased density on right, extends to heart boarder but right heart boarder is sharp



Area of increased density is posterior,

below major fissure, overlies vertebral bodies,

which look whiter lower down



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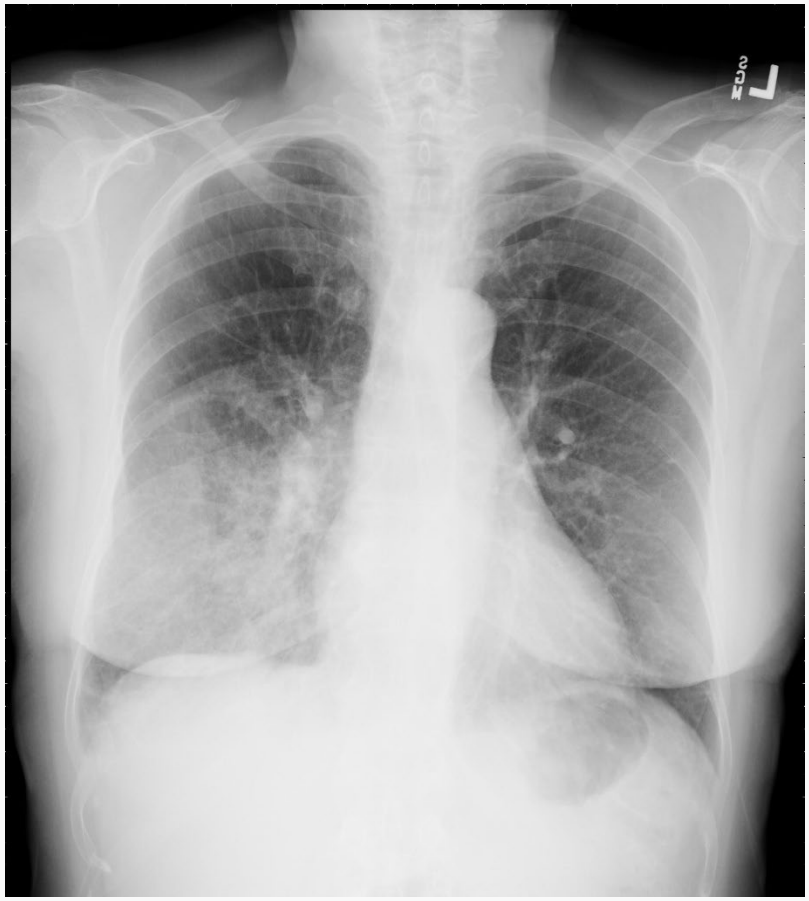
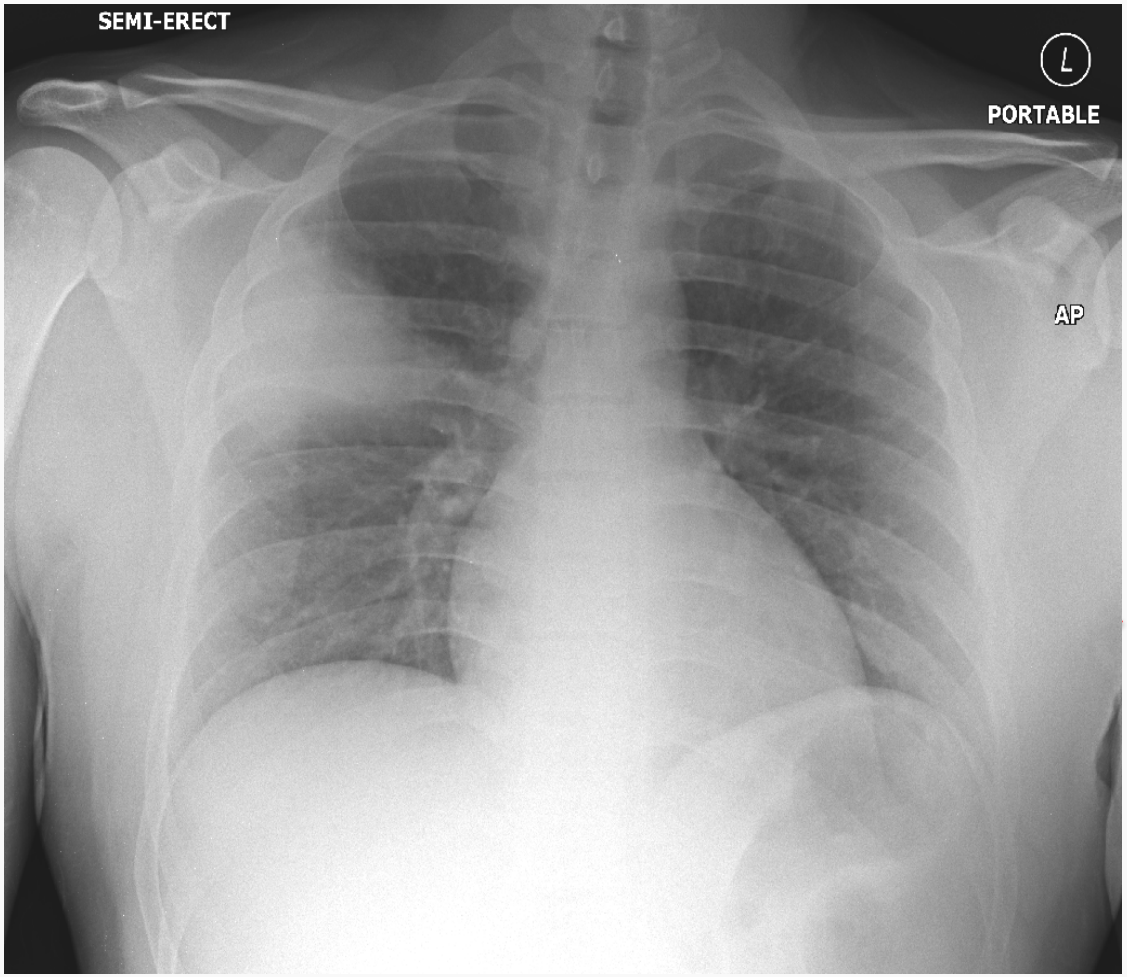
RLL pneumonia



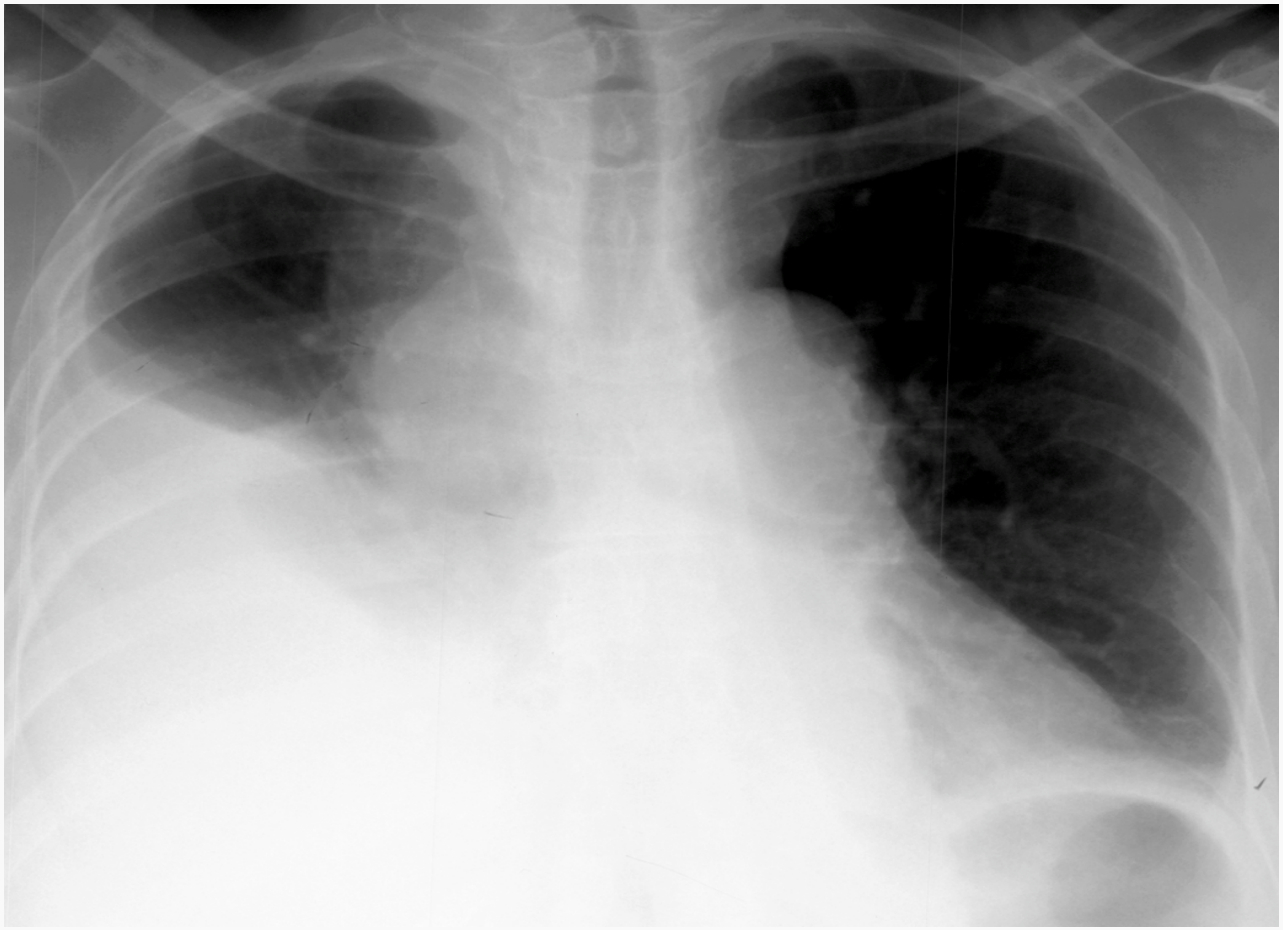
Right upper lobe pneumonia



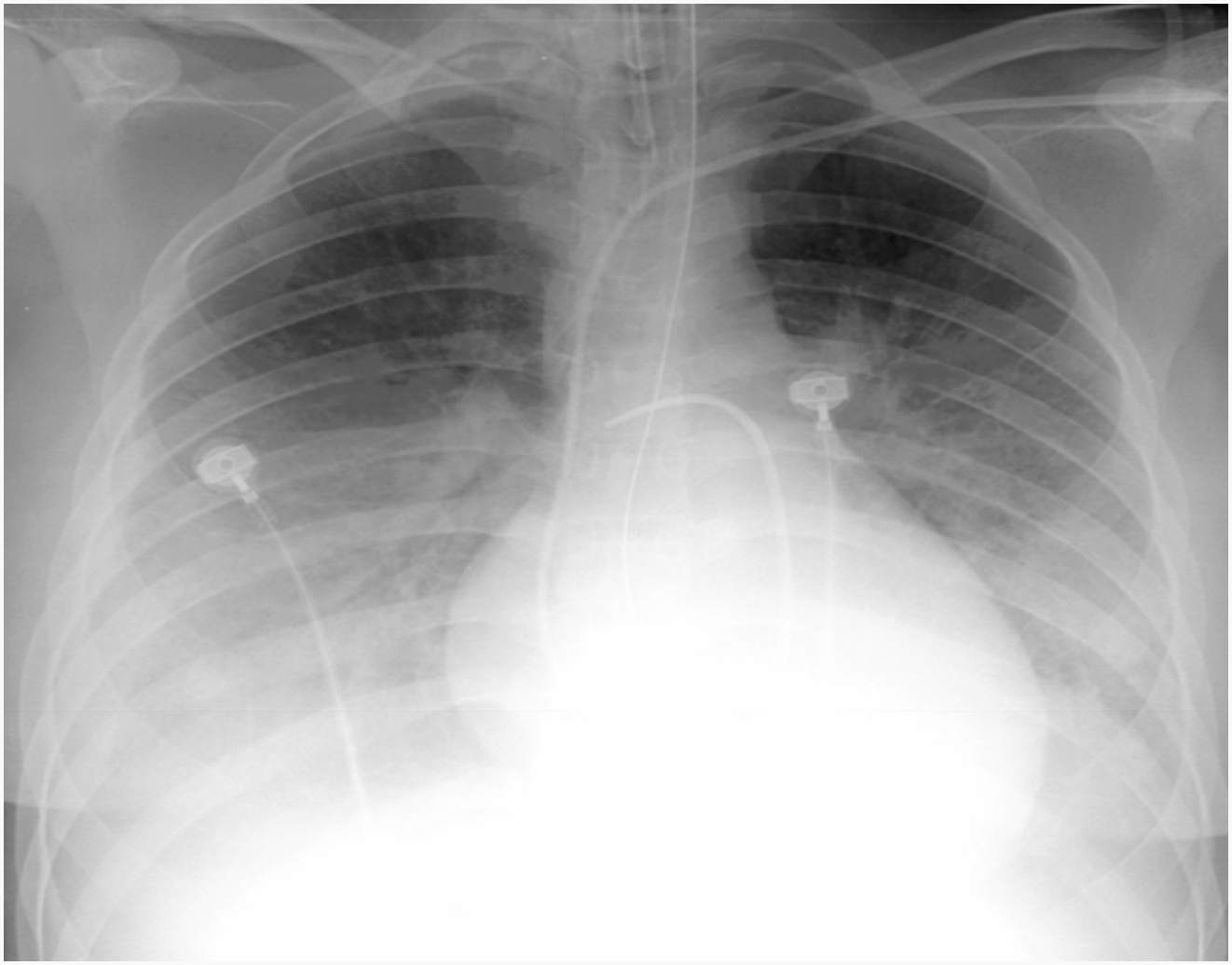
Pleural effusions

- Chest radiographs are the most commonly used examination to assess for the presence of a pleural effusion; however, it should be noted that on a routine erect chest x-ray as much as 250-600 mL of fluid is required before it becomes evident.
- Features include on a frontal film:
  - blunting of the costophrenic angle
  - blunting of the cardiophrenic angle
  - fluid within the horizontal or oblique fissures
- Eventually, a meniscus will be seen, on frontal films seen laterally and gently sloping medially
- Lateral films are able to identify a smaller amount of fluid as the costophrenic angles are deepest posteriorly.

*Large pleural effusion

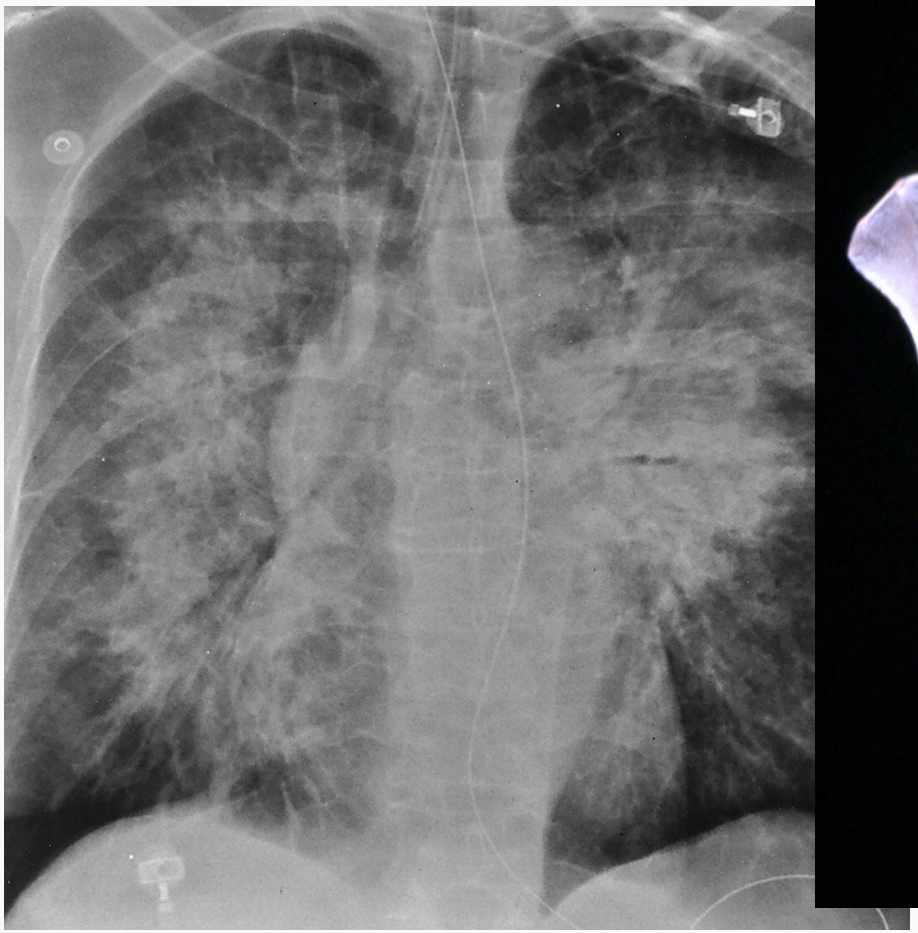


Large layering pleural effusions on a semiupright radiograph



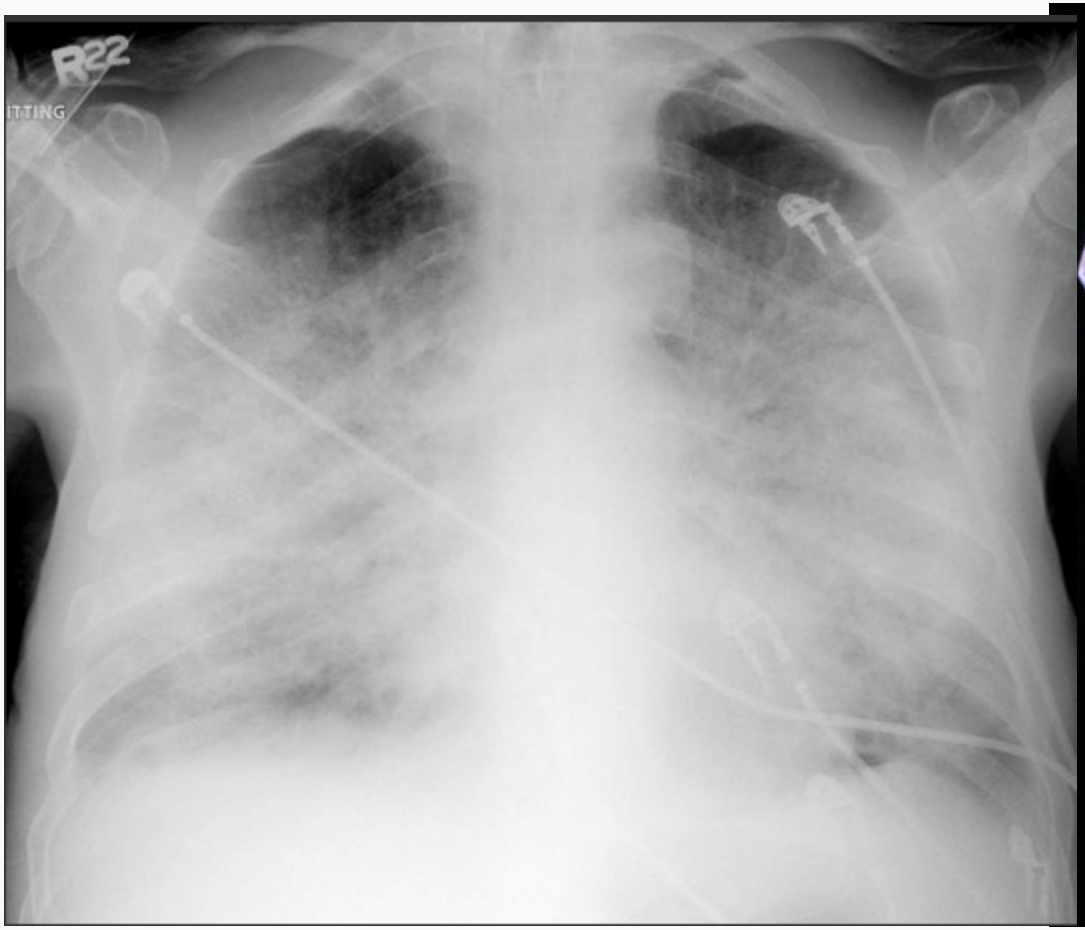
- Large amounts of fluid can be present on supine films with minimal imaging changes, as the fluid is dependent and collects posteriorly.
- There is no meniscus, and only a veil-like increased density of the hemithorax may be visible.

*Pulmonary edema-classically "batwing" in appearance



Gluecker T et al. Radiographics 1999;19:1507-1531

*Pulmonary edema-classically "batwing" in appearance



Gluecker T et al. Radiographics 1999;19:1507-1531

*Pulmonary edema-before and after lasix

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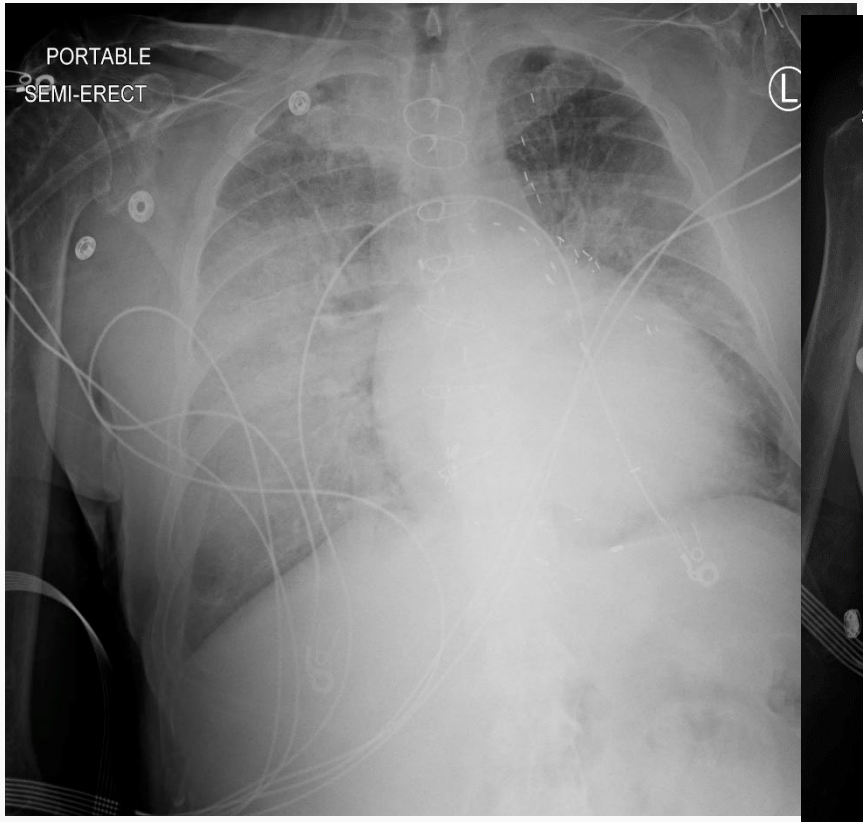

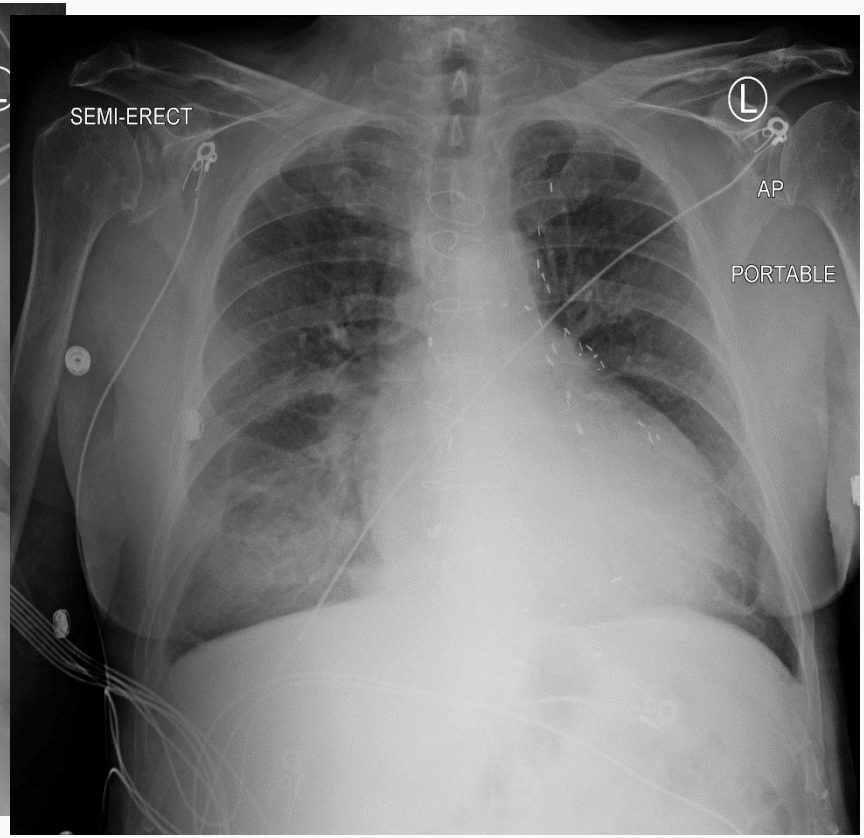

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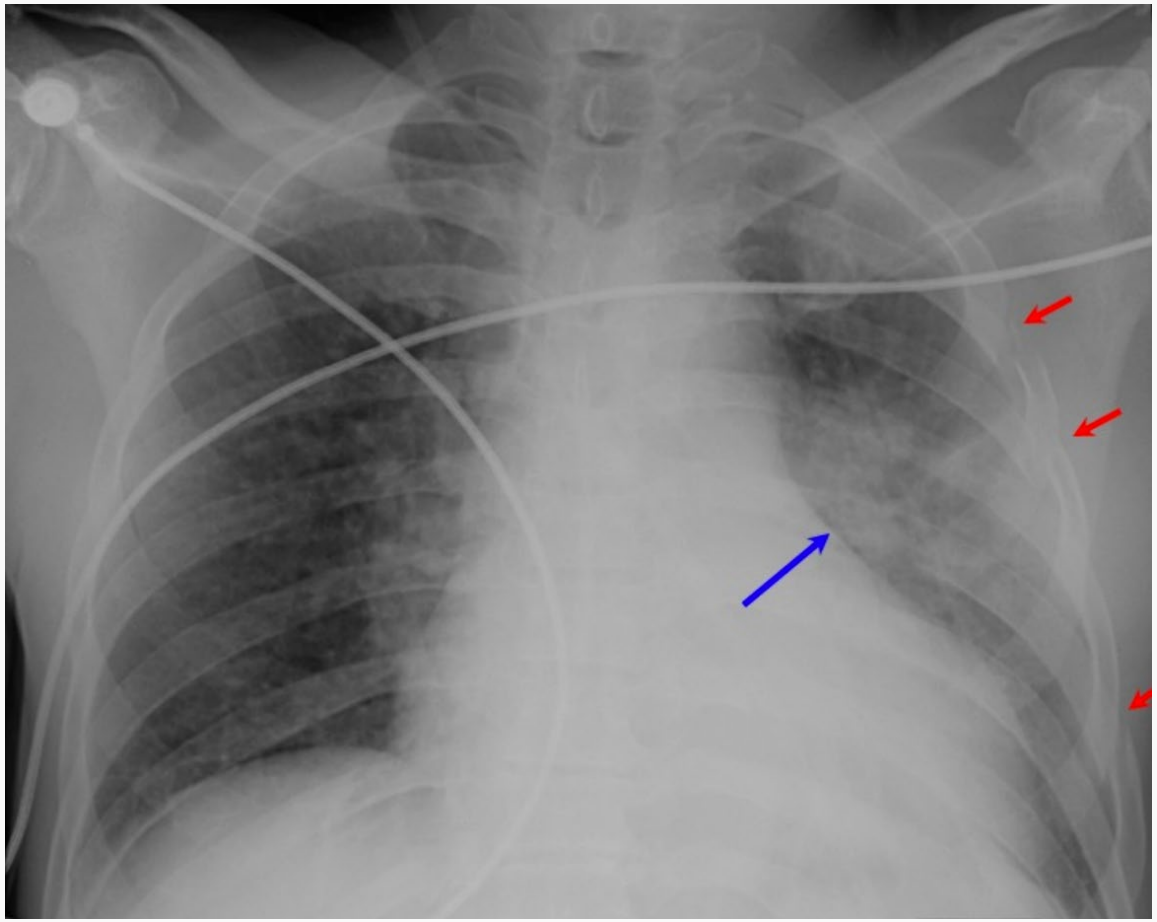
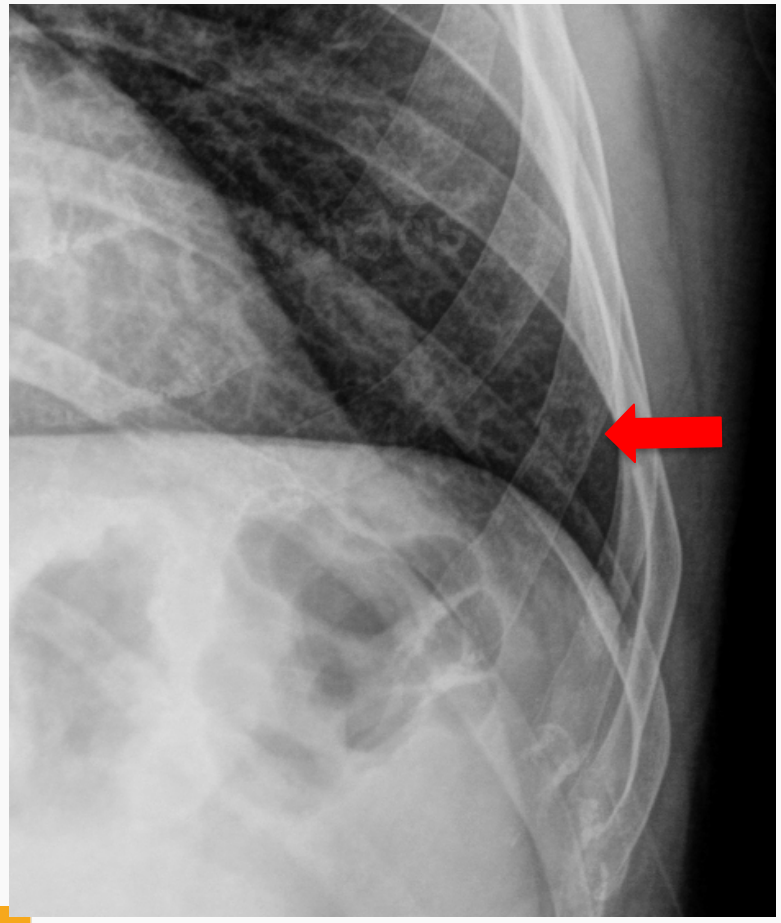
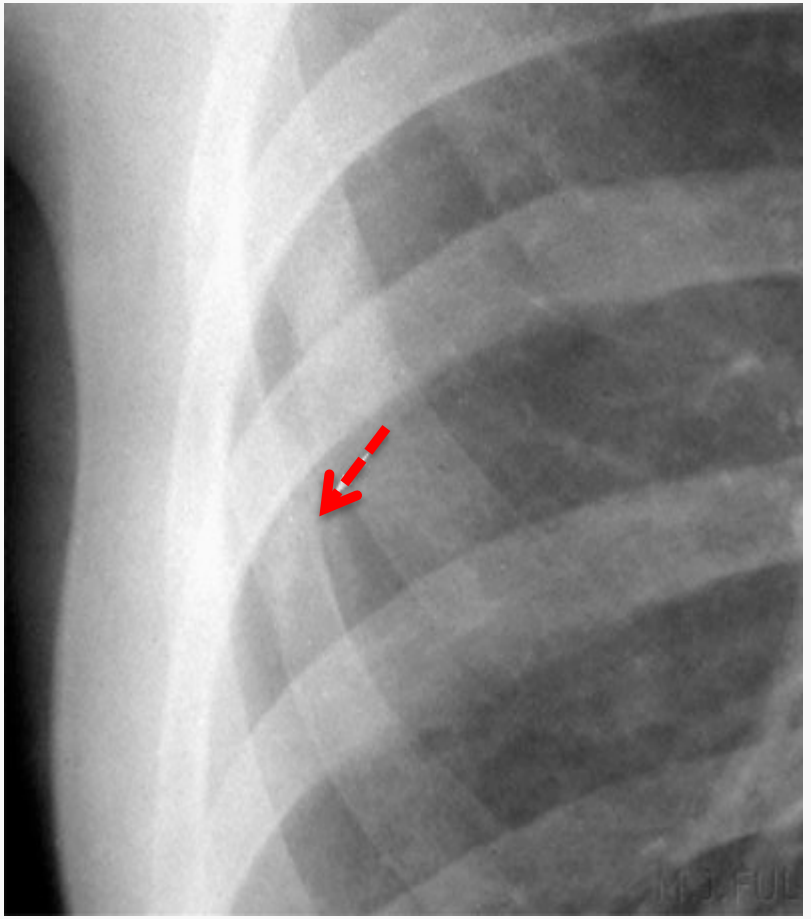
Rib fractures

- Look at all views! A fracture may only be well seen on one view
- Look for associated pneumothorax, pulmonary contusion, and other fractures especially of scapula and shoulder. Pulmonary contusions may however be present without rib fractures.
- Look for callus-many patients have OLD rib fractures
- Patients with more that 3 acute rib fractures generally need to be admitted/observed
- Fractures of first through third ribs may be hard to see on plain radiographs, but are associated with high impact injuries

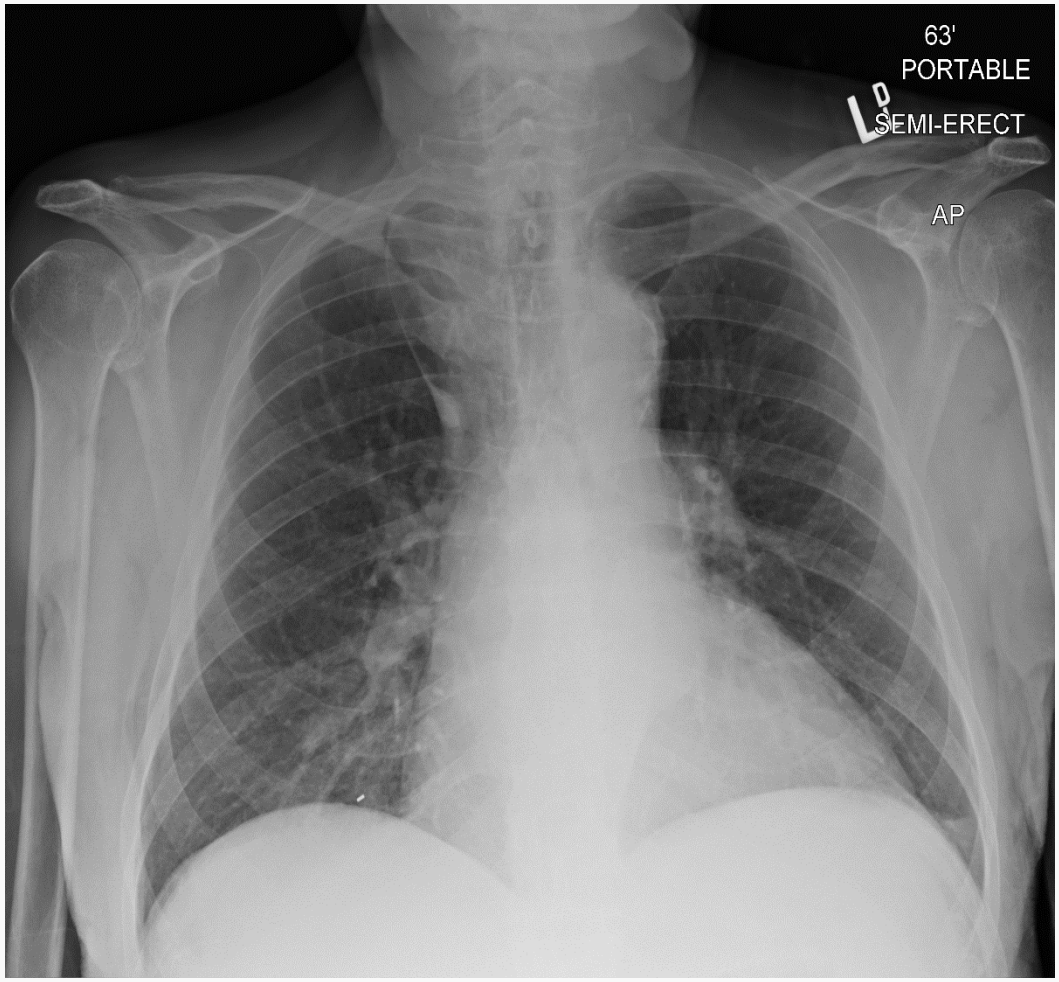
Rib fractures with pulmonary contusion

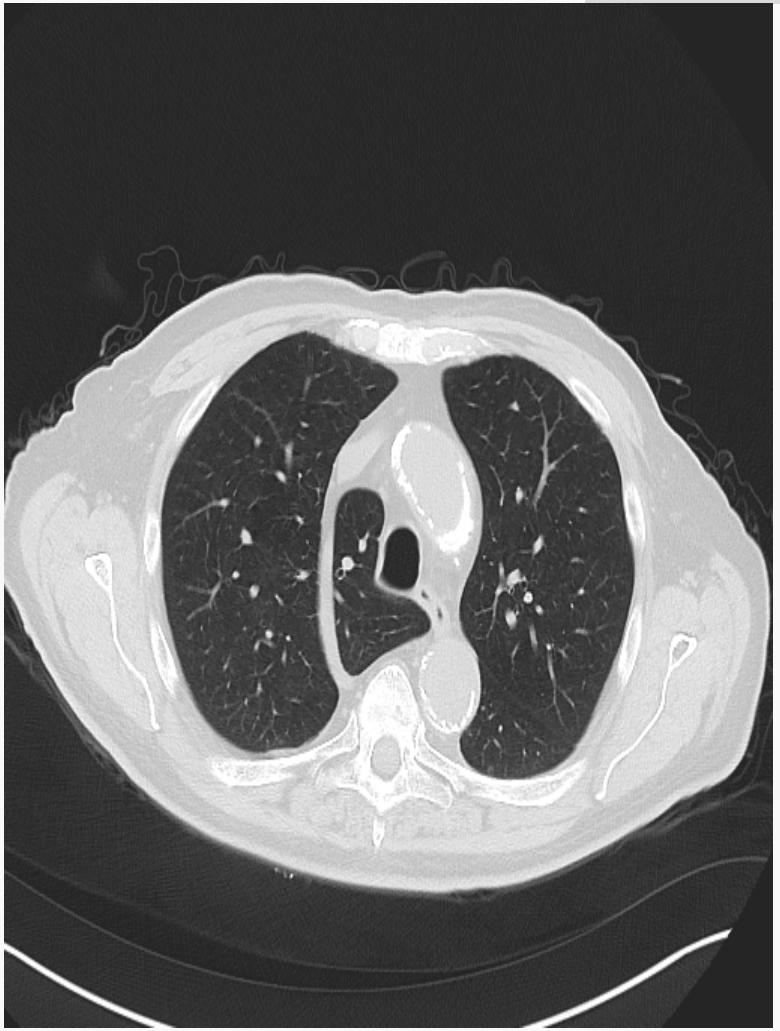






Azygous fissure





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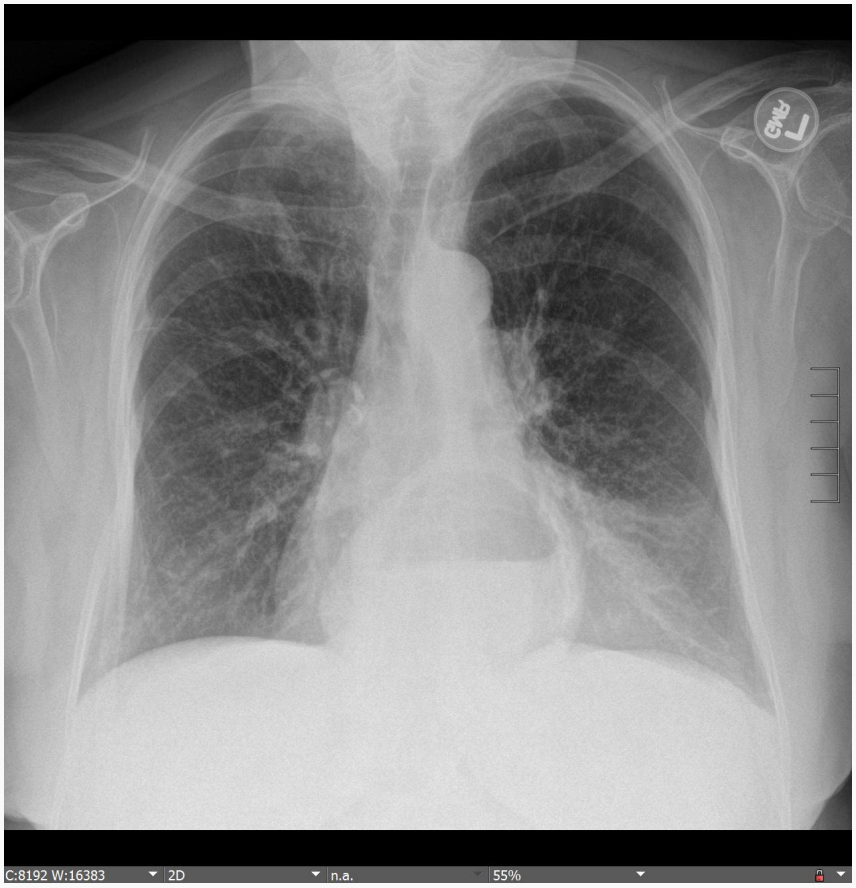
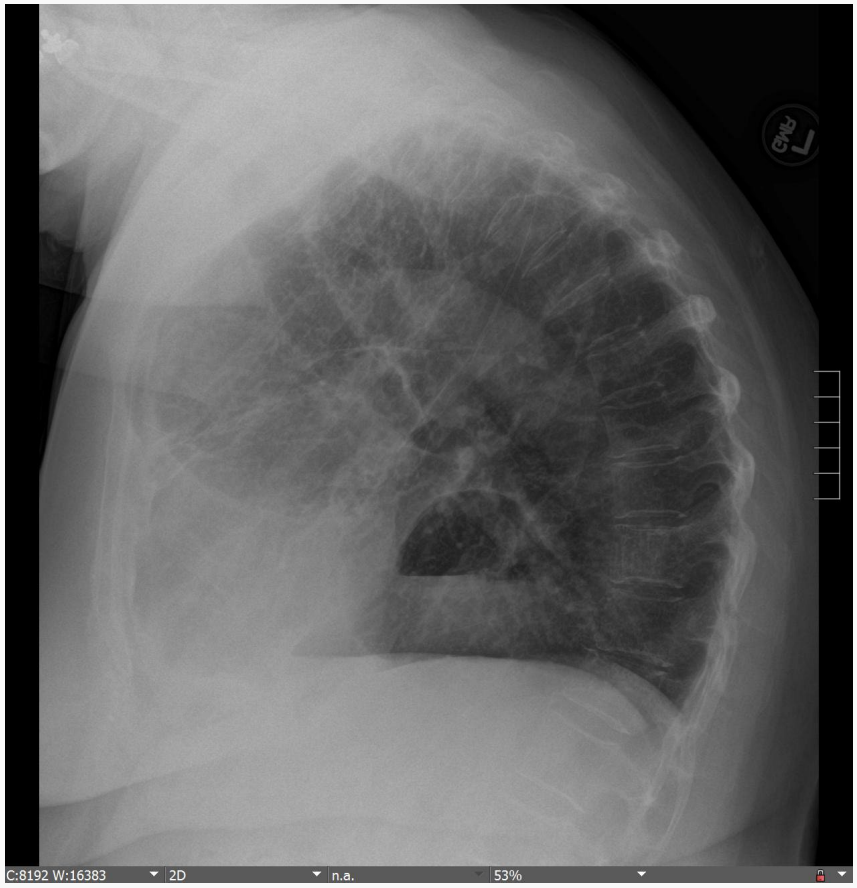
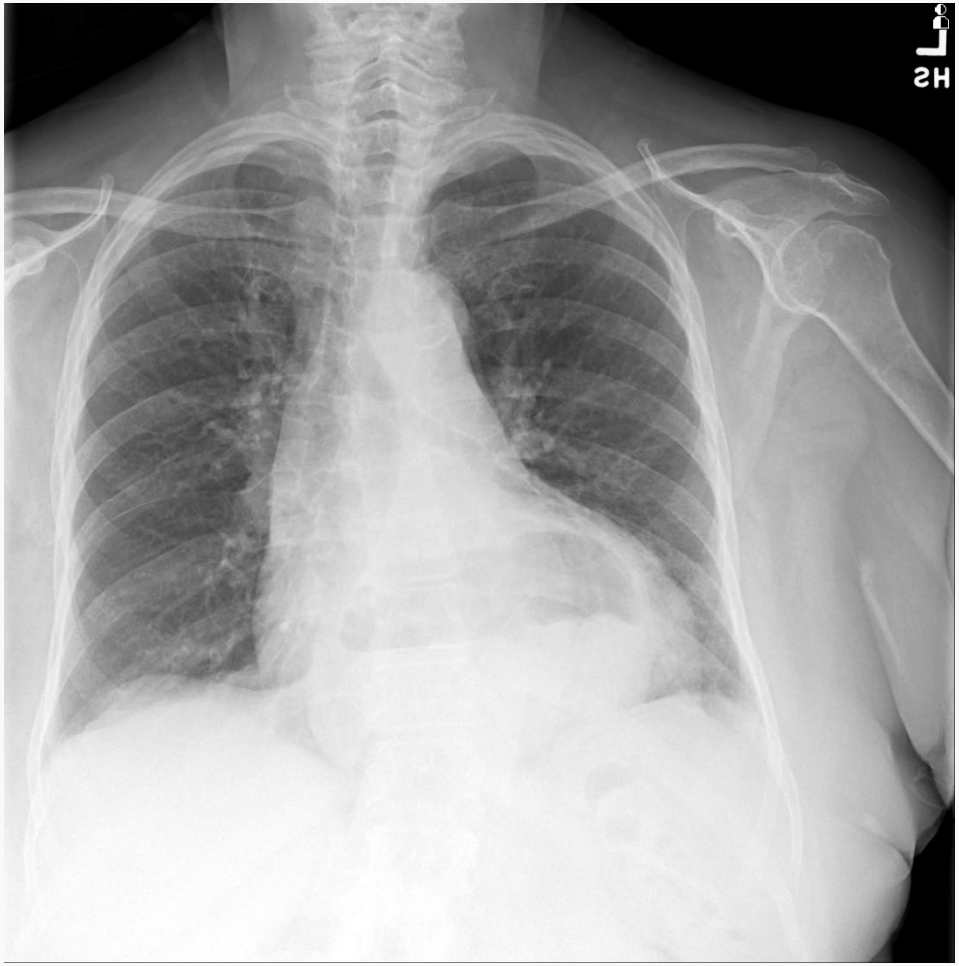
Large Hiatal hernia with air fluid level

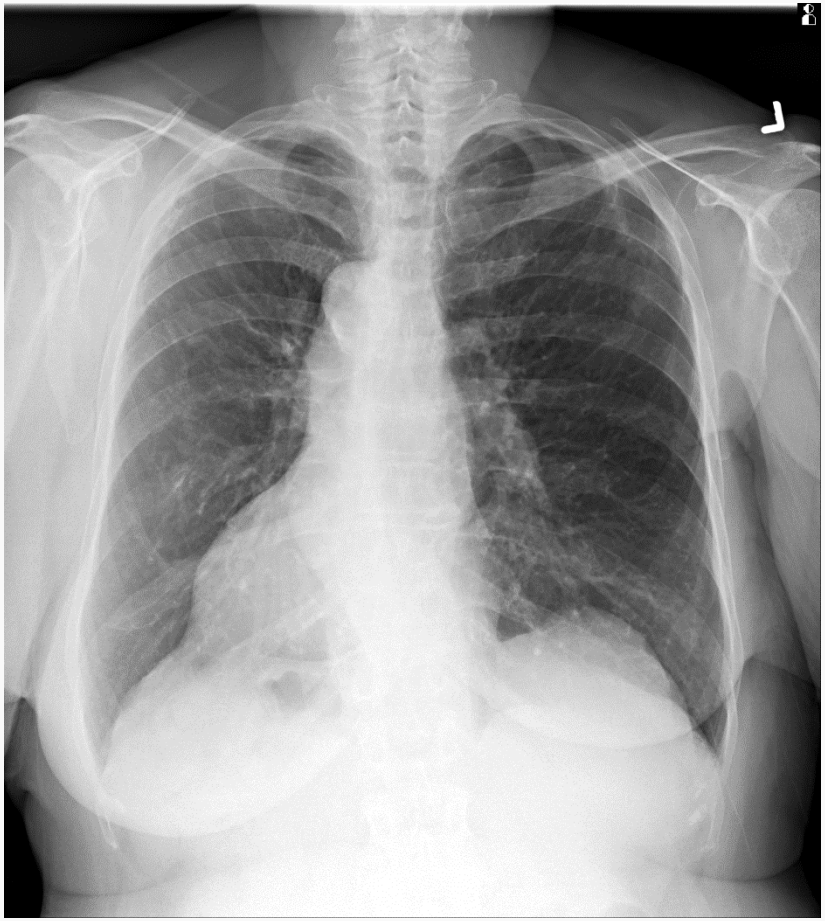




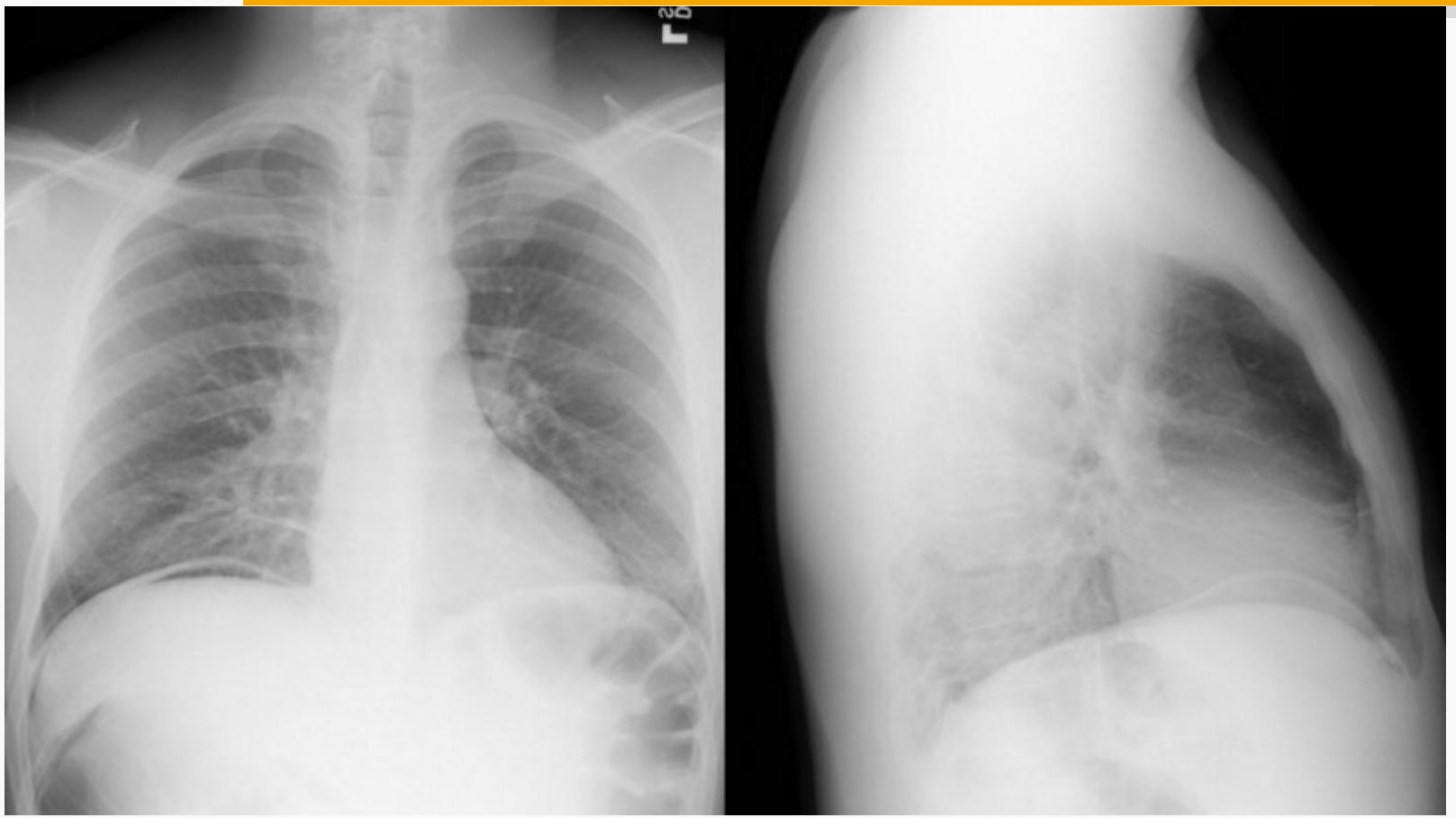
Hiatal hernia



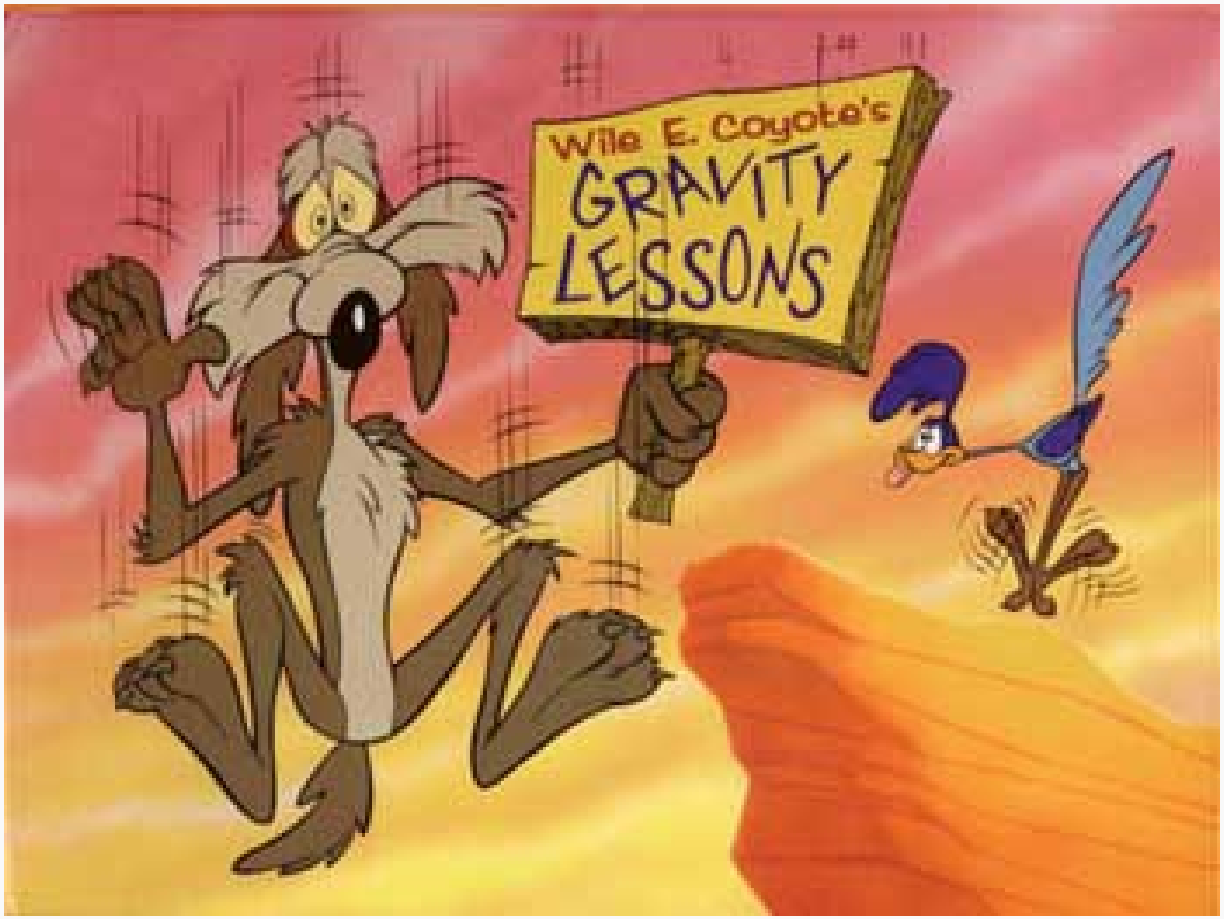
Situs inversus (*much less common than mislabeled film)



*Free intra-peritoneal gas



Hopefully all this will keep you from falling into trouble …



…as you continue on your journey to become outstanding physicians!

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Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R